BELIEFS AND HETEROGENEITY: CREDIBILITY UNDER LARGE SHOCKS

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Abstract. Inflation can persist after a large shock for reasons that go beyond the shock itself. One such source is the monetary policy reaction, when a central bank allows its short-run inflation target to shift with the size of the disturbance. In a two-sector model, that contingent shift that extends the relative convergence horizon improves the output–inflation trade-off, reducing output losses while tolerating a more sustained inflation path. The channel works only through monetary policy credibility, since the target movement reaches the economy through private expectations of future policy path. When the nominal anchor is credible, extending the policy horizon increases short-run inflation and lowers output losses together, but when it is not, the same extension cumulates inflation but deepens losses. The last-mile inflation problem thus reflects, in part, a belief-mediated displacement of the nominal anchor.

JEL Classification: E31; E32; E52
Keywords: Inflation Persistence; Large Shocks; Monetary Policy Credibility; Sectoral Heterogeneity

1. Introduction

The inflationary episode that followed the COVID-19 pandemic has proven unusually difficult to bring to a close (BIS, 2024; IMF, 2024; Federal Reserve Board, 2025). Median inflation across advanced and emerging economies rose to nearly ten percent in 2021 and 2022, then fell quickly but not completely. Inflation came down from its peak yet settled above pre-pandemic levels, and since late 2023 it has stayed above the targets of most central banks (OECD, 2024; ECB, 2025). This residual phase, in which the final return to target remains incomplete, constitutes a *last-mile convergence problem*. What stands out is how slowly the final inflation adjustment happens. This persistence is only partly explained by accounts that link it mainly to the duration of the original shock.

The aggregate pattern also masks an important sectoral asymmetry. Much of the initial surge was concentrated in goods-related prices, while the later phase of persistence was increasingly associated with services inflation, which adjusted more slowly after goods inflation receded (Eo et al., 2023; Amatyakul et al., 2024; Martínez Hernández et al., 2025). This sectoral separation is therefore relevant for the last-mile inflation convergence problem.

One possible source of this persistence lies in the conduct of monetary policy itself. Standard frameworks usually treat the policy reaction as independent of shock size, so that large and small shocks receive responses that differ mainly in scale. The post-pandemic record suggests a more nuanced pattern. Central banks appear to have accommodated the surge to an unusual degree (Harding et al., 2023; English et al., 2024), allowing inflation to remain elevated rather than forcing a faster return to target under a fixed-objective benchmark. This accommodation operates through two distinct margins. The first is the degree of tolerance granted, reflected in how far the operational objective is allowed to rise during the episode. The second is the intended convergence horizon, reflected in the speed at which that tolerance is gradually withdrawn. Observed policy conduct bundles these margins into a single stance, and the question pursued here is whether a shock-size-dependent policy reaction, transmitted through the credibility-mediated belief formation of private agents, can account for the persistence that follows large supply shocks.

This pattern motivates the structural exercise, which provides one interpretation of the observed policy conduct. Across a panel of economies, monetary-policy conduct during the episode is consistent with a more accommodative short-run inflation objective. The model

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provides one interpretation of this pattern. A two-sector economy is augmented with a shock-size mechanism, so that sufficiently large disturbances shift the persistent component of the short-run inflation target. This shift reaches the rest of the economy only through the signal-extraction problem that agents solve when the short-run target is imperfectly observed, which makes policy credibility the channel through which the mechanism works.

Allowing the target to shift with the size of the shock improves the output–inflation trade-off. Cumulative output losses fall while inflation follows a more sustained path, so the mechanism tolerates higher inflation rather than removing it, with the largest gain arising under cost-push shocks. This advantage depends entirely on credibility. Under a credible anchor, a mild extension of the policy horizon shortens the inflation-convergence horizon and lowers output losses at the same time. Under an impaired anchor, the same extension lengthens convergence and raises real-side costs. The intended accommodation does not pass into expected inflation, real rates tighten, and output absorbs the adjustment the mechanism was meant to spare it. Horizon flexibility is therefore no substitute for credibility, since what matters is whether the target path reaches beliefs, not how far it is extended.

The mechanism operates through beliefs rather than through conventional easing. When a disturbance exceeds the threshold, the persistent component of the target is displaced upward, and the policy rule, responding to the gap between inflation and this higher objective, tolerates a more sustained inflation path. The force of the adjustment runs through private expectations. Agents observe the target but not its decomposition into persistent and transitory parts, and they re-infer the persistent component through a Kalman filter whose gain measures the speed of learning and, with it, the credibility of the monetary authority. As expectations of future inflation rise, the real interest rate falls even as the nominal rate is higher. This real-rate channel, not nominal easing, relieves the output cost of the shock. Credibility is therefore the hinge. The same target movement that stabilizes activity under a credible anchor leaves output to absorb the adjustment when the authority is non-credible.

Literature Review. The post-pandemic inflation literature has emphasized several sources for the surge and its persistence. Some accounts stress demand pressures, fiscal expansion, and wage dynamics (Bianchi et al., 2023; Jordà and Nechio, 2023), while others emphasize supply disruptions, sectoral bottlenecks, and relative-price shocks (Guerrieri et al., 2022; Afrouzi et al., 2024; Fornaro and Wolf, 2023). Cross-country decompositions suggest that the dominant driver varies across countries and periods (Bernanke and O. Blanchard, 2025; Dao et al., 2024; Shapiro, 2026). Monetary policy response is typically treated as invariant to the magnitude of the shock, leaving the shock-dependence of the policy reaction itself comparatively unexplored.

Sectoral heterogeneity also matters for inflation persistence and monetary transmission. Differences in price stickiness amplify the real effects of monetary disturbances (Carvalho, 2006; Nakamura and Steinsson, 2010; Pastén et al., 2020), and stabilizing the stickier sector can be closer to optimal than stabilizing a broad aggregate (Aoki, 2001; La'O and Tahbaz-Salehi, 2022). The services-goods division is the relevant empirical counterpart, since trend inflation has increasingly been driven by the slower-adjusting services sector (Eo et al., 2023; Amatyakul et al., 2024).

A related literature links inflation dynamics and persistence to credibility and beliefs about the monetary-policy regime. When the inflation objective is imperfectly observed, agents infer it through a filtering problem, and imperfect credibility can generate persistent inflation and sizable disinflation costs without departing from rational expectations (Erceg and Levin, 2003; Céspedes and Soto, 2005; Melosi, 2017). In these frameworks, however, the object being inferred is usually exogenous, whereas recent evidence suggests that the inflation anchor itself may drift with realized inflation during large-shock episodes (Reis, 2021; Carvalho et al., 2023; Coibion et al., 2022).

Finally, large shocks generate different dynamics from small shocks. State-dependent pricing makes the Phillips curve nonlinear in shock size (Golosov and Lucas, 2007; Vavra, 2014; Cavallo et al., 2024), while threshold effects and trend-inflation nonlinearities can steepen the

output-inflation trade-off once inflation is elevated (Harding et al., 2023; Benigno and Eggertsson, 2023; Ascari and Sbordone, 2014). Monetary policy effects are also state-dependent (Tenreyro and Thwaites, 2016). This literature shows that shock size matters, but it does not usually locate size-dependence inside the policy objective itself.

Taken together, these strands leave open the role of the policy objective itself. The literature shows that large shocks, sectoral composition, and credibility matter for inflation dynamics, yet the inflation objective that is being transmitted is usually taken as given. The remaining margin is whether the inflation objective can itself move with the size of the shock, and whether the effects of that movement depend on credibility.

Inflation can persist after a large shock for reasons that have little to do with the shock itself. Beyond the duration of the originating disturbance and the sectoral composition of prices, persistence can come from the monetary-policy reaction, when a large shock shifts the operational target and that shift is transmitted through credibility-dependent belief formation. Within the channel, the response of the target to a tail event sets the magnitude of accommodation while the persistence of the target component sets the horizon over which convergence is intended. Observed conduct combines them into a single implicit gap, which the model separates. The last-mile problem is thereby read not as the mechanical tail of the originating shock alone, but as the residue of a state-dependent and belief-mediated displacement of the nominal anchor.

The structure is as follows. Section 2 documents the stylized facts of the episode and recovers the implicit target gap that motivates the analysis. Section 3 develops the model, and Section 4 describes its solution and calibration. Section 5 examines the model's responses to large disturbances under alternative configurations of accommodation, credibility, and policy horizon. Section 6 presents model robustness, while Section 7 concludes.

2. Key Stylized Facts

Three stylized facts are presented. First, post-pandemic inflation surged sharply and remained above pre-pandemic levels after the initial disinflation. Second, this persistence is mainly sectoral, with services inflation remaining elevated after goods inflation receded. Third, the recovered implicit target gap increased broadly during 2021–2022 and is positively associated with large supply-like shocks. These facts motivate a model in which large shocks can affect the inflation path in a two-sector economy through shock-dependent monetary policy and credibility.

2.1. *Dynamics of Inflation and its Sectoral Components*

The disruptions triggered by the COVID-19 pandemic resulted in a significant increase in inflation, followed by an incomplete disinflation dynamic.[1] Figure 1 depicts the evolution of the median inflation gap between year-on-year inflation and each country's inflation target for a panel of advanced and emerging economies from 2000 onward. The pandemic-related disruptions, including supply bottlenecks and fiscal stimulus,[2] generated a price surge that was substantially larger both in absolute terms and relative to announced inflation objectives. At first, the median inflation gap declined to negative values, albeit with heterogeneity across countries. Beginning in 2021, the gap increased sharply, reaching values close to 7.5 percentage points at the median.

The subsequent disinflation was rapid but incomplete. The gap narrowed substantially from its peak, yet the median remained positive through the later phase of the adjustment, with the inflation gap above the levels prevailing before the pandemic. Since late 2023, inflation has therefore been closer to target, but not fully back to target for the median economy. This residual positive gap constitutes a last-mile convergence problem, in which the initial inflation

[1] Details regarding the panel composition can be found in Appendix A.

[2] Explanations of the initial inflation surge lie beyond the scope of this work. See Bernanke and O. Blanchard (2025), Giovanni et al. (2023) and Guerrieri et al. (2022) for additional sources of the pandemic-related inflation surge.

surge has largely receded but monetary policy still faces the challenge of bringing inflation fully back to its objective.

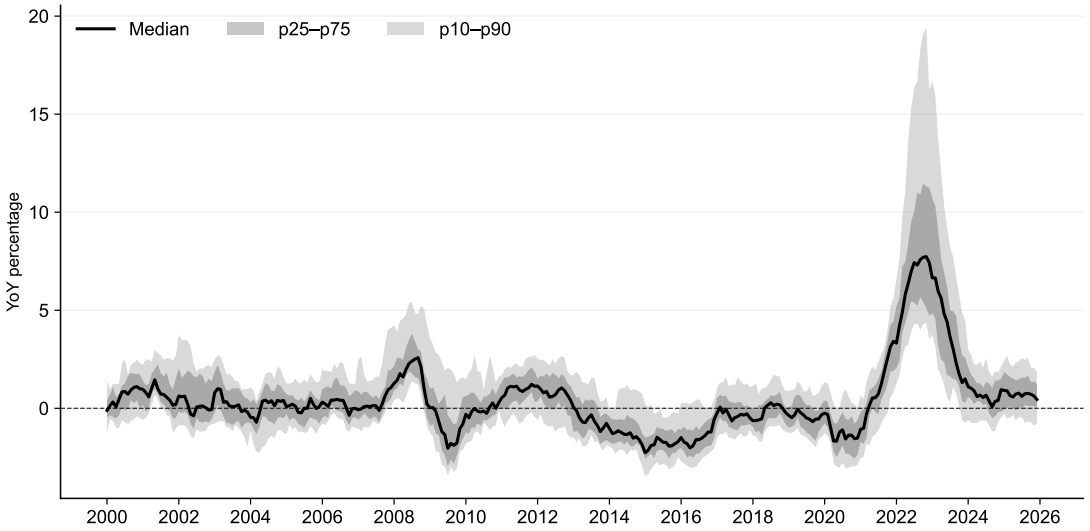


FIGURE 1. Median Inflation Gap Relative to Country-Specific Inflation Targets.

Sectoral inflation evolved asymmetrically over this episode, and the interaction of sector-specific dynamics helps account for the aggregate path described above. As shown in Figure 2, the last-mile inflation problem is mainly driven by price changes in the service-providing sector. The onset of the pandemic coincided with a pronounced sectoral divergence. In particular, the goods-producing sector inflation rose rapidly and sharply, while services inflation exhibited a lagged response to these initial shocks. As goods inflation subsequently declined, services inflation remained elevated, generating a positive services-goods gap during the last-mile phase. This pattern suggests that the persistence of aggregate inflation after the initial disinflation is closely related to the slower adjustment of services prices.

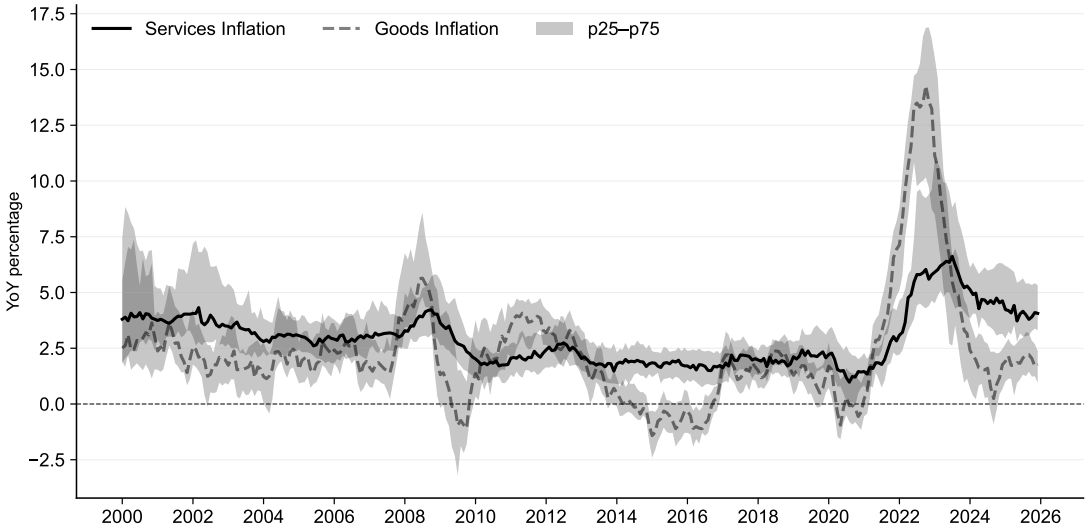


FIGURE 2. Median Inflation for Services and Goods-Related Prices.

The sectoral decomposition clarifies the nature of the last-mile problem. During the inflation episode, goods inflation accounts for much of the initial surge, but its subsequent decline

does not fully close the aggregate inflation gap to pre-pandemic range. Instead, services inflation remains systematically above goods inflation after the peak, and its post-peak decline is comparatively slow. Aggregate persistence therefore increasingly reflects services-price adjustment rather than the mechanical tail of the initial goods-price disturbance.

2.2. *Implicit Inflation Target and State-Dependent Accommodation under Large Shocks*

Monetary policy conduction can be represented as depending not only on the formal inflation target, but also on the short-run inflation path that is consistent with the observed policy stance, contemporaneous inflationary pressures, and the intended convergence of prices toward the announced target. This distinction is particularly relevant for the 2021–2022 inflation episode, when inflation rose sharply across economies. Under a constant-target Taylor-rule benchmark, the observed stance during this period leaves a sizable implied gap between the formal target and the short-run target consistent with policy conduct. The following empirical exercise recovers this implicit target gap and evaluates whether it is associated with large supply-like disturbances. The purpose is to assess whether large shocks are associated with a higher implicit short-run inflation target path, which can be interpreted as a slower intended convergence toward the formal target.

To recover the short-run inflation target consistent with observed policy conduct, the monetary-policy rule[3] is solved for the target term. The resulting expression for country c at quarter t is given by

$$\widehat{\pi}^{T}_{c,t} = \pi_{c,t} - \frac{1}{\phi_\pi}\left[\frac{i_{c,t} - a_c - \gamma_i i_{c,t-1}}{1-\gamma_i} - \phi_y\left(\Delta y_{c,t} - g^{y}_{c}\right)\right]. \tag{1}$$

The recovered implicit target gap is defined as $g^{T}_{c,t} = \widehat{\pi}^{T}_{c,t} - \pi^{*}_{c,t}$, where $\widehat{\pi}^{T}_{c,t}$ is the target implied by observed policy conduct and $\pi^{*}_{c,t}$ is the formal announced target. A positive value of $g^{T}_{c,t}$ indicates that, conditional on the imposed rule, the observed policy stance is consistent with a short-run operational target above the announced target. This does not imply a legal target revision. Instead, it identifies a higher short-run inflation path consistent with policy conduct, and mechanically implies a more gradual convergence toward the formal target.[4]

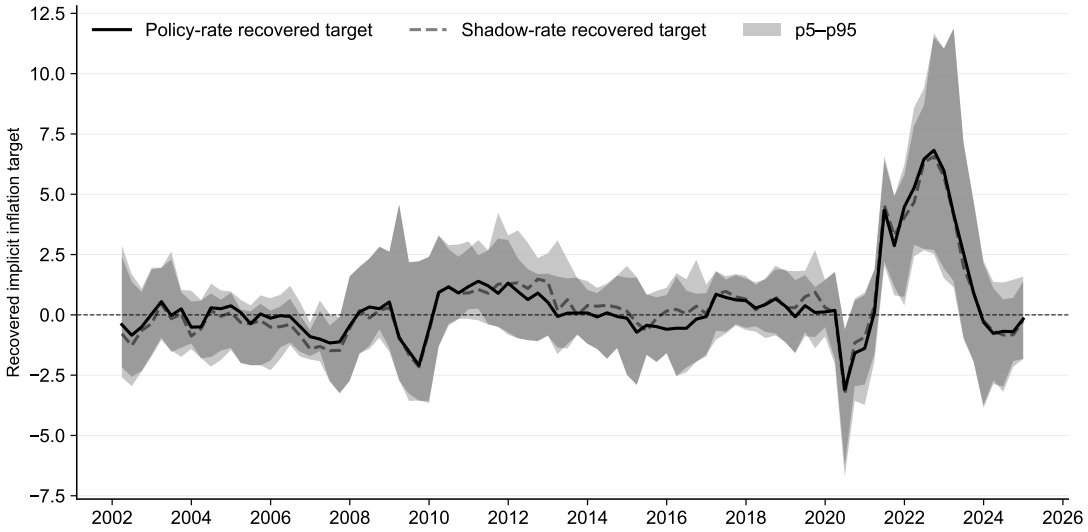


FIGURE 3. Median Recovered Implicit Target Gap for Monetary Policy Rate and Shadow Interest Rate.

[3]See equation (26), introduced in Subsection 3.3, and calibrated in Subsection 4

[4]The exercise uses the same Taylor-rule coefficients employed in the calibration ($\gamma_i = 0.50$, $\phi_\pi = 2.00$, and $\phi_y = 0.50$) on a sample of 24 economies covering 2002Q1 to 2024Q4. The policy stance is measured using both the effective monetary policy rate and a shadow interest rate (Krippner, 2013; Wu and Xia, 2016). The country-specific intercept anchors the pre-COVID recovered target around the formal target, while trend output growth is obtained from a pre-2020 Hodrick-Prescott filter. Further details are provided in Appendix A.

Figure 3 shows the recovered gap median across policy stance variables. The series moves negatively during episodes of deflationary pressure, such as the Global Financial Crisis and the pandemic recession, when policy accommodated weak inflation and activity. The central pattern for the present analysis begins in 2021, as the gap rises rapidly with inflation, reaches its maximum during 2022, remains positive through the 2023 disinflation, and then declines into the 2024 last-mile phase.

Table 1 reports summary statistics for the recovered implicit target gap, clarifying the timing of the movements. During the 2021–2022 inflation episode, the mean recovered gap reaches 4.43 and 4.25 percentage points under the two stance measures, with positive values in more than 90 percent of observations. The post-peak phase shows that the gap remained positive during the 2023 disinflation, with average values of 2.13 and 2.01 percentage points. By contrast, the recovered gap turns negative during the 2024 last-mile period, reaching average values of -0.54 and -0.68 percentage points under the policy-rate and shadow-rate measures, respectively. This negative value is consistent with inflation having largely normalized, although remaining above target on average, while policy rates remained restrictive. Conditional on the Taylor-rule inversion, the last-mile period no longer appears as a continuation of implicit-target accommodation. It instead corresponds to a later phase of the adjustment, in which inflation had already moved substantially toward target, but not enough for policy to abandon its restrictive stance before the final part of the disinflation was complete.

TABLE 1. Recovered Implicit Target Gap by period: Summary Statistics

	Obs.	Mean	Median	p05	p95	Positive obs. (%)
Policy Rate gap						
Pre-COVID	1533	0.07	0.10	-1.97	2.04	53.16%
GFC 2007-2009	240	-0.34	-0.61	-3.25	2.42	35%
Euro-Area Debt Crisis 2009-2012	348	0.64	0.78	-2.38	3.09	71%
COVID 2020	96	-1.66	-1.42	-5.54	1.08	19.79%
Inflation Episode 2021-2022	192	4.43	4.2	0.05	9.19	93.75%
Disinflation 2023	96	2.13	1.72	-2.40	7.25	75.00%
Last Mile 2024	96	-0.54	-0.66	-2.37	1.50	31.25%
Shadow Rate gap						
Pre-COVID	1533	0.10	0.13	-2.12	2.28	53.75%
GFC 2007-2009	240	-0.50	-0.76	-3.44	2.42	31.67%
Euro-Area Debt Crisis 2009-2012	348	0.72	0.89	-2.58	3.33	70.40%
COVID 2020	96	-1.39	-1.14	-5.11	1.1	21.88%
Inflation Episode 2021-2022	192	4.25	4.17	-0.17	8.69	92.71%
Disinflation 2023	96	2.01	1.45	-2.29	7.22	73.96%
Last Mile 2024	96	-0.68	-0.72	-2.38	1.24	23.96%

Note: The target gap is defined as the difference between the implicit short-run target $\widehat{\pi}^{T}_{c,t}$ obtained from a calibrated Taylor-rule inversion and the legal inflation target $\pi^{*}_{c,t}$. The 5th and 95th percentiles refer to the country-quarter distribution within each period. Sample composed of 24 economies, ranging from 2002Q1 to 2024Q4. Periods: Pre-COVID = 2002Q1–2019Q4; GFC = 2007Q1-2009Q4; Euro-Area Debt Crisis = 2009Q1-2012Q4; COVID 2020 = 2020Q1–2020Q4; Inflation Episode 2021–2022 = 2021Q1–2022Q4; Disinflation 2023 = 2023Q1–2023Q4, Last Mile 2024 = 2024Q1-2024Q4.

The positive recovered gap during the 2021–2022 inflation episode is robust to several alternative recovery procedures.[5] Varying the inflation response coefficient in the Taylor rule does not eliminate the result. Across alternative values of ϕ_π, the average gap remains between 3.7 and 4.2 percentage points, with positive country-average gaps in 22 economies. The result is also robust to alternative anchoring rules. Anchoring the intercept using $r^* + \pi^*$ yields an average gap of 4.4 percentage points, while removing the anchor altogether still produces an average gap of 3.2 percentage points, with 23 economies displaying positive values. Replacing

[5]For further details, see Appendix A. All robustness alternatives modify components of the inversion while retaining the Taylor-rule specification.

contemporaneous inflation with a four-quarter-ahead forecast raises the average gap to 9.12 percentage points. Given that inflation was persistent throughout the episode, the forward inflation forecast exceeded contemporaneous inflation, so a forward-looking rule would have prescribed even tighter policy and implied an even larger gap. Finally, using a rolling real-time HP trend for output growth produces an average gap of 4.47 percentage points. Across the checks, the recovered gap is directionally the same and quantitatively at least as large as in the baseline. The procedure remains descriptive, but it consistently identifies a short-run target path above the formal target during the inflation episode.

The recovered implicit target gap can be used to evaluate the nonlinear association with supply shocks. The empirical specification takes the form

$$g^T_{c,t} = \alpha_c + \lambda_t + \rho g^T_{c,t-1} + \theta_S z^S_{c,t} + \theta_D z^D_{c,t} + \mathcal{N}_{c,t} + \varepsilon_{c,t}. \tag{2}$$

where $z^S_{c,t}$ and $z^D_{c,t}$ denote standardized supply-like and demand-like shocks, constructed from pseudo-real-time forecast errors and standardized using pre-2020 moments.[6] The coefficients θ_S and θ_D capture the linear association between ordinary shocks and the recovered implicit target gap. The nonlinear term $\mathcal{N}_{c,t}$ captures the additional association generated by unusually large supply-like shocks. Fixed effects are included, so the nonlinear term is interpreted as the upper-tail association between large shocks and the recovered gap, conditional on country heterogeneity, common time variation, and persistence in the gap itself.

Two versions of $\mathcal{N}_{c,t}$ are considered. The first is a one-sided trigger, based on $(z^S_{c,t} - \tau)_+$, which focuses on large positive supply-like shocks. This specification aims to answer whether inflationary supply shocks are associated with higher recovered gaps, so it is the closest empirical counterpart to the post-pandemic inflation episode and its supply-side nature. The second is a symmetric trigger, based on $(|z^S_{c,t}| - \tau)_+$, which is closer to the model's shock-size function $g(|z_t|;\tau)$, and aims to ask whether unusually large shock realizations, regardless of sign, carry higher implicit target gaps.[7]

Two alternative forecast error shock-construction procedures are used. In the main procedure, the forecasting equations are selected using only pre-2020 information, and the resulting lag orders are kept fixed throughout the post-2020 period while the coefficients are updated recursively. As an alternative exercise, the lag structure is annually updated after 2020, while preserving the coefficient update of the forecast-error construction. The regressions are estimated for both the policy-rate and shadow-rate versions of the recovered gap, with standard errors clustered at the country level.

Table 2 reports the main estimates of equation (2). The coefficient θ_S captures the association between supply-like shocks and the recovered implicit target gap below the threshold, while β_S captures the additional marginal association once the shock exceeds the threshold. The total upper-tail marginal effect is therefore given by $\theta_S + \beta_S$ in the one-sided specification and by $\theta_S^{\text{sym}} + \beta_S^{\text{abs}}$ in the symmetric specification. Across both stance measures, all thresholds, and both shock-construction panels, the upper-tail marginal effect is positive and statistically significant. This indicates that large supply-like shocks are associated with higher recovered implicit target gaps. Moreover, the positive values of the upper-tail coefficients (β_S or β_S^{abs}), especially under the annual lag-update procedure and at higher thresholds, suggest that the association between supply-like shocks and the recovered gap becomes stronger in the upper tail of the shock distribution.

Central banks may not immediately interpret a shock as persistent or policy relevant, so movements in the recovered implicit target may occur with delay as the policy stance adjusts to a more persistent inflation path. For this reason, distributed-lag specifications are estimated for the nonlinear term $\mathcal{N}_{c,t}$.[8] The results indicate that the nonlinear supply association becomes more visible when delayed adjustment is allowed. Allowing the one-sided supply hinge

[6] Following C. D. Romer and D. H. Romer (2004) for the forecasting-error logic for identifying shocks, and Shapiro (2026) procedure for supply-like and demand-like shock identification.

[7] Since the shocks are standardized, $\tau \in \{1.0, 1.5, 2.0\}$ is measured in standard-deviation units.

[8] For more details, refer to Appendix A and Table A.16.

TABLE 2. Upper-tail Supply Slope: One-Sided and Symmetric Triggers

Panel A. Fixed lag-order specification

	One-sided trigger			Symmetric trigger		
τ	θ_S	β_S	$\theta_S + \beta_S$	θ_S^{sym}	β_S^{abs}	$\theta_S^{\text{sym}} + \beta_S^{\text{abs}}$
Policy Rate gap						
1.0	0.109^{***}	0.024	0.134^{**}	0.136^{***}	-0.009	0.126^{***}
1.5	0.097^{***}	0.046^{*}	0.142^{**}	0.135^{***}	0.005	0.140^{***}
2.0	0.092^{***}	0.051^{*}	0.144^{**}	0.135^{***}	0.012^{*}	0.147^{***}
Shadow Rate gap						
1.0	0.080^{**}	0.077	0.157^{***}	0.124^{***}	0.001	0.126^{***}
1.5	0.069^{**}	0.103	0.171^{***}	0.124^{***}	0.016^{*}	0.140^{***}
2.0	0.066^{**}	0.109^{*}	0.175^{***}	0.124^{***}	0.023^{**}	0.147^{***}

Panel B. Annual lag-update specification

	One-sided trigger			Symmetric trigger		
τ	θ_S	β_S	$\theta_S + \beta_S$	θ_S^{sym}	β_S^{abs}	$\theta_S^{\text{sym}} + \beta_S^{\text{abs}}$
Policy Rate gap						
1.0	0.071^{***}	0.091^{*}	0.162^{***}	0.125^{***}	0.032	0.157^{***}
1.5	0.060^{**}	0.116^{**}	0.176^{***}	0.125^{***}	0.054^{**}	0.179^{***}
2.0	0.057^{**}	0.127^{**}	0.183^{***}	0.126^{***}	0.070^{**}	0.196^{***}
Shadow Rate gap						
1.0	0.045^{*}	0.127^{**}	0.173^{***}	0.112^{***}	0.039^{*}	0.151^{***}
1.5	0.035	0.158^{**}	0.193^{***}	0.112^{***}	0.061^{**}	0.173^{***}
2.0	0.033	0.170^{**}	0.203^{***}	0.113^{***}	0.077^{***}	0.190^{***}

Note: Estimates are obtained from fixed-effects regressions with country and year fixed effects. Standard errors are clustered by country. All specifications use 1,719 quarterly observations from 24 countries. The values of θ_S under the one-sided and symmetric specifications are not directly comparable, as they are estimated within separate regressions and absorb different upper-tail variation. One-sided and symmetric coefficients are estimated in separate regressions. The reported sums $\theta_S + \beta_S$ and $\theta_S^{\text{sym}} + \beta_S^{\text{abs}}$ measure the upper-tail marginal effect of a supply-like shock on the implicit target gap. Statistical significance is assessed using Wald tests. Panel A fixes forecasting lag orders at their pre-2020 selection, while Panel B updates lag orders annually from 2020 onward. Demand-side coefficients are estimated but omitted. Significance level given by $^{*}\, p < 0.10$, $^{**}\, p < 0.05$, and $^{***}\, p < 0.01$. Complete statistics reported in Table A.14 and Table A.15 in Appendix A.

to enter with up to two lags yields larger cumulative effects, with two-lag estimates ranging roughly from 0.23 to 0.31, a pattern consistent with a gradual adjustment of the recovered gap after large shocks, given an implicit target deviation.

For Euro-Area countries, a common policy rate is heterogeneously transmitted across member countries. To account for this issue, the implicit target gap is reconstructed using lending rates to non-financial corporations (NFC) as an alternative stance measure, in the spirit of Arnold (2024).[9] The exercise is implemented using both raw NFC lending rates and NFC rates net of the ten-year sovereign spread. Upper-tail supply level and marginal effects remain positive and statistically significant across both new measures, construction panels, and thresholds, sustaining the previous findings.

Taken together, the previous results provide descriptive evidence of a state-dependent pattern in the recovered implicit target gap. The gap increased sharply and broadly during the 2021–2022 inflation episode, and supply-like shocks are positively associated with it, with stronger marginal effects in the upper tail of the shock distribution. The evidence should be interpreted as reduced-form: it links large supply-like disturbances to the short-run target

[9]For further discussion of bank lending-rate heterogeneity and the pass-through of monetary policy in the Euro-Area, see ECB (2017).

gap implied by observed monetary-policy conduct, without establishing a legal target revision, a structural policy shock, or the exact timing and functional form of the policy response. Combined with the sectoral evidence in Subsection 2.1, these results suggest that the last-mile inflation problem may partly reflect a higher implicit short-run inflation path after large shocks. Section 3 formalizes one structural interpretation of this pattern, through a shock-size mechanism in which sufficiently large disturbances shift the persistent component of the operational inflation target and affect the economy through expectations and credibility.

3. Model

A two-sector New Keynesian economy is built around a simple distinction between two categories of consumption goods: services and goods. Both product and labor markets are characterized by monopolistic competition, allowing for nominal price and wage rigidities.

Sectoral heterogeneity is introduced along four dimensions. First, the service-providing sector displays higher nominal price rigidity than the goods-producing sector, consistent with the empirical evidence on sectoral price adjustment. Second, wage rigidity is allowed to differ across sectors, with services exhibiting less persistent nominal wage adjustment. Third, the service-providing sector concentrates a larger share of labor, both in relative and absolute terms. Fourth, marginal costs in services depend more strongly on wages than marginal costs in goods, making service inflation more sensitive to labor-market conditions.

The monetary policy block builds on Erceg and Levin (2003) by introducing a shock-size mechanism into the inflation-targeting structure. The inflation target contains a persistent and a transitory component, but private agents cannot directly observe this decomposition and infer the persistent component through a signal-extraction problem. The main departure from the benchmark framework is that sufficiently large structural shocks can temporarily shift the persistent component of the inflation target.

3.1. *Production*

3.1.1. *Hybrid Sectoral Phillips Curve*

Each sector, indexed by $s \in \{S, G\}$, is populated by a final-goods producer. This agent uses a continuum of differentiated intermediate goods to produce a sectoral final output good using the following *Dixit-Stiglitz inspired* technology

$$Y_{s,t} = \left[\int_0^1 (Y_{s,t}(j))^{-\frac{1}{1+\theta_s}} \, \mathrm{d}\, j \right]^{1+\theta_s}.$$

where $\theta_s > 0$ governs the elasticity of substitution between intermediate varieties.

Assuming perfect competition and taking as given the price for each input, $P_{s,t}(j)$, and factor production, $Y_{s,t}(j)$, the following demand for each variety is derived

$$Y_{s,t}(j) = \left[\frac{P_{s,t}(j)}{P_{s,t}} \right]^{-\frac{1+\theta_s}{\theta_s}} \cdot Y_{s,t}.$$

where $P_{s,t}$ and $Y_{s,t}$ denote the sectoral aggregate price and sectoral final output, respectively.

Intermediate production firms operate in each sector and set prices under sector-specific nominal rigidities. Following Calvo (1983), in each period t firms in sector s face a probability $1-\alpha_s$ of being able to reset prices, so that a fraction α_s of firms cannot reoptimize and instead keep last period's price $P_{s,t-1}(j)$. Among the firms that can reset prices, pricing behavior is heterogeneous in the spirit of Galí and Gertler (1999). A fraction ω_s follows a backward-looking rule of thumb, while the remaining firms set prices according to a forward-looking optimal pricing rule.

The sectoral price index for sector s is given by

$$P_{s,t}^{1-\epsilon_s} = \int_0^1 P_{s,t}(j)^{1-\epsilon_s} \, \mathrm{d}\, j. \tag{3}$$

This aggregate sectoral price is composed of the composite price of price-adjusting firms, as well as the past price set by non-adjusting firms. The weighted composite price, $p^C_{s,\,t}$, for price-adjusting firms is given by

$$p^C_{s,\,t} \;=\; (1-\omega_s)\cdot p^*_{s,\,t} + (\omega_s)\cdot p^b_{s,\,t}. \tag{4}$$

Log-linearizing equation (3) yields the following sectoral inflation relation

$$\pi_{s,\,t} \;=\; (1-\alpha_s)\cdot (p^C_{s,\,t} \;-\; p_{s,\,t-1}). \tag{5}$$

This equation implies that sectoral inflation arises from the adjustment of a subset of firms, either to the optimal price or to an indexed rule, while non-adjusting firms keep their previous price.

Forward-looking firms account for the share $1-\omega_s$ of all price-adjusting firms in sector s. The optimization problem of this type of firms is given by a pricing problem with α_s as the sectoral price rigidity parameter. The problem of the forward-looking firm j in sector s is

$$\max_{P_{s,\,t}(j)} \sum_{l=0}^{\infty} Q_{t,\,t+l}\,\alpha_s^l \left[P_{s,\,t}(j)\cdot Y_{s,\,t+l}(j) - MC_{s,\,t+l}(j)\cdot Y_{s,\,t+l}(j)\right],$$

$$S.t.\;\; Y_{s,\,t+l}(j) \;=\; \left[\frac{P_{s,\,t}(j)}{P_{s,\,t+l}}\right]^{-\frac{1+\theta_s}{\theta_s}} \cdot Y_{s,\,t+l}.$$

where $Q_{t,\,t+l}$ is the stochastic discount factor, $Y_{s,\,t}(j)$ and $MC_{s,\,t}(j)$ are the firm's production and marginal cost, respectively. Finally, $P_{s,\,t}$ is the sectoral aggregate price index.

The rearranged first-order condition of the problem is

$$\mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l \left[\left(\frac{P_{s,\,t}(j)}{1+\theta_s}\right) - MC_{s,\,t+l}(j)\right]\cdot Y_{s,\,t+l}(j) = 0. \tag{6}$$

Equation (6) states that the forward-looking intermediate producer in sector s chooses its optimal price so that the expected discounted nominal marginal revenue equals expected discounted nominal marginal cost.

Solving for prices yields the following result

$$P^*_{s,\,t}(j) = (1+\theta_s)\cdot \frac{\mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l \cdot MC_{s,\,t+l}(j)\cdot Y_{s,\,t+l}(j)}{\mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l \cdot Y_{s,\,t+l}(j)}.$$

Taking log-differences of the expression, the following pricing decision is derived[10]

$$p^*_{s,\,t}(j) \;=\; (1-\beta\,\alpha_s)\cdot \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l\, mc_{s,\,t+l}(j). \tag{7}$$

This equation states that the optimal reset price chosen by forward-looking firm j in sector s is the discounted value of current and future marginal costs. Note that marginal costs are common across firms within a sector under the production function described below. Equation (7) results in the pricing rule for forward-looking firms:

$$p^*_{s,\,t} = (1-\beta\,\alpha_s)\cdot mc_{s,\,t} + (\beta\,\alpha_s)\cdot \mathbb{E}_t\, p^*_{s,\,t+1}. \tag{8}$$

Backward-looking firms in sector s change prices following a rule of thumb that relies on the previous pricing decisions made by competitors within the same industry. In contrast to a direct sectoral application of Galí and Gertler (1999), the rule uses the aggregate inflation rate rather than sectoral inflation, given by

$$p^b_{s,\,t} \;=\; p^C_{s,\,t-1} + l_s\,\pi_{t-1}. \tag{9}$$

Backward-looking firms choose their prices using a simple rule that indexes the weighted-composite price of price-adjusting firms to past aggregate inflation. This pass-through is mediated by $l_s \in [0,\,1]$.

[10]Henceforth, lower case variables represent the first-order expansion around the steady state of the logarithmic transformation of the upper case variable.

Combining equations (4), (5), (8) and (9) yields the *Hybrid Sectoral Phillips Curve*

$$\pi_{s,t} = \gamma_{f,s}\mathbb{E}_t\,\pi_{s,t+1} + \gamma_{b,s}\,\pi_{s,t-1} + \gamma_{\text{agg},s}\,\pi_{t-1} - \gamma_{\text{lag agg},s}\,\pi_t + \gamma_{mc}\,(mc_{s,t} - p_{s,t}). \tag{10}$$

Equation (10) implies a hybrid sectoral Phillips curve in which inflation dynamics reflect both forward-looking expectations and intrinsic persistence. Higher expected sectoral inflation translates into higher current sectoral inflation, while both past sectoral inflation and lagged aggregate inflation contribute to persistence through backward-looking price-setting and indexation mechanisms. Nonetheless, contemporaneous aggregate inflation negatively affects sectoral inflation. Real marginal cost pressures are translated positively into sectoral inflation, capturing a standard cost-push channel whereby increases in production costs are passed through into prices. Overall, sectoral inflation depends on expected future inflation, inherited inflation inertia, and current pricing pressure from real marginal costs, with an additional correction coming from indexation to aggregate inflation.

3.1.2. *Sectoral Marginal Costs*

The production function of the intermediate firm j in sector s at time t is given by

$$Y_{s,t}(j) \;=\; \big(A_t \cdot X_{s,t}\big)\left[(\omega_{s,L})\cdot\big(L_{s,t}(j)\big)^{\frac{\sigma_s-1}{\sigma_s}} + (\omega_{s,K})\cdot\big(K_{s,t}(j)\big)^{\frac{\sigma_s-1}{\sigma_s}}\right]^{\frac{\sigma_s}{\sigma_s-1}}. \tag{11}$$

Both A_t and $X_{s,t}$ are productivity factors, the former is common across sectors, while the latter is sector-specific. The producer uses capital and labor services, $K_{s,t}$ and $L_{s,t}$ respectively. Sectoral elasticity of substitution between factors is denoted by σ_s, and $\omega_{s,K}$ and $\omega_{s,L}$ represent the sectoral factor weights [11] of each factor in the production function.

Given the production function in equation (11), the firm's cost-minimization problem implies the following real marginal cost for firm j of sector s

$$MC_{s,t}(j) \;=\; \left(\frac{1}{A_t \cdot X_{s,t}}\right)\cdot\left[(\omega_{s,L}^{\sigma_s})\cdot W_{s,t}^{1-\sigma_s} + (\omega_{s,K}^{\sigma_s})\cdot\big(R_{K,t}^{s}\big)^{1-\sigma_s}\right]^{\frac{1}{1-\sigma_s}}. \tag{12}$$

3.2. *Households*

A continuum of monopolistically competitive households is assumed, indexed on the unit interval as $h \in [0,\,1]$. The household consumes an aggregate consumption index composed of service and goods, saves through state-contingent claims and accumulates sectoral capital through investment, and holds real money balances. In addition, each household supplies differentiated labor services to firms, which gives rise to wage-setting power.

For convenience, an external representative labor aggregator is introduced in each sector $s \in \{S, G\}$. The aggregator combines household-specific labor services, $N_{s,t}(h)$, into sectoral labor demand, $N_{s,t}$. The sectoral labor demand, $N_{s,t}$, is given by the following aggregation technology

$$N_{s,t} \;=\; \left[\int_0^1 N_{s,t}(h)^{\frac{1}{1+\theta_n}}\,\mathrm{d}\,h\right]^{1+\theta_n},$$

The labor aggregator agent solves a cost minimization problem that yields the demand for each household's labor service, which the household internalizes when setting wages.

$$N_{s,t}(h) \;=\; \left[\frac{W_{s,t}(h)}{W_{s,t}}\right]^{-\frac{1+\theta_n}{\theta_n}} N_{s,t}. \tag{13}$$

The utility function of household h is given by

$$\mathbb{E}_t \sum_{j=0}^{\infty}\beta^j\left[\frac{C_{t+j}(h)^{1-\sigma}}{1-\sigma} - \chi_0\frac{N_{t+j}(h)^{\chi}}{\chi} + \frac{\mu_0}{1-\mu}\left(\frac{M_{t+j}(h)}{P_{t+j}}\right)^{1-\mu}\right]. \tag{14}$$

where $\beta \in (0,\,1)$ is the discount factor. Moreover, $C_{t+j}(h)$ is a CES aggregator for sectoral consumption, $N_{t+j}(h)$ represents the hours to work taking a CES sectoral aggregator form.

[11] This implies that $\omega_{s,L} = 1 - \omega_{s,K}$.

Both objects defined below. Finally, $M_{t+j}(h)/P_{t+j}$ represents real money balances.

The consumption aggregator, $C_t(h)$, is defined by

$$C_t(h) \;=\; \left[(\omega_{C,\,S}^{1/\eta_c}) \cdot (C_{S,\,t}(h)\,)^{\frac{\eta_c-1}{\eta_c}} + (\omega_{C,\,G}^{1/\eta_c}) \cdot (C_{G,\,t}(h)\,)^{\frac{\eta_c-1}{\eta_c}}\right]^{\frac{\eta_c}{\eta_c-1}}. \tag{15}$$

where $C_{s,t}(h)$ denotes the consumption allocated in the sector $s \in [S,\, G]$. Additionally, $\omega_{C,\,s}$ denotes the steady-state expenditure share allocated to sector s.

In addition, the household's labor aggregator, $N_t(h)$, is defined by

$$N_t(h) \;=\; \left[\omega_{N,S}^{1/\eta_n}\;(N_{S,\,t}(h)\,)^{\frac{\eta_n-1}{\eta_n}} + \omega_{N,G}^{1/\eta_n}\;(N_{G,\,t}(h)\,)^{\frac{\eta_n-1}{\eta_n}}\right]^{\frac{\eta_n}{\eta_n-1}}. \tag{16}$$

where $N_{s,t}(h)$ denotes household h labor supply in sector s. Note that $\omega_{N,\,s}$ is the proportion of hours worked in the $s = [S,\, G]$ sector in steady state.[12]

In period t, household h's expenditures include consumption in services and goods, direct investment in capital and its associated adjustment cost following Christiano et al. (2005), insurance against future states of nature, and nominal money balances. Household income includes labor income from both sectors, capital rents, previous-period nominal money balances, payoffs from state-contingent claims, and profits from firm ownership. Given that, the budget constraint of the household h in period t is

$$\begin{aligned}
\sum_{s=S,\,G}\Big[P_{s,\,t}\,C_{s,\,t}(h)\Big] + \sum_{s=S,\,G}\Big[P_{s,\,t}\,I_{s,\,t}(h)\Big] + \\
M_{t+1}(h) - M_t(h) + \int \xi_{t+1}\,B_{t+1}(h) - B_t(h) \\
= \sum_{s=S,\,G}\Big[W_{s,\,t}(h)\,N_{s,\,t}(h)\Big] + \sum_{s=S,\,G}\Big[\Gamma_{s,\,t}(h)\Big] + \\
\sum_{s=S,\,G}\left[R^s_{K,\,t}\,K_{s,\,t}(h) - 0.5\,\phi^s_I\,I_{s,\,t}(h)\,P_{s,\,t}\left(\frac{I_{s,\,t}(h)}{I_{s,\,t-1}(h)} - 1\right)\right].
\end{aligned} \tag{17}$$

For simplicity, investment is sector-specific, and each sectoral capital stock evolves according to a linear law of motion with sector-specific depreciation

$$K_{s,\,t+1}(h) \;=\; (1-\delta_s)\,K_{s,\,t}(h) + I_{s,\,t}(h). \tag{18}$$

Financial asset accumulation is composed of nominal money holdings and state-contingent claims. Note that ξ_t is the price of an asset that pays a unit of currency in a specific state of nature, and $B_t(h)$ represents the household total holdings of state-contingent claims. In general asset accumulation terms, households also incur costs when adjusting their stock of sectoral physical capital. The adjustment cost increases quadratically with investment growth.

Household maximizes its utility function (14) subject to its nominal budget constraint (17), its labor demand equation (13), its labor aggregator equation (16), its consumption aggregator equation (15), and both sectoral capital laws of motion (18). This implies the following first-order conditions[13] [14] for a generic $t+l$ period

$$\begin{aligned}
C_{s,\,t+l}(h) \;&:(C_{t+l}(h))^{1/\eta_c-\sigma}\,\omega_{C,\,s}^{\frac{1}{\eta_c}}\,(C_{s,\,t+l}(h))^{\frac{\eta_c-1}{\eta_c}-1} - \lambda^1_{t+l}(P_{s,\,t+l}) = 0, \\
N_{s,\,t+l}(h) \;&:(-\chi_0)\,(N_{t+l}(h))^{\chi+1/\eta_n-1}(\omega_{N,\,s}^{\frac{1}{\eta_n}})(N_{s,\,t+l}(h))^{\frac{\eta_n-1}{\eta_n}-1} + \lambda^1_{t+l}(W_{s,\,t+l}) = 0, \\
B_{t+l+1}(h) \;&:\beta^{t+l}\left(-\xi_{t+l+1}\,\lambda^1_{t+l}\right) + \beta^{t+l+1}\mathbb{E}_t\left[\lambda^1_{t+l+1}\right] = 0,
\end{aligned}$$

[12] Additionally, note that $\sum_{s=[S,\,G]}\omega_{i,\,s} = 1$, for $i \in [C,\, N]$.

[13] Note that the following is the summarized system of first order conditions. For more information in this regard see Appendix C.2.

[14] Note that λ^1_t is the Lagrange multiplier associated with the Budget Constraint restriction, moreover, λ^s_t is the Lagrange multiplier of the associated sector s capital law of motion.

$$I_{s,\,t+l}(h)\,:(-\lambda^1_{t+l})P_{s,\,t+l}\left(1+(\phi^s_I/2)\left(\frac{I_{s,\,t+l}}{I_{s,\,t+l-1}}-1\right)^2+\phi^s_I\left(\frac{I_{s,\,t+l}}{I_{s,\,t+l-1}}\right)\left(\frac{I_{s,\,t+l}}{I_{s,\,t+l-1}}-1\right)\right)$$
$$+\lambda^s_{t+l}+\beta\left[(-\lambda^1_{t+l+1})\,P_{s,\,t+l+1}\,I_{s,\,t+l+1}\,\phi^s_I\left(\frac{I_{s,\,t+l+1}}{I_{s,\,t+l}}\right)^2\left(\frac{I_{s,\,t+l+1}}{I_{s,\,t+l}}-1\right)\right]=0,$$
$$K_{s,\,t+l+1}(h)\,:(-\lambda^s_{t+l})+\beta\left[\lambda^1_{t+l+1}\,R^s_{K,\,t+l}+\lambda^s_{t+l+1}(1-\delta_s)\right]=0.$$

The first-order conditions for aggregate consumption and state-contingent claims yield the usual Euler Equation for Consumption, which links present marginal benefit of consumption with expected discounted future cost derived from the savings opportunity cost

$$C_t^{-\sigma}\;=\;\beta\,\mathbb{E}_t\,(1+R_t)\,C_{t+1}^{-\sigma}. \tag{19}$$

where R_t is a risk-free real interest rate, representing the rate of return of an asset that pays a unit of output in every state of nature. Market completeness for state-contingent claims assumption allows us to suppress household identifiers.

Sectoral investment adjustment costs imply a sectoral *Tobin's q*

$$q_{s,\,t}=\beta\,\mathbb{E}_t\left[\frac{\Lambda_{s,\,t+1}}{\Lambda_{s,t}}\left(\frac{R^s_{K,\,t+1}}{P_{s,\,t+1}}+q_{s,\,t+1}(1-\delta_s)\right)\right]. \tag{20}$$

where $\Lambda_{s,t}$ is the marginal utility of consumption associated with sector s consumption.

Sectoral investment first-order condition leads to the Sectoral Euler Equation for Investment[15],

$$q_{s,\,t}=1+\frac{\phi^s_I}{2}\left(\frac{I_{s,\,t}}{I_{s,\,t-1}}-1\right)^2+\phi^s_I\left(\frac{I_{s,\,t}}{I_{s,\,t-1}}-1\right)\frac{I_{s,\,t}}{I_{s,\,t-1}}$$
$$-\beta E^*_t\left[\frac{\Lambda_{s,\,t+1}}{\Lambda_{s,\,t}}\phi^s_I\left(\frac{I_{s,\,t+1}}{I_{s,\,t}}-1\right)\left(\frac{I_{s,\,t+1}}{I_{s,\,t}}\right)^2\right]. \tag{21}$$

The previous relation states that the real shadow value of installed capital. equals the marginal cost of investing today, so that the purchase price in addition to an additional wedge that rises when investment changes sharply relative to last period, i.e. investment adjustment costs exist. Given that today's investment also affects next period's adjustment costs, the condition includes an expected discounted term capturing those future cost effects, which smooths investment dynamics.

Consumption allocation between sectors is given by

$$\frac{C_{S,\,t}(h)}{C_{G,\,t}(h)}=\left(\frac{\omega_{C,\,S}}{\omega_{C,\,G}}\right)\left(\frac{P_{S,\,t}}{P_{G,\,t}}\right)^{-\eta_c} \tag{22}$$

Relative demand for services increases with its steady-state expenditure weight and decreases with its relative price, with the strength of this substitution governed by the elasticity η_c.

3.2.1. *Wage Setting Problem*

Households set nominal wages in staggered contracts, as in Erceg et al. (2000), individually in two segmented labor markets. Following the renegotiation probability framework of Calvo (1983), the model is extended to include wage indexation to past aggregate inflation in the spirit of Christiano et al. (2005) to account for wage inertia. As the utility function is additively separable, the labor supply decision of the household can be isolated from the utility maximization problem previously described.

Households are subject to nominal wage rigidities, thus, in every period t, they are subject to a $(1-\alpha^W_s)$ probability to reset wages to the optimal nominal wage, $W^*_{s,\,t}$. Households that are unable to renegotiate wages instead index them to past aggregate inflation: $W_{s,\,t}=W_{s,\,t-1}(h)\,(1+\pi_{t-1})^{l^s_W}$.

[15]Note that an Euler Equation for Investment for each sector is needed because, in the present model, there is no aggregate investment. The opposite case is given by the Euler Equation for Consumption, where the aggregate does exist, so sectoral equations become redundant, and an intra-temporal allocator is needed.

The sectoral indexation parameter $l_W^s \in [0,1]$, governs the degree of pass-through from lagged aggregate inflation into the nominal wages of households that cannot renegotiate. When $l_W^s = 0$, non-adjusting wages remain fixed in nominal terms between reoptimization dates, whereas $l_W^s = 1$ implies full indexation to past inflation. Accordingly, conditional on a wage contract not being reset, the nominal wage follows a predetermined path driven by realized inflation, which is summarized by $W_{s,\,t+l}(h) = W^*_{s,\,t} \prod_{k=0}^{l-1} (1+\pi_{t+k})^{l_W^s}$.

The wage aggregator for sector s is given by

$$W_{s,\,t}^{1-\epsilon_s^W} = \int_0^1 W_{s,\,t}^{1-\epsilon_s^W}(h)\,\mathrm{d}\,h.$$

which yields the following sectoral wage equation

$$\pi^W_{s,\,t} = \alpha_s^W\, l_W^s\, \pi_{t-1} + (1-\alpha_s^W)\left(w^*_{s,\,t} - w_{s,\,t-1}\right). \tag{23}$$

Conditional on being able to reset wages at time t, the household selects $W^*_{s,\,t}$ to maximize expected discounted utility, taking the sectoral labor-demand functions as given.

The problem for sector s wage optimization can written as

$$\max_{W^*_{s,\,t}} \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s^W)^l \Big[W_{s,\,t+l}(h)\, N_{s,\,t+l}(h) - MRS_{s,\,t+l}(h)\, N_{s,\,t+l}(h) \Big],$$

$$S.t.\;\; N_{s,\,t+l}(h) = \left(\frac{W_{s,\,t+l}(h)}{W_{s,\,t+l}} \right)^{-\frac{1+\theta_n}{\theta_n}} N_{s,\,t+l},$$

$$W_{s,\,t+l}(h) = W^*_{s,\,t} \prod_{k=0}^{l-1} (1+\pi_{t+j})^{l_W^s}.$$

Equivalently,

$$\max_{W^*_{s,\,t}} \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s^W)^l \Big[W^*_{s,\,t}\mathcal{I}_{t,\,l} - MRS_{s,\,t+l}(h) \Big] \left(\frac{W^*_{s,\,t}\mathcal{I}_{t,\,l}}{W_{s,\,t+l}} \right)^{-\frac{1+\theta_n}{\theta_n}} N_{s,\,t+l}.$$

solving for the optimal wage in the firs-order condition

$$\frac{W^*_{s,\,t}}{1+\theta_n} \sum_{o}^{\infty} (\beta\,\alpha_s^W)^l\, N_{s,\,t+l}(h) = \sum_{o}^{\infty} (\beta\,\alpha_s^W)^l\, N_{s,\,t+l}(h) \frac{MRS_{s,\,t+l}(h)}{\mathcal{I}_{t,l}}.$$

which yields, in log-linearized form

$$w^*_{s,t} = (1-\beta\alpha_s^W) mrs_{s,t} + \beta\alpha_s^W E_t w^*_{s,t+1} - \beta\alpha_s^W l_W^s \pi_t. \tag{24}$$

Finally, combining the wage sectoral aggregation equation (23) and the optimal wage equation (24), the sectoral Hybrid Wage Phillips Curve is obtained

$$\pi^W_{s,\,t} - l_W^s\,\pi_{t-1} = \beta\big(\mathbb{E}_t\pi^W_{s,\,t+1} - l_W^s\,\pi_t\big) + \big((1-\alpha_s^W)\,(1-\beta\,\alpha_s^W)/\alpha_s^W\big)\,\big(mrs_{s,\,t} - w_{s,\,t}\big). \tag{25}$$

The sectoral Hybrid Wage Phillips Curve can be interpreted as a forward-looking wage-inflation relation augmented by intrinsic persistence induced by indexation. Wage inflation in sector s depends positively on expected future wage inflation, reflecting the intertemporal nature of staggered wage setting, and on the contemporaneous wedge $\big(mrs_{s,t} - w_{s,t}\big)$, which captures the pressure to increase wages when the household's marginal valuation of labor rises relative to the prevailing real wage.

The indexation mechanism introduces an additional direct role for aggregate inflation in the sectoral wage setting decision. Past aggregate inflation, $l_W^s\pi_{t-1}$, positively affects current wage inflation, while the term $-\beta l_W^s\pi_t$ reflects that contracts are indexed to lagged inflation but evaluated in a forward-looking optimal-reset condition. Accordingly, higher l_W^s strengthens inflation spillovers into wage dynamics and increases persistence. In the limiting case $l_W^s = 0$, the expression collapses to the standard Calvo wage Phillips curve, whereas $l_W^s = 1$ delivers full lagged-inflation indexation.

3.3. *Monetary Policy and Shock-Size Mechanism*

The monetary authority that governs this economy follows a rule that aims to adjusts the short-term nominal interest rate so as to affect the real interest rate, responding to deviations of inflation from the target and to deviations of output growth from its long-run rate. Given that, monetary policy is described by the following *Taylor-rule-inspired* nominal interest rate function

$$i_t = \gamma_i \, i_{t-1} + (1-\gamma_i)\left(\phi_\pi \left(\pi_t - \pi_{T,\,t}\right) + \phi_y(\Delta y_t - g_y)\right). \tag{26}$$

where i_t represents the short-run nominal interest rate and γ_i is the policy inertia parameter. Moreover, π_t represents inflation and $\pi_{T,\,t}$ represents the mentioned inflation target. Additionally, Δy_t represents the output growth and g_y the long-term growth. Both ϕ_π and ϕ_y are interpreted as the central bank's sensitivity to the respective rule's components.

3.3.1. *Imperfectly Observed Target and Shock Size*

Based on the analytical framework of Erceg and Levin (2003), an extended version is proposed. This mechanism allows for an interaction between the inflation target scheme and the size of the shock that impacts this economy, motivated by the look-through behavior for small shocks of the monetary authority.

Private agents are assumed to have rational expectations, but hold imperfect information regarding the process that underlies the inflation targeting scheme, which leads to partial anticipation of the monetary authority's behavior. As an assumption, agents can infer the current inflation target from the central bank's reaction function, however, they cannot directly observe the components that govern its dynamics.

As in Erceg and Levin (2003), the inflation target, $\pi_{T,\,t}$, is assumed to have a persistent component, $\pi_{p,\,t}$, and a transitory component, $\pi_{q,\,t}$. Thus,

$$\left(\pi_{T,\,t} - \bar{\pi}\right) = \left(\pi_{p,\,t} - \bar{\pi}\right) + \pi_{q,\,t}. \tag{27}$$

The long-run value of the inflation target, $\bar{\pi}$, is normalized to zero. Moreover, $\pi_{q,\,t}$ is a zero-mean white noise process with variance σ_q^2, and the persistent component of the inflation target follows a first-order autoregressive process

$$\pi_{p,\,t} = \rho_p \, \pi_{p,\,t-1} + \phi_z \, g\left(|z_t|;\, \tau\right) + \epsilon_{p,\,t}. \tag{28}$$

where ρ_p is the autoregressive coefficient, assumed close to unity, and $\epsilon_{p,\,t}$ is a zero-mean innovation with σ_p^2 variance. This component represents highly persistent deviations of the inflation target, although the inflation target eventually returns to its steady-state value.

Note that equation (28) introduces the novel term $\phi_z g(|z_t|;\tau)$, which captures the effects of large shocks over the persistent component of the inflation target. This shock-size block consists of a sensitivity parameter, ϕ_z, and a shock-size function, $g\left(|z_t|;\, \tau\right)$, where z_t is a disturbance metric and τ is a size threshold. Both objects are defined below.

Private agents infer the inflation target, but not the components of it. This implies that whenever a change in the inflation target is announced, the public will not know whether it is a long-lasting change of the target, or whether it will be quickly reversed. Given that, agents will need to form expectations of the persistent component of the inflation target[16], and for that purpose, it is assumed that agents use a Kalman filter to obtain an estimate of this component.

Following Céspedes and Soto (2005), who build on Erceg and Levin (2003), the optimal estimate of the persistent component at time t, based on the information available at that moment, is given by

$$\mathbb{E}_t\left[\pi_{p,\,t}\right] = \mu_g + \rho_p \, \mathbb{E}_{t-1}\left[\pi_{p,\,t-1}\right] + \left(\frac{k_g}{\rho_p}\right)\left(\pi_{T,\,t} - \mathbb{E}_{t-1}\left[\pi_{T,\,t}\right]\right). \tag{29}$$

[16]Agents will not be able to form expectations of the transitory component of the inflation target given its zero-mean nature.

where k_g denotes the calibrated *Kalman gain parameter*, and μ_g is the conditional mean of the shock-size component of equation (28), i.e. $\mu_g = \mathbb{E}_t\left[g(|z_t|;\,\tau)\right]$.[17]

The Kalman gain parameter is the core parameter of the shock-size issue of the central bank. The parameter is interpreted as an exogenously determined learning speed coefficient. A larger value of this parameter represent a greater control over inflation expectations by the central bank. This is explained by the fact that even though the persistent component is still unobserved, the expected value of it will incorporate the observed inflation target information more rapidly compared to a case in which the value of the parameter is lower. In the same spirit, lower levels of k_g, implies a lower level of credibility, given the lower level of expectation control that the central bank has.

In addition to equation (29), private agents need to form an expectation for the future evolution of the inflation target, given that they only infer the contemporaneous value of the inflation target. Taking equations (27) and (28), the optimal J-steps ahead forecast is given by

$$\mathbb{E}_t\big[\pi_{T,\,t+J}\big] = \frac{1-\rho_p^J}{1-\rho_p}\,\mu_g + (\rho_p^J)\,\mathbb{E}_t\left[\pi_{p,\,t}\right]. \tag{30}$$

3.3.2. *Shock-Size Block*

This subsection formalizes the threshold-dependent behavior of the monetary authority with respect to the inflation target. The mechanism captures the idea that small perturbations are treated as transitory noise, consistent with a look-through policy approach, whereas sufficiently large shocks may trigger a strategic temporary reorientation of the targeted policy, which can be accompanied by a convergence extension. In such episodes, the central bank may permit a transitory deviation of the implicit short-run target from its long-run level in order to limit real activity loss, thereby generating an endogenous, shock-contingent shift in the target that interacts with the policy rule and private-sector expectations.

Formally, let z_t denote a one-dimensional, policy-relevant measure of the contemporaneous structural disturbance at time t.[18] In the baseline calibration, z_t is defined as the realization of the policy-relevant exogenous shock in the model, although the framework can accommodate alternative scalar constructions. The regime and relevance of the disturbance are determined by the magnitude of the shock z_t relative to a calibrated threshold τ. The key modeling premise is that this disturbance becomes relevant for the inflation-target process only to the extent of its realized size is unusually large relative to its typical scale.

To operationalize the unusually large characteristic in the same units as the shock itself, the threshold is parametrized as $\tau = \kappa\,\sigma_z$, where σ_z is the unconditional standard deviation of the underlying innovation and $\kappa > 0$ controls how exceptional a realization must be to be treated as policy-relevant. This construction allows the model to treat small realizations as transitory disturbances that should be looked through, while preserving the possibility that sufficiently large realizations are interpreted as events that warrant a temporary shift in the implicit short-run target.

The functional form of the shock-size mechanism, $g(|z_t|;\tau)$, is specified as a hard-threshold function given by

$$g(|z_t|;\,\tau) = \big(|z_t| - \tau\big)_+ \tag{31}$$

where $\tau > 0$ denotes the size threshold parameter and $\big(x_t\big)_+$ maps its argument to the maximum of that argument x_t and zero. This implies that the persistent component of the inflation target is given by

$$\pi_{p,\,t} = \rho_p\,\pi_{p,\,t-1} + \phi_z\,\big(|z_t| - \tau\big)_+ + \epsilon_{p,\,t}. \tag{32}$$

[17]Note that under a gaussian standardized assumption for the shock variable, this conditional mean can be transformed to the unconditional mean of the function, with the following mathematical form: $2\left(\phi(\tau) - \tau(1-\Phi(\tau))\right)$. This expression tends to zero when the value of τ increases.

[18]Further discussion of the shock choice is provided in the following subsection.

The mechanism introduces a crucial state-dependent feature into the model, because it produces a piecewise-linear response based on two regimes. It is identically zero below the threshold and increases linearly with shock magnitude above the threshold, implying a kink at $|z_t| = \tau$ and a nonlinear mapping from structural disturbances to the persistent component of the inflation target. When the mechanism is active, $|z_t| - \tau$ measures the amount by which the shock realization exceeds the standard deviation threshold given by $\kappa\,\sigma_z$.

The shock-size mechanism implies two distinct effects through which structural shocks propagate. The direct effect operates through the standard transmission channels of the model framework, shaping real activity conditional on the realized disturbance. The indirect effect operates through the modification of the inflation target dynamics, and consequently, the nominal anchor of the economy. When the shock magnitude exceeds the threshold, the shock-size block shifts the persistent component of the target. Since the monetary authority follows a Taylor-type rule that responds to deviations of inflation from the target, a shift in the target changes the path of the nominal interest rate relative to a fixed-target benchmark. In particular, a higher implicit short-run target during a large-shock episode relaxes the policy stance and delivers partial accommodation.

This interaction is pivotal for understanding inflation persistence under large shocks. The direct effect of the shock captures the mechanical adjustment of the economy to the disturbance, but the indirect effect changes the policy-relevant reference point against which inflation is stabilized, thereby reshaping expectations and the subsequent disinflation path. As a result, last-mile persistence does not reflect only the intrinsic duration of the original shock. It can arise because a temporary, state-dependent displacement of the inflation target modifies the nominal interest rate trajectory and induces a gradual re-anchoring process, as agents update their beliefs about the persistent component through the signal extraction mechanism.

3.4. *Discussion*

The model considers two structural disturbances capable of generating inflationary pressure, which differ fundamentally in their transmission channels. The first is an aggregate productivity shock. An adverse realization of this shock reduces effective productive capacity in both sectors, raises the input requirements associated with a given level of production, and increases marginal costs. The second is a cost-push shock. This disturbance enters sectoral marginal costs directly and generates inflationary pressure without shifting the production frontier. Hence, although both shocks raise inflation, they differ in the sources of the output cost they generate.

The shock-size mechanism is designed to respond to the severity of the structural disruption rather than to inflation outcomes. Inflation responds endogenously to the underlying disruption and therefore acts as a symptom rather than a signal.[19] The disturbance metric is therefore defined as the primitive structural shock itself. In the quantitative analysis, this is implemented through two separate specifications. In one case, the metric is anchored in the productivity shock. In the other, the metric is anchored in the cost-push shock. Both choices preserve the structural interpretation of the trigger. In each case, the threshold τ operationalizes the notion of a tail event, separating normal fluctuations from episodes that might justify a temporary reorientation of the nominal anchor.

The absolute value trigger is a reduced form representation of tail events.[20] This formulation recognizes that extreme realizations of the underlying disturbance may occur in either direction. Although the baseline motivation emphasizes adverse supply episodes that generate inflationary pressure, sufficiently large favorable shocks could also produce sharp disinflationary forces and persistent undershooting through real interest rate dynamics and expectation formation, which might require monetary policy reorientation. The maintained specification

[19] A proxy-based approach could instead define shock size using functions of inflation (e.g., inflation itself, inflation gap or inflation surprises). However, such objects are equilibrium outcomes that reflect policy, expectations, and the target process, which blurs the intended direction of causality.

[20] Since z_t can be defined by a_t or cps_t, the activation criterion on equation (31) is based on $|a_t|$ or $|cps_t|$ relative to τ.

does not model different target responses across tails, but sign-contingent variants can be accommodated within the same framework.[21] The quantitative analysis conditions on the inflationary tail, consistent with the COVID-type motivation, and the analysis of the opposite tail is left for future research.

The signal-extraction problem connects the shock-size block to expectations. When the structural disturbance exceeds the threshold, the mechanism shifts the persistent component of the inflation target. Private agents can infer the contemporaneous target from the policy rule, but they cannot observe whether the movement reflects the persistent or transitory component. The Kalman updating rule maps the observed target surprise into a revision of beliefs about the latent persistent component. Once the belief state is updated, agents form the expected future path of the target and, through the policy rule, the expected path of monetary policy, thereby linking tail events in the real disturbance, whether originating in the supply side or in the cost structure, to a belief driven re-anchoring process.

The distinction between the shocks has important implications for the scope and effectiveness of the accommodation mechanism. Under an adverse productivity shock, part of the output loss reflects a direct reduction in effective productive capacity. Monetary accommodation can smooth the adjustment, but it cannot restore the lost production frontier, implying that part of the total output loss is irreducible regardless of the policy stance. The mechanism can attenuate the belief driven amplification of the shock, but it cannot eliminate its supply-side component. Under the cost-push shock, productive capacity is not directly affected by the disturbance. The associated output contraction arises mainly because monetary policy compresses demand in response to higher marginal costs and inflation. The output loss is therefore a mainly stabilization cost and is more directly shaped by the monetary policy response. The accommodation mechanism should therefore have greater leverage in the cost-push case, since a temporary target adjustment can reduce the output losses associated with monetary stabilization while preserving long run re-anchoring through the signal extraction process.

Overall, the mechanism can be understood as a tail-event adjustment of the nominal anchor driven by primitive structural disturbances. By defining the trigger in fundamentals, the framework preserves a structural interpretation of the policy response. By combining this trigger with imperfect observability of the inflation target, the model links large shocks to expectations, policy paths, and the gradual re-anchoring of inflation.

4. Solution and Calibration

To analyze the behavior of this model, we log-linearize the model's equations around the non-stochastic steady-state. The log-linearized model consists of seventeen core equilibrium equations, from which twelve come from the private-sector block (two sectoral hybrid Phillips curves (10), two sectoral marginal costs (12), the Euler equation for consumption (19), two sectoral Tobin's q (20), two sectoral Euler equations for investment (21), the intersectoral consumption allocation (22), and two sectoral hybrid wage Phillips curves (25)), three determining the monetary authorities behavior (monetary policy rule (26), target process (27), and the threshold dependent persistent component (32)), and two determining private sector expectations (expected persistent target (29), and the forecast expected target (30)).

4.1. *Solution and Methodologies*

Conditional on a realized path of the shock-size sequence $g(|z_t|;\tau)$, the model is solved using a first-order perturbation around the deterministic steady state. Under this conditioning, the equilibrium system is linear in the endogenous variables and can be represented in the canonical rational-expectations form of O. J. Blanchard and Kahn (1980). Because the shock-size function introduces a piecewise-linear mapping from structural disturbances to the inflation-target process, the nonlinearity is handled outside the solution step. The realized path of the underlying disturbance z_t is first constructed by simulating its law of motion, the function

[21] A sign-contingent shock-size mechanism is out of the scope of the present work.

$g(|z_t|;\tau) = (|z_t| - \tau)_+$ is evaluated at each date, and the resulting sequence is injected into the simulation as a pre-computed exogenous input. Within each period the model therefore admits a standard linear solution, and the state-dependent activation of the mechanism is captured through the exogenous variation in the $g(\cdot)$ sequence across the reference and perturbed paths.

Since the shock-size block makes the transmission from primitive disturbances to endogenous outcomes state-dependent, standard impulse response functions are no longer appropriate for characterizing the model's dynamics. The analysis therefore relies on Generalized Impulse Response Functions (GIRFs; Koop et al., 1996). The procedure is implemented as follows. A total of $T = 400$ periods are simulated, of which the first $T_0 = 200$ serve as a burn-in phase. The impulse is introduced at $t_0 = T_0 + 1$, and responses are recorded over the subsequent $T = 100$ periods. For each replication $r = 1, \ldots, 2000$, a common draw of innovations $\{\boldsymbol{\varepsilon}_t^{(r)}\}_{t=1}^{T}$ is generated for all structural shocks. A *reference path* is then constructed by simulating the full model over T periods using these innovations. The relevant disturbance process z_t is built from its AR(1) law of motion, and the shock-size function $g_t^0 = (|z_t| - \tau)_+$ is evaluated at each date and passed to the model as exogenous input. In the *perturbed path*, the same innovations sequence is used, except that at date t_0 the innovation to the shocked process, z_t, is increased by ν standard deviations, with $\nu = 5$ in the baseline exercises. The AR(1) path of the disturbance is recomputed with this augmented innovation, yielding a new shock-size sequence g_t^1.[22] The GIRF for each endogenous variable x at horizon h is then defined as the average of the difference between the two paths,

$$\text{GIRF}_x(h) = \frac{1}{R}\sum_{r=1}^{R}\left[x_{t_0+h}^{1,(r)} - x_{t_0+h}^{0,(r)}\right], \qquad h = 1, \ldots, H. \tag{33}$$

The burn-in phase ensures that the economy at the time of the impulse reflects the model's ergodic distribution rather than the deterministic steady state, so that the GIRF is computed conditional on a representative stochastic environment. In addition, averaging over a large number of replications integrates out the influence of any particular sequence of future shocks.

4.2. *Calibration*

The model is calibrated at a quarterly frequency using parameter values drawn from the literature, complemented by assumptions consistent with the characteristics of an advanced economy. The calibration of the model proceeds in two stages. Grid values are given to the shock-size block mechanism. The remaining parameters of the model are calibrated at point values consistent with the literature.

4.2.1. *Shock-Size Block and Monetary Policy Parameters*

The parameters governing the shock-size mechanism, the Kalman gain k_g, the persistence of the permanent component of the inflation target ρ_p, and the pass-through coefficient from the disturbance to the target ϕ_z, are explored over a grid rather than fixed at point values, since the model is designed precisely to characterize how the dynamic response of the economy varies with the credibility of the nominal anchor, the horizon over which the target component is expected to revert, and the strength of the shock-target linkage.

The Kalman Gain, k_g, ranges from 0.15 to 0.95, governing the speed at which private agents update their expectations about the persistent component of the inflation target after observing an inflation target realization, and thus, is interpreted as the credibility stock of the Monetary Authority. A value of $k_g = 0.15$ corresponds to a no-credibility regime, in which agents place little weight on each new observation, so that information about a change in the target is completely incorporate in more than a year. On the opposite, a value of $k_g = 0.95$ corresponds to a full-credibility regime in which agents incorporate almost the entirety of each observation of the target into their expectations in a period, so that any reorientation of the nominal anchor is quickly internalized.

[22] Note that at this point, all other innovations of the model remain unchanged.

The persistence of the permanent target component, ρ_p, ranges from 0.65 to 0.85, which determines the length of the inflation target deviation from its' steady-state value, and thus, the monetary policy horizon for inflation convergence. A lower ρ_p implies that any deviation of the persistent target component from its long-run value is undone relatively quickly, while a higher value for the parameter implies that such deviations are nearly permanent and therefore have stronger consequences for medium-term inflation dynamics.

The pass-through coefficient of the inflation target, ϕ_z, ranges from 0.0 to 0.9, and it governs how strongly the realized magnitude of the underlying disturbance, once it passes the threshold, is transmitted into the persistent component of the inflation target. A value of $\phi_z \approx 0.0$ corresponds to a regime in which the central bank looks through tail shocks almost entirely, leaving the nominal anchor almost unchanged, responding with a normal monetary policy. A value of $\phi_z = 0.9$ corresponds to a regime in which the target is reoriented by close to the full magnitude of the excess disturbance. Intermediate values capture partial accommodation. Given the selected shock-size, values over a 30% retro-alimentation generate abnormal real interest rate dynamic, so they are excluded from the exercise.

The activation threshold is set to $\tau = 2\sigma_z$, so that the shock-size mechanism remains dormant under normal cyclical fluctuations and becomes active only when the realized magnitude of the disturbance exceeds two standard deviations of its unconditional distribution. This convention is consistent with the macroeconomic literature on rare disasters and large shocks (Barro, 2006; Gourio, 2012) and with the analysis of nonlinear responses to outlier realizations (Ascari and Sbordone, 2014; Aruoba et al., 2018). The baseline impulse used to compute the GIRFs is set to $\nu = 5\sigma_z$, a magnitude motivated by the empirical record of recent crises. The COVID-19 pandemic generated quarterly contractions on the order of 8 to 10 percent in major economies (Eichenbaum et al., 2021; Guerrieri et al., 2022; Giovanni et al., 2023), corresponding to deviations of roughly four to six standard deviations relative to the pre-pandemic distribution of the relevant shock processes, with comparable magnitudes characterizing the 2008 financial crisis. The chosen value of ν accordingly lies in the middle of this empirical range.

Regarding the interest rate reaction function, standard literature values are used. Erceg and Levin (2003), consistent with Céspedes and Soto (2005), determine the value for the reaction parameters, namely ϕ_π and ϕ_y, as 2.0 and 0.5 respectively. For the inertia coefficient, $\gamma_i = 0.5$, closely matching Erceg and Levin (2003).

4.2.2. *Firms Parameters*

The two sectors are modeled with CES production functions that differ in the elasticity of substitution between capital and labor. The elasticity is set to $\sigma_S = 0.5$ in the service-providing sector and $\sigma_G = 0.8$ in the good-producing sector. Both values are below unity, as Herrendorf et al. (2015) empirically finds, implying gross complementary between factors. The parameter values are consistent in range with the plant-level evidence for U.S. manufacturing reported by Oberfield and Raval (2021), who find elasticity values concentrated between 0.5 and 0.7, and with the DSGE estimates of Cantore et al. (2021). The order between elasticities, $\sigma_S < \sigma_G$, is based upon the sectoral asymmetries work of Acemoglu and Guerrieri (2008), later deepened by Alvarez-Cuadrado et al. (2017) and Eden and Gaggl (2018). The sectoral labor shares, in line with the sectoral factor income decompositions of Valentinyi and Herrendorf (2008) and Schwellnus et al. (2018), are 60% for service sector and 40% for goods sector.

The depreciation rates of capital are set to $\delta_S = \delta_G = 0.025$ in both sectors, the standard quarterly value in the literature. Investment adjustment costs are set to $\phi_I^S = \phi_I^G = 5$. The chosen value lies within the range estimated by Smets and Wouters (2007) and Justiniano et al. (2010), who report values in estimated DSGE models for the U.S. economy. Erceg and Levin (2003) and Christiano et al. (2005) parameters closely match the selected values.

The Calvo probabilities of price rigidity are set to $\alpha_S = 0.75$ for services and $\alpha_G = 0.50$ for goods, implying average price contracts of four and two quarters, respectively. Both the levels and the ordering between them are grounded in the micro-data evidence of Bils and Klenow

(2004) and Nakamura and Steinsson (2008), who document that the median frequency of price adjustment is substantially higher in goods than in services. The asymmetry $\alpha_s^p > \alpha_g^p$ is also the one adopted by Carvalho (2006), Nakamura and Steinsson (2010) and Pastén et al. (2020), among others, in heterogeneous-sector New Keynesian models, where this kind of sectoral dispersion in price stickiness has been shown to amplify the real effects of monetary policy shocks relative to a one-sector benchmark. The remaining parameters of the hybrid New Keynesian Phillips curve are calibrated symmetrically across sectors. The share of backward-looking firms is set to $\omega_s^p = \omega_g^p = 0.30$, following the original estimations of Galí and Gertler (1999). Finally, the indexation parameter for backward-looking firms is set to $l_S = l_G = 0.9$, indicating near-complete indexation of non-optimized prices to lagged inflation. This value is consistent with the estimated indexation parameters in Smets and Wouters (2003), Smets and Wouters (2007), and the seminal specification of Christiano et al. (2005), who find that lagged-inflation indexation is essential for matching the observed persistence of aggregate inflation.

4.2.3. *Households Parameters*

The discount factor is set to $\beta = 0.99$, the standard quarterly value implying an annualized real interest rate of approximately 4%. The inverse elasticity of intertemporal substitution is set to $\sigma = 1.5$, which implies an elasticity of 0.67, in line with the estimated DSGE values of Smets and Wouters (2007). The inverse Frisch elasticity is set to $\chi = 2.0$, elasticity value in the upper end of the range supported by the micro-level evidence surveyed in Chetty et al. (2011), and is consistent with the values used in Erceg et al. (2000).

The Calvo wage rigidity parameters are set to $\alpha_S^W = 0.40$ for the services sector and $\alpha_G^W = 0.75$ for the goods sector, implying average wage contract durations of approximately two and four quarters, respectively. The goods-sector value is closely aligned with the aggregate U.S. estimates of Smets and Wouters (2007), who report probability close to 0.73, and lies somewhat above the 0.64 wage non-reoptimization probability estimated in Christiano et al. (2005). The sectoral asymmetry $\alpha_S^W < \alpha_G^W$ is treated as a calibrated baseline rather than as a direct empirical estimate. Some evidence is consistent with stronger wage rigidity in goods-producing or manufacturing activities (Anderton et al., 2017; Jackson and Kurt, 2025). However, the broader micro evidence is more mixed, finding limited cross-sectoral heterogeneity (Le Bihan et al., 2012; Barattieri et al., 2014). Cross-country European evidence also indicates that sectoral patterns are mediated by country institutions, collective bargaining, employment protection, firm size, workforce composition, and the wage concept used (Messina et al., 2010; Babecký et al., 2010; Druant et al., 2012). The calibration therefore uses a sectoral ordering that is consistent with part of the evidence, while the robustness exercises evaluate whether the mechanism depends on this particular wage-rigidity ranking. The wage indexation parameters are set to $l_S^W = l_G^W = 0.50$, indicating symmetrical partial indexation of non-optimized wages to lagged inflation. This value is consistent with the estimated indexation parameters in Smets and Wouters (2007).

The elasticity of substitution between services and goods consumption is set to $\eta_c = 0.9$, a value just below unity that places the consumption aggregator close to the Cobb-Douglas case while preserving mild gross complementarity between sectoral consumption. This value is intended to reconcile two strands of the literature that have reached different conclusions on the appropriate value of this parameter. The structural change literature, working at low frequencies, consistently finds substantial complementarity between services and goods. Herrendorf et al. (2013) estimate η_c close to zero using long U.S. expenditure data, Moro (2012) calibrates a value near 0.45 in a two-sector model, and Comin et al. (2021) document that complementarity is robust under a non-homothetic CES specification. Rogerson (2008) and Duarte and Restuccia (2010) reach similar empirical conclusions. The business-cycle literature, by contrast, has typically employed higher values. Bouakez et al. (2009) and Bouakez et al. (2014) use elasticities close to or moderately above unity, on the grounds that short-run substitutability between sectoral consumption may exceed the slow secular pattern documented in long-run data. The chosen value of 0.9 takes a middle position, sufficiently close to maintain consistency with both strands of literature.

The elasticity of substitution between sectoral labor types, which governs the household's willingness to reallocate hours between sectors when relative wages move, is set to $\eta_n = 1.8$. This value falls within the range used in Horvath (2000), and in the multi-sector New Keynesian framework of Bouakez et al. (2014). The ordering $\eta_n > \eta_c$, with labor more mobile across sectors than consumption, is consistent with the view that workers reallocate across sectors at higher frequency than households revise their consumption baskets, an asymmetry supported by the micro evidence on intersectoral labor mobility in Lee and Wolpin (2006).

5. Results

This section analyzes the quantitative implications of the model through simulated responses to large supply-side disturbances. The exercises compare alternative configurations of the shock-size mechanism, credibility, and policy-horizon persistence, isolating how temporary movements in the inflation target affect general dynamics. A five standard deviation shock impulse is assumed, motivated by the empirical record of recent crises.

5.1. *The Shock-Size Mechanism and the Output-Inflation Trade-Off*

Allowing large shocks to shift the inflation target through its persistent component reduces the associated output costs while tolerating a more sustained inflation path. Comparing the system dynamics with and without the shock-size mechanism, given by equation (28), isolates the effects of the proposed inflation targeting regime.

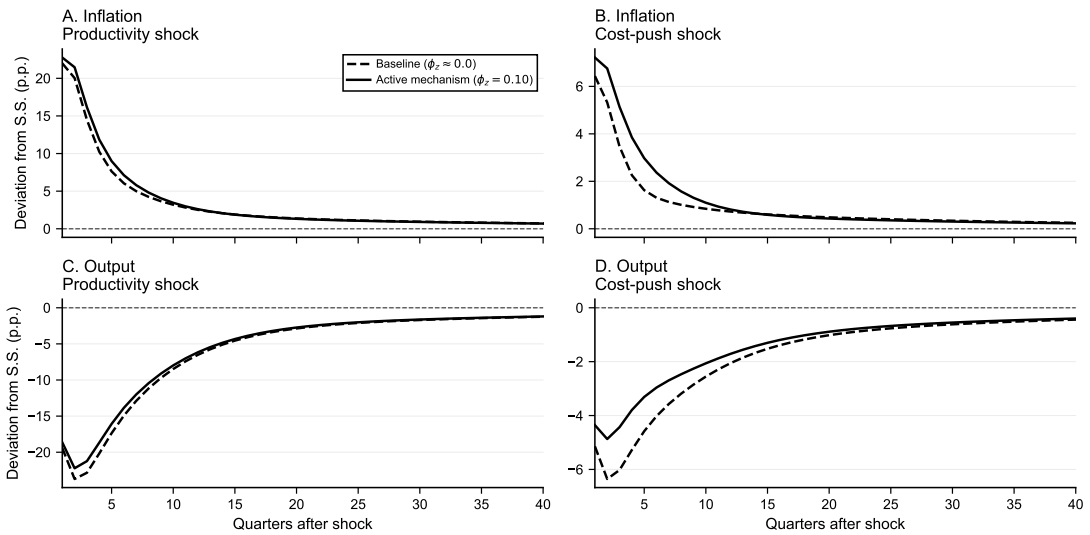


Figure 4. *Generalized Impulse Response Functions* under the shock-size mechanism.

Figure 4 presents the GIRFs of inflation and output to a productivity and cost-push shock, under the two described scenarios. The active mechanism produces a more sustained inflation path and higher cumulative response relative to the baseline scenario. Nevertheless, the inflation peak is not meaningfully higher when the mechanism is active. In terms of output loss, the initial trough is similar across scenarios for the productivity shock, but shallower under the active mechanism for the cost-push shock. However, the output recovers faster and more completely under the active mechanism for both shocks. The output difference between scenarios is more visible at medium horizons, with both scenarios converging to a near-common path from mid-periods onwards.

Table 3 presents summary statistics for inflation and output for the given shocks. Even though the active mechanism implies higher cumulative inflation, at least 6.9% and 24.7% more for productivity and cost-push shocks, respectively, the output losses are reduced by at least 5.9% and 19.6%. In addition to the output loss reduction, the percentage points of output sacrificed for each percentage point of cumulative inflation pressure are reduced by

TABLE 3. Shock-Size Mechanism: Summary Statistics

	$\sum^{40} \pi_t$	$\sum^{40} y_t$	$H^*_{5\%}$	$H^{\pi-\pi_T}_{5\%}$	SR^{40}_{rec}	SR^{40}_{vol}
Productivity Shock						
Baseline	1.350	-2.424	26	26	-1.796 $(-,+)$	2.221
Active Mechanism	1.443	-2.282	24	27	-1.582 $(-,+)$	1.670
Cost-Push Shock						
Baseline	0.372	-0.716	32	32	-1.924 $(-,+)$	2.364
Active Mechanism	0.464	-0.576	25	42	-1.239 $(-,+)$	0.854

Note: The variables π_t and y_t denote inflation and output, respectively. All cumulative statistics and sacrifice ratios are computed over 40 periods. $H^*_{5\%}$ denotes the first horizon at which inflation falls to at most 5% of its initial response and remains there for four consecutive periods, while $H^{\pi-\pi_T}_{5\%}$ denotes the analogous relative convergence horizon for the inflation gap $(\pi_t - \pi_{T,t})$. Sacrifice ratios measure output loss per unit of cumulative inflation pressure, formal definitions are provided in Appendix B. The sign pair in parentheses reports the sign of the ratio for the numerator and denominator. Complete statistics are reported in Table B.1 in Appendix B.

11.9% for the productivity shock, and 35.6% for the cost-push shock. This implies that, for both shocks, the recession-associated costs and the volatility of the adjustment path are significantly reduced, as reflected in the improvement of both ratio measures. The mechanism allows higher cumulative and sustained inflation, but delivers lower output costs, resulting in a better output-inflation trade-off.

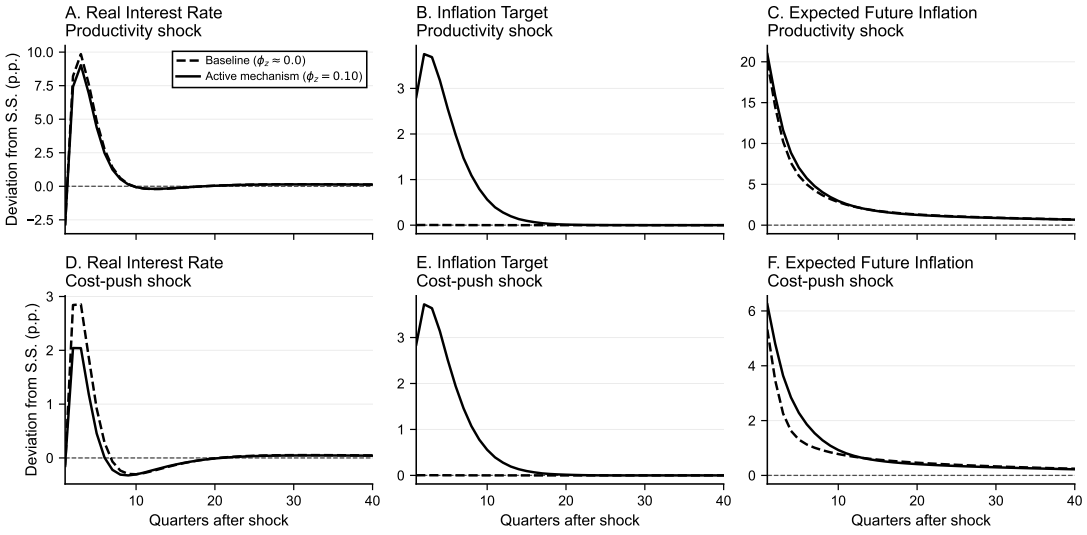


FIGURE 5. *Generalized Impulse Response Functions* under the shock-size mechanism.

Figure 5 presents the GIRFs for the real interest rate, the inflation target and the expected future inflation to the selected shocks. The shock-dependent mechanism shifts the inflation target upwards, sustaining a higher-than-steady state value for an extended period. Accompanying the target movement, the activation of the mechanism leads to higher private sector expected inflation, with a greater peak and more expectations persistence than the baseline scenario. The upward movement of the target directly feeds into future inflation expectations, which rise above the baseline path and remain elevated for a sustained period. This expectations channel is the primary transmission mechanism through which the active regime affects real activity.

Table 4 presents summary statistics for monetary policy related variables for both shocks. The cumulative nominal interest rate is higher under the active mechanism, increasing by 2.5%and 9.6% for the productivity and cost-push shock respectively. This higher nominal

TABLE 4. Shock-Size Mechanism: Policy-Stance Summary Statistics

	$\sum^{40} i_t$	$\sum^{40} r_t$	$\sum^{40} \mathbb{E}_t \pi_{t+1}$	$\sum^{40} \pi_{T,t}$
Productivity Shock				
Baseline	1.479	0.342	1.137	—
Active Mechanism	1.516	0.309	1.207	0.232
Cost-Push Shock				
Baseline	0.383	0.073	0.310	—
Active Mechanism	0.420	0.040	0.380	0.230

Note: The variables i_t, r_t, $\mathbb{E}_t \pi_{t+1}$, and $\pi_{T,t}$ denote the nominal interest rate, real interest rate, expected inflation, and inflation target, respectively. Expected inflation is inferred from a *Fisher* equation. All cumulative statistics are computed over 40 periods. Complete statistics are reported in Table B.2 in Appendix B.

response does not imply tighter real conditions. This higher nominal response does not imply tighter real conditions. The cumulative real rate falls by 9.7% for the productivity shock and by 45.2% for the cost-push shock. The difference is accounted for by the expectations channel, since the implied cumulative expected inflation response rises by 6.2% for the productivity shock and by 22.5% for the cost-push shock. The accommodation therefore operates through the real interest rate rather than through conventional nominal easing, supporting the expectations channel.

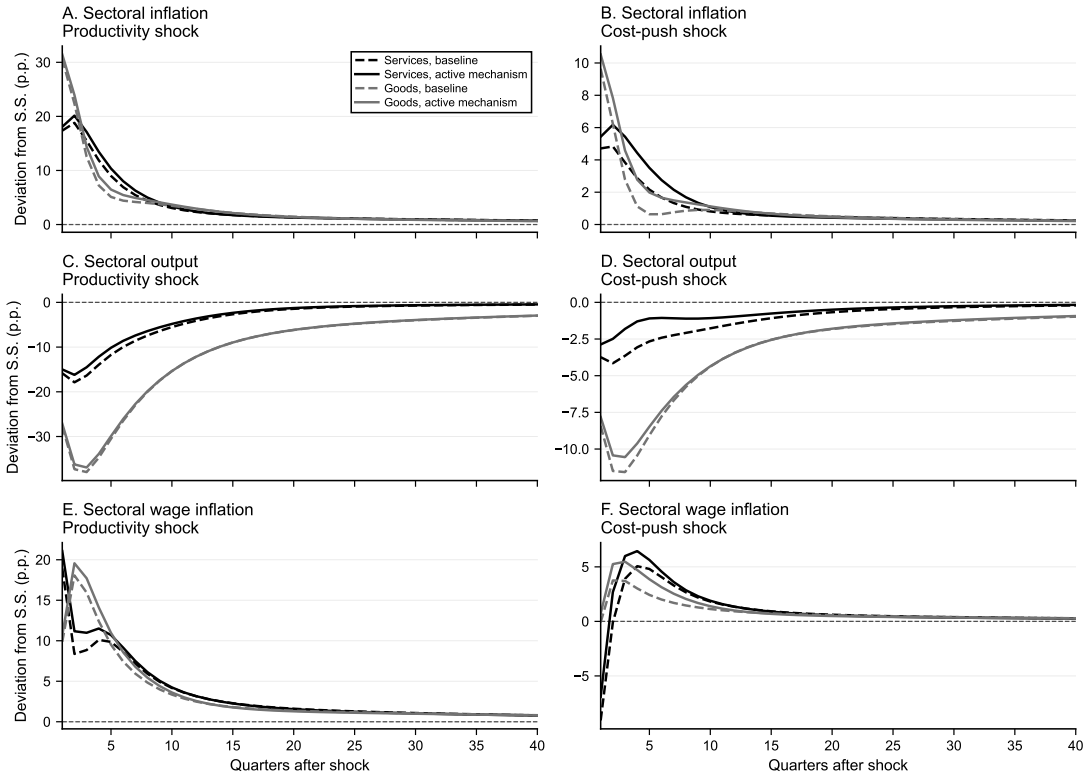


FIGURE 6. *Generalized Impulse Response Functions* under the shock-size mechanism.

Figure 6 presents the GIRFs for the sectoral inflation, sectoral output and sectoral wage inflation for the described scenarios.[23] On the nominal side, the active mechanism raises both price and wage inflation in the services and goods sectors. The active paths remain above the baseline ones in the short run, and the gap progressively closes along the convergence path, indicating that the nominal effect is broad across sectors. On the real side, the gains are

[23]For the corresponding cumulative statistics, please refer to Table B.3 in Appendix B.

service-led rather than evenly distributed across sectors. Services output losses are reduced under both shocks, while the corresponding improvement in goods output is considerably lower. Hence, the service sector is the main real-side margin through which the mechanism improves the aggregate output-inflation trade-off, mainly by accelerating the recovery of services activity, rather than by producing a symmetric real adjustment across sectors.

This sectoral-asymmetric real-side recovery follows from the price-setting structure of the model. The active mechanism lowers the real interest rate and adds a demand impulse on top of the supply contraction, and price rigidity governs how it splits between output and inflation in each sector. In the stickier sector prices adjust slowly, so the relative price falls and the additional demand is met through output rather than through inflation, while in the more flexible sector the same impulse is absorbed mainly through prices. The wage side works against this gain, since more flexible wages in services raise their marginal cost faster as activity recovers, which erodes part of that output gain. Two further features of the model reinforce this concentration once the impulse reaches each sector. Because the impulse is lead by consumption, it favors services, whose output depends more on consumption, over goods, whose output depends more on investment and is slowed by investment adjustment costs. As the impulse raises labor demand, the larger labor share in services converts that recovery into a larger output gain. The real relief still concentrates in services, so the price-setting advantage outweighs the wage-side offset.

The mechanism relaxes policy during large shocks by raising the inflation target contingently, allowing part of the adjustment to occur through cumulative sustained inflation rather than entirely through output. The mechanism does not eliminate inflation persistence. Instead, it tolerates sustained inflation, at a higher relative convergence horizon, to reduce the output cost of large shocks. This results in a better output-inflation trade-off when monetary policy is allowed to react to large supply shocks.

5.2. *Policy-Horizon Flexibility under High Credibility*

As the previous subsection shows, under large disturbance episodes, there are clear gains from a state-dependent monetary policy. Given the benefits of the mechanism, extending the policy horizon reduces output losses at the cost of higher cumulative inflation pressure and, at sufficiently long horizons, slower inflation convergence. Comparing the system dynamics, conditional on that the mechanism is active, when changing the policy horizon, governed by ρ_p, separates the horizon effects when credibility is high.

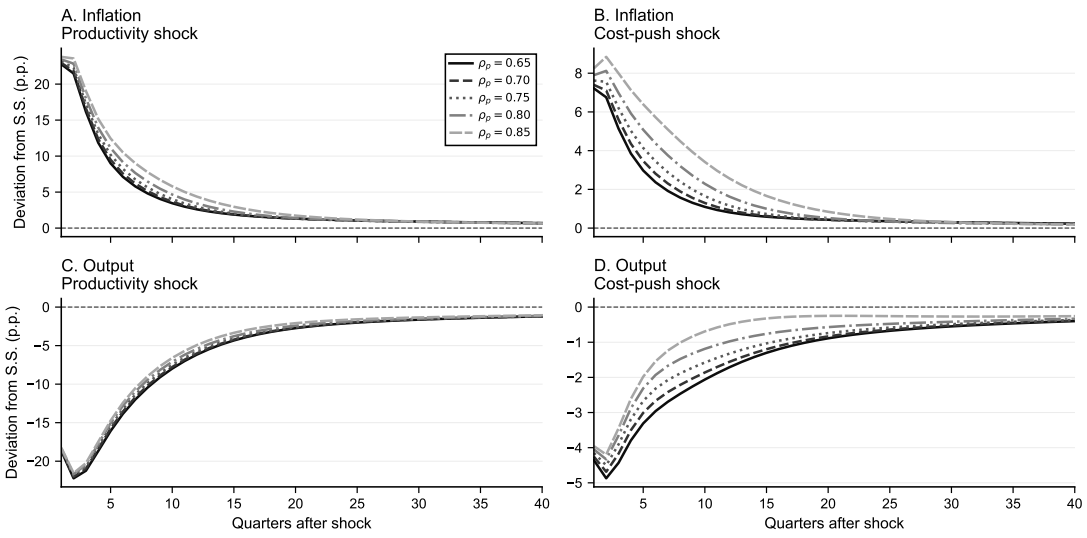


FIGURE 7. *Generalized Impulse Response Functions* under alternative policy horizon.

Figure 7 presents the GIRFs of inflation and output to the selected shocks for the active mechanism with high credibility but different policy-horizon scenarios. Inflation shows progressive fanning with similar initial peaks for both shocks. A longer policy horizon yields a larger cumulative response, but it increases inflation persistence under sufficiently long horizons. In terms of output, the initial troughs vary little across horizon scenarios within each shock, showing faster initial recoveries as the policy horizon increases. For the cost-push shock, high values of ρ_p substantially reduce output losses by around period 15. However, at lower horizon values, the same values are reached at period 35. The output path differences come entirely from the recovery path, with longer horizon diminishing output losses faster.[24]

TABLE 5. Policy-Horizon Flexibility: Extensions Under High Credibility

Panel A. Summary Statistics

	H (yr)	$\sum^{40}\pi_t$	$\sum^{40} y_t$	$H^*_{5\%}$	$H^{\pi-\pi_T}_{5\%}$	SR^{40}_{rec}	SR^{40}_{vol}
Productivity Shock							
$\rho_p = 0.65$	3.2	1.443	−2.282	24	27	−1.582 (−, +)	1.670
$\rho_p = 0.70$	3.8	1.481	−2.242	23	27	−1.514 (−, +)	1.545
$\rho_p = 0.75$	4.8	1.542	−2.189	23	26	−1.419 (−, +)	1.395
$\rho_p = 0.80$	6.2	1.644	−2.114	23	24	−1.286 (−, +)	1.218
$\rho_p = 0.85$	8.5	1.834	−2.007	25	23	−1.094 (−, +)	1.012
Cost-Push Shock							
$\rho_p = 0.65$	3.2	0.464	−0.576	25	42	−1.239 (−, +)	0.854
$\rho_p = 0.70$	3.8	0.503	−0.536	23	39	−1.066 (−, +)	0.643
$\rho_p = 0.75$	4.8	0.563	−0.483	23	35	−0.858 (−, +)	0.443
$\rho_p = 0.80$	6.2	0.664	−0.409	23	28	−0.616 (−, +)	0.272
$\rho_p = 0.85$	8.5	0.852	−0.303	27	23	−0.355 (−, +)	0.146

Panel B. Marginal gains from a one-year horizon extension

	Base$H$	ΔH	ΔSR^{40}_{rec}	
Scenario	(yr)	(yr)	Productivity	Cost-Push
Low base (ρ_p: 0.650 → 0.720)	3.2	1.0	0.102	0.252
Mid base (ρ_p: 0.760 → 0.795)	5.0	1.0	0.095	0.170
High base (ρ_p: 0.820 → 0.842)	7.0	1.1	0.088	0.116
Decline: low → mid			−7.43%	−32.47%
Decline: mid → high			−7.41%	−32.08%
Decline: low → high			−14.26%	−54.14%

Note: The variables π_t and y_t denote inflation and output, respectively. All cumulative statistics and sacrifice ratios are computed over 40 periods. H denotes the implied policy horizon in years. $H^*_{5\%}$ denotes the first horizon at which inflation falls to at most 5% of its initial response and remains there for four consecutive periods, while $H^{\pi-\pi_T}_{5\%}$ denotes the analogous relative convergence horizon for the inflation gap $(\pi_t - \pi_{T,t})$. The sign pair in parentheses reports the sign of the ratio for the numerator and denominator. Panel A reports cumulative summary statistics by shock and policy scenario. Panel B compares pairs of configurations that differ by approximately one year of policy horizon, anchored at three different base levels. ΔSR^{40}_{rec} is expressed as an absolute improvement. Complete statistics are reported in Table B.4 in Appendix B.

Panel *A* of Table 5 presents summary statistics for inflation and output for the given policy extensions and shocks. In spite of higher accumulated inflation as the policy horizon is extended, the output losses are reduced, which leads to less percentage points of output sacrificed per point of cumulative inflation pressure, for both shocks. From the minimum to maximum policy horizon, the output points sacrificed per point of cumulative inflation pressure decrease a 30.8% and 71.3% for the productivity and cost-push shock, respectively. The sacrifice ratios

[24]As in Subsection 5.1 and previously explained, cost-push shocks present more gains from the mechanism, compared to the productivity shock.

improve monotonically as the policy horizon is extended for both shocks.

Panel *B* of Table 5 presents recession-cost statistics for three different horizon scenarios. To isolate the effect of the base horizon, three pairs of configurations are compared, each extending the horizon by approximately one year, but anchored at different starting levels.[25] The marginal benefit of horizon flexibility declines as the initial policy horizon increases. For the cost-push shock, the gain from a one-year extension falls by approximately 54.1% when moving from the low-base to the high-base horizon. For the productivity shock, the corresponding decline is approximately 14.3%. Central banks starting from short horizon gain substantially more per unit of additional flexibility than one that already is operating at a long horizon.

Under high credibility, a longer policy horizon allows the central bank to smooth the real effects of a large shock, so that the extension generates a monotonic improvement in the output-inflation trade-off. Horizon extension reallocates adjustment from output to inflation pressure, reducing the cumulative cost of inflation. Crucially, the gain from each additional year of horizon is larger when the central bank starts from a short base than from a long one, implying that the marginal value of flexibility is highest when the stabilization problem is most constrained by policy horizon.

5.3. *The Cost of Low Credibility*

The gains from the shock-size mechanism documented above rely on the complete incorporation of target adjustments into private-sector expectations. Holding the policy horizon fixed, a decline in credibility reduces the cumulative inflation but increases output losses, thereby eroding the gains of the shock-size mechanism. Holding the policy horizon fixed and varying credibility, governed by k_g, isolates the credibility channel of the model.

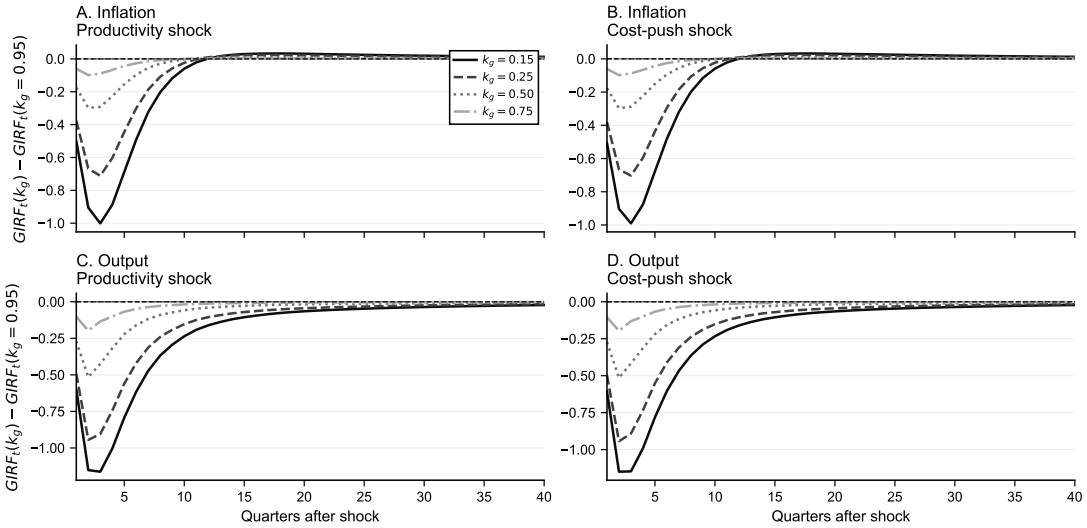


FIGURE 8. *Generalized Impulse Response Functions* relative to the high-credibility benchmark.

Figure 8 presents, for inflation and output, the difference between each lower-credibility GIRF and the high-credibility benchmark, $GIRF_t(k_g) - GIRF_t(k_g = 0.95)$, for both shocks. Across shocks, inflation gap shows deeper troughs as credibility falls, roughly reaching a one p.p. GIRF difference at the minimum for the lowest possible credibility. In terms of output loss, lower credibility never helps for any shock, progressively worsening the output path. Common across variables, the cost is front-loaded, where inflation rapidly converges to a common path, differently for the output one. The cost of credibility loss is concentrated in the early horizons rather than in the long-run convergence path.

[25]Note that ρ_p is a non-linear measure of the monetary policy horizon.

TABLE 6. Cost of Low Credibility: Summary Statistics

	$\sum^{40} \pi_t$	$\sum^{40} y_t$	$H^*_{5\%}$	$H^{\pi-\pi_T}_{5\%}$	SR^{40}_{rec}	SR^{40}_{vol}
Productivity Shock						
$k_g = 0.15$	1.397	-2.368	25	29	-1.695 $(-,+)$	1.985
$k_g = 0.25$	1.412	-2.344	25	29	-1.660 $(-,+)$	1.893
$k_g = 0.50$	1.431	-2.310	24	28	-1.614 $(-,+)$	1.764
$k_g = 0.75$	1.439	-2.291	24	28	-1.592 $(-,+)$	1.701
$k_g = 0.95$	1.443	-2.282	24	27	-1.582 $(-,+)$	1.670
Cost-Push Shock						
$k_g = 0.15$	0.419	-0.660	29	48	-1.575 $(-,+)$	1.592
$k_g = 0.25$	0.434	-0.637	28	47	-1.466 $(-,+)$	1.346
$k_g = 0.50$	0.453	-0.603	26	44	-1.330 $(-,+)$	1.046
$k_g = 0.75$	0.461	-0.585	25	43	-1.268 $(-,+)$	0.914
$k_g = 0.95$	0.464	-0.576	25	42	-1.239 $(-,+)$	0.854

Note: The variables π_t and y_t denote inflation and output, respectively. All cumulative statistics and sacrifice ratios are computed over 40 periods. $H^*_{5\%}$ denotes the first horizon at which inflation falls to at most 5% of its initial response and remains there for four consecutive periods, while $H^{\pi-\pi_T}_{5\%}$ denotes the analogous relative convergence horizon for the inflation gap $(\pi_t - \pi_{T,t})$. The sign pair in parentheses reports the sign of the ratio for the numerator and denominator. Complete statistics are reported in Table B.6 in Appendix B.

Table 6 presents summary statistics for inflation and output across credibility levels. For both shocks, cumulative inflation increases with credibility, while the cumulative output loss deepens as credibility falls. In terms of percentage units of output loss per unit of cumulative inflation pressure, the cost-push shock implies 27.1% more units sacrificed when moving from the highest to the lowest credibility. For the same metric but for the productivity shock, the units sacrificed increase by 7.1%. A shock asymmetry arises at this point. Credibility loss costs roughly four times more for cost-push shocks than for productivity shocks in lost output per point of cumulative inflation pressure. This reverses the pattern from previous subsections. The shock that depended most on the mechanism for its trade-off improvement is the same that suffers the most when the mechanism fails to transmit.

TABLE 7. Cost of Low Credibility: Policy-Stance Summary Statistics

	$\sum^{40} i_t$	$\sum^{40} r_t$	$\sum^{40} \mathbb{E}_t \pi_{t+1}$	$\sum^{40} \pi_{T,t}$
Productivity Shock				
$k_g = 0.15$	1.512	0.336	1.176	0.232
$k_g = 0.25$	1.519	0.331	1.188	0.232
$k_g = 0.50$	1.522	0.322	1.200	0.232
$k_g = 0.75$	1.519	0.314	1.205	0.232
$k_g = 0.95$	1.516	0.309	1.207	0.232
Cost-Push Shock				
$k_g = 0.15$	0.416	0.066	0.350	0.230
$k_g = 0.25$	0.423	0.062	0.361	0.230
$k_g = 0.50$	0.426	0.053	0.373	0.230
$k_g = 0.75$	0.423	0.045	0.378	0.230
$k_g = 0.95$	0.420	0.040	0.020	0.230

Note: The variables i_t, r_t, $\mathbb{E}_t \pi_{t+1}$, and $\pi_{T,t}$ denote the nominal interest rate, real interest rate, expected inflation, and inflation target, respectively. Expected inflation is inferred from a *Fisher* equation. All cumulative statistics are computed over 40 periods. Complete statistics are reported in Table B.7 in Appendix B.

Table 7 presents summary statistics for monetary policy variables for the given shocks under different credibility values. The monetary policy intent, the target movement, is the same across credibility levels within the shocks, since credibility shapes how private agents react to

monetary policy, not what monetary policy is.[26] As credibility is lost, cumulative real-rate tightening increases. Moving from high to low credibility, the cumulative real rate rises by 8.7% for the productivity shock and by 65.0% for the cost-push shock, while cumulative expected inflation falls by 2.6% and 7.9%, respectively. Nominal rate differences are small and non-monotonic, so the relevant policy margin is the credibility-mediated real-rate response. When policy is not credible, expected inflation is lower because private agents do not internalize the central bank's intended accommodation. This leads to higher real rates, which finally impacts output loss convergence. The cumulative real rate path, together with the implied expected inflation, shows how the mechanism fails when the central bank is not credible.

Credibility loss erodes the benefits of the shock-size mechanism monotonically. Cost-push shocks, which concentrated the benefits of the mechanism, are precisely the ones that lose the most when credibility is reduced. The mechanism's value and its core fragility flow through the same channel, inflation expectations, which ultimately shapes real rate dynamics. The active mechanism is only as effective as the credibility on which it relies. Without a credible monetary strategy, the central bank moves the inflation target on paper but does not move expectations, so the output absorbs the adjustment the mechanism was designed to spare it.

5.4. *Horizon Flexibility under Credibility Stress*

Low credibility weakens the gains of the active mechanism. A monetary authority that responds to such an environment by extending its operational horizon may fail to recover the stabilization gains obtained under a credible nominal anchor. A high-credibility short-horizon configuration is compared with a low-credibility extended-horizon case, providing a stress comparison between a credible policy stance and a more flexible but less credible alternative.[27]

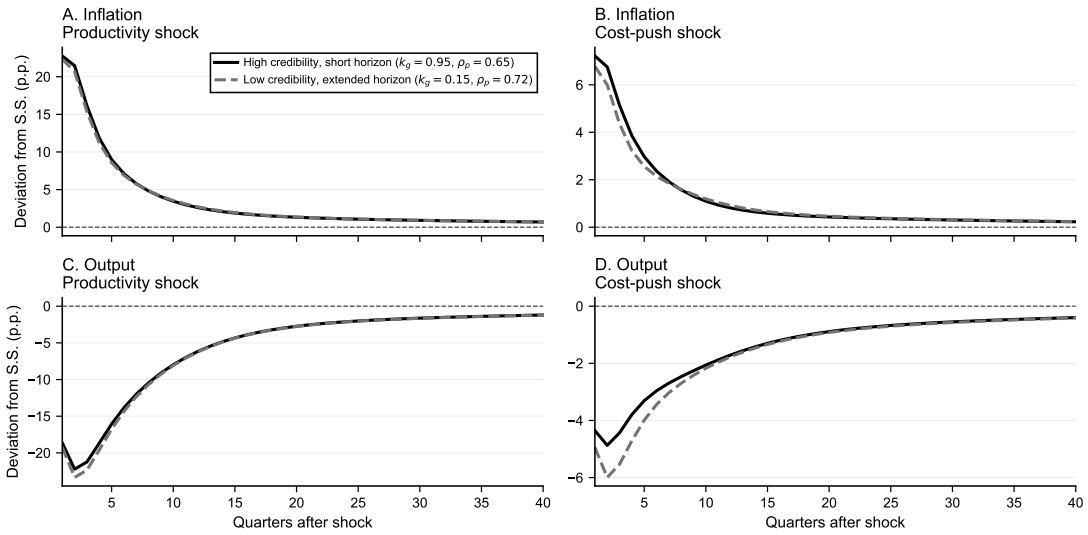


FIGURE 9. *Generalized Impulse Response Functions* under high credibility with short horizon and low credibility with extended horizon.

Figure 9 presents the GIRFs of inflation and output to the selected shocks for a high-credibility short-horizon scenario and a low-credibility extended-horizon scenario.[28] Productivity shock dynamics are relatively similar across scenarios, with output losses slightly deeper in the low-credibility extended-horizon scenario, but no clearly distinguishable difference in inflation. The cost-push shock displays a sharper contrast. Inflation under the low-credibility extended-horizon scenario falls more rapidly, but then displays a slightly longer tail, implying

[26] A credibility-level contingent monetary policy response is out of the scope of this work.

[27] This subsection provides an illustrative comparison of the interaction between horizon flexibility and credibility stress. The comparison is not intended to characterize the full credibility-horizon parameter space or to decompose the separate marginal effects of credibility and policy horizon.

[28] The high-credibility short-horizon scenario is ($k_g = 0.95$, $\rho_p = 0.65$), while the low-credibility extended-horizon scenario is ($k_g = 0.15$, $\rho_p = 0.72$), implying an approximately one-year horizon extension.

larger inflation convergence. Output losses are deeper, and the recovery is slower. Under low credibility, the extended-horizon configuration does not deliver the real-side accommodation that the longer target path is intended to provide.

TABLE 8. Horizon Flexibility and Credibility Stress: Summary Statistics

	$\sum^{40} \pi_t$	$\sum^{40} y_t$	$H^*_{5\%}$	$H^{\pi-\pi_T}_{5\%}$	SR^{40}_{rec}	SR^{40}_{vol}
Productivity Shock						
High Credibility and Short Horizon	1.443	-2.282	24	27	-1.582 $(-,+)$	1.670
Low Credibility and Extended Horizon	1.421	-2.349	25	29	-1.653 $(-,+)$	1.914
Cost-Push Shock						
High Credibility and Short Horizon	0.464	-0.576	25	42	-1.239 $(-,+)$	0.854
Low Credibility and Extended Horizon	0.443	-0.641	28	47	-1.449 $(-,+)$	1.400

Note: The variables π_t and y_t denote inflation and output, respectively. All cumulative statistics and sacrifice ratios are computed over 40 periods. $H^*_{5\%}$ denotes the first horizon at which inflation falls to at most 5% of its initial response and remains there for four consecutive periods, while $H^{\pi-\pi_T}_{5\%}$ denotes the analogous relative convergence horizon for the inflation gap $(\pi_t - \pi_{T,t})$. The sign pair in parentheses reports the sign of the ratio for the numerator and denominator. Complete statistics are reported in Table B.8 in Appendix B.

Table 8 presents summary statistics for inflation and output for the outlined scenarios.[29] The active mechanism reallocates adjustment toward sustained inflation only when the target movement is sufficiently transmitted through expectations. Under credibility stress, this reallocation is weakened. Despite the policy horizon being one year longer, cumulative inflation in the low-credibility extended-horizon scenario is 1.5% lower for the productivity shock and 4.5% lower for the cost-push shock. At the same time, output losses increase by 2.9% and 11.3%, respectively. This implies that under the low-credibility extended-horizon configuration, the points of output sacrificed per cumulative inflation pressure increase by 4.5% for the productivity and 17.0% for the cost-push shock. The additional accommodation associated with the longer target path does not materialize in real activity. The convergence-horizon comparison reinforces this interpretation. Under high credibility, moderate horizon extensions can shorten the inflation convergence horizon, because the target movement is sufficiently transmitted into expectations and supports a smoother real-rate adjustment.[30] Under credibility stress, this mechanism breaks down. The low-credibility extended-horizon scenario raises the convergence horizon one and three quarters for the productivity and cost-push shock, respectively. The longer horizon therefore does not operate as controlled flexibility, but instead produces a less favorable adjustment with larger output losses and a more persistent inflation tail. In this sense, inflation persistence becomes counterproductive once credibility is too weak to transmit the intended accommodation.

Table 9 presents summary statistics for monetary policy variables under the given scenarios. The cumulative inflation target increases by 16.5% under the low-credibility extended-horizon scenario. This implies that the longer horizon mechanically produces greater policy intent, yet, as Table 8 shows, the outcomes deteriorate. Despite somewhat similar cumulative nominal tightening, the cumulative real rate diverges. When credibility is lower, the cumulative real rate increases by 65.0% following a cost-push shock, whereas it rises by only 8.4% for a productivity shock. Driven by cumulative inflation expectations, the real tightening reflects the weak transmission of the target path into expected inflation. The authority is not credible enough to transmit that the policy stance will be more accommodative, so agents treat target movements as imperfectly informative about future inflation. The relevant mechanism is therefore not the size of the target movement by itself, but the weak pass-through from the observed target adjustment to private-sector expectations when credibility is low. Beyond the model, one possible interpretation is that large deviations announced by a low-credibility authority may be perceived as signs of policy stress rather than as reliable accommodation.

[29]Since the comparison uses one representative pair of credibility-horizon configurations, the results should be read as an illustrative rather than a general monotonic result over all values of credibility and policy horizon.

[30]See Subsection 5.2 for more details.

TABLE 9. Horizon Flexibility and Credibility Stress: Policy-Stance Summary Statistics

	$\sum^{40} i_t$	$\sum^{40} r_t$	$\sum^{40} \mathbb{E}_t \pi_{t+1}$	$\sum^{40} \pi_{T,t}$
Productivity Shock				
High Credibility and Short Horizon	1.516	0.309	1.207	0.232
Low Credibility and Extended Horizon	1.533	0.335	1.198	0.270
Cost-Push Shock				
High Credibility and Short Horizon	0.420	0.040	0.380	0.230
Low Credibility and Extended Horizon	0.437	0.066	0.371	0.268

Note: The variables i_t, r_t, $\mathbb{E}_t \pi_{t+1}$, and $\pi_{T,t}$ denote the nominal interest rate, real interest rate, expected inflation, and inflation target, respectively. Expected inflation is inferred from a *Fisher* equation. All cumulative statistics are computed over 40 periods. Complete statistics are reported in Table B.9 in Appendix B.

Horizon flexibility is not a substitute for low credibility. The stabilizing value of a longer policy horizon depends on whether the nominal anchor remains credible enough for the target path to be incorporated into expectations. When credibility is weak, a longer target path may fail to operate as reliable accommodation, making the policy response both less effective and more costly. A central bank may therefore obtain the formal flexibility of a longer horizon without securing its stabilizing effect. In that case, the adjustment becomes less efficient, with larger output losses despite a smaller cumulative inflation response, while inflation convergence is delayed. The resulting persistence is therefore counterproductive rather than stabilizing.

6. Robustness Exercise

Parameter sensitivity is examined over different structural calibrations grouped into seven parameter blocks.[31] The blocks cover policy and nominal transmission, the sectoral structure behind the sectors, and the ease of reallocation across sectors. Each calibration is evaluated under three configurations, an inactive mechanism under high credibility and a short horizon, an active mechanism under the same regime, and an active mechanism under low credibility and an extended horizon. Performance is evaluated using the recession sacrifice ratio, cumulative inflation, and cumulative output for both the productivity and cost-push shocks.

The qualitative findings hold across the calibrations examined. For every parameter block and both shocks the active mechanism under high credibility and a short horizon remains preferable to the inactive mechanism and to the active mechanism under low credibility and an extended horizon. The improvement in the output-inflation trade-off and the ordering that favors credibility over horizon flexibility are not features of the baseline calibration alone.

Table 10 gathers the perturbations that act most directly on the transmission of the mechanism. The first three blocks vary the policy rule and the degree of nominal rigidity in prices and wages, the following two vary the sectoral composition of output and the distribution of labor across sectors, and the final two vary the ease with which consumption and labor reallocate. The selected blocks correspond to the margins through which the target displacement is transmitted to activity, so they provide the most direct stress test of the mechanism. For each block, the table reports the applied perturbations, the case that produces the thinnest margin, and the corresponding improvement of the active credible configuration over each of the other two scenarios. Every reported margin is positive, the smallest equal to 0.022 in the monetary policy block. The advantage over the inactive configuration exceeds the advantage over the impaired extended one in every block, which makes the comparison between the two active configurations the binding one rather than the value of the mechanism itself.

The margins narrow most under a dovish policy rule and under inverted price rigidity, with the productivity shock proving the most demanding for the robustness purpose. A less aggressive rule already delivers part of the accommodation in the inactive economy, which reduces the additional gain from the mechanism. An inverted rigidity ordering weakens the services channel, through which the real-side gains of the mechanism operate. The ranking

[31] Please refer to Appendix E for complete results.

TABLE 10. Robustness of the Main Mechanism

Parameter block	Perturbations included	Thinnest-margin case	min $\Delta SR^{rec}_{2\text{-}1}$	min $\Delta SR^{rec}_{2\text{-}3}$
Monetary policy	Low and high for γ_i, ϕ_π, ϕ_y	$\phi_\pi = 1.50$	0.109	0.022
Price rigidity	Weaker, symmetric, stronger, and inverted for (α_S, α_G)	$(\alpha_S, \alpha_G) = (0.50, 0.75)$	0.160	0.032
Wage rigidity	Weaker, symmetric, and inverted for (α^W_S, α^W_G)	$(\alpha^W_S, \alpha^W_G) = (0.50, 0.65)$	0.218	0.051
Sectoral output composition	Weaker, symmetric, stronger, and inverted for $\left(\omega^{c/y}_S, \omega^{c/y}_G\right)$	$\left(\omega^{c/y}_S, \omega^{c/y}_G\right) = (0.55, 0.70)$	0.211	0.048
Production labor shares	Weaker, symmetric, stronger, and inverted for (B_S, B_G)	$(B_S, B_G) = (0.40, 0.60)$	0.174	0.039
Consumption substitution	Low and high for η_c	$\eta_c = 0.50$	0.202	0.046
Labor substitution	Low and high for η_n	$\eta_n = 2.50$	0.212	0.048

Note: $S1$ denotes the inactive mechanism under high credibility and the short policy horizon. $S2$ denotes the active mechanism under high credibility and the short policy horizon. $S3$ denotes the active mechanism under low credibility and the extended policy horizon. The table reports the minimum difference in the signed recession sacrifice ratio within each parameter block, across the productivity and cost-push shocks and across both the fixed 40-period horizon and the convergence horizon $H^*_{5\%}$. The differences are defined as $\Delta SR^{rec}_{2-1} = SR^{rec}(S2) - SR^{rec}(S1)$ and $\Delta SR^{rec}_{2-3} = SR^{rec}(S2) - SR^{rec}(S3)$. A positive difference value means that $S2$ has a less negative sacrifice ratio. The thinnest-margin case is the perturbation attaining the minimum $\Delta SR^{rec}_{2\text{-}3}$ within the block, which coincides with the minimizer of $\Delta SR^{rec}_{2\text{-}1}$ in every block. Complete cumulative inflation, cumulative output, policy-stance variables, sectoral statistics, and sacrifice ratios for the full robustness grid are reported in Appendix E.

nevertheless holds in each of these cases.

The real relief produced by the mechanism is governed by the sectoral rigidity structure. Under the baseline ordering of price rigidity, where service prices are more rigid than good prices, the output gain is concentrated in services. This concentration survives variation in sectoral output composition, production labor shares, and in the ease with which consumption and labor reallocate across sectors. The relief shifts toward goods as the rigidity gap narrows, and once goods becomes the more rigid sector the mechanism deepens the contraction in services rather than easing it. The channel therefore rests on nominal rigidity.

The qualitative results hold throughout the exercise, while their magnitudes weaken under particular calibrations. The trade-off improvement, the dominance of credibility over horizon flexibility, and the concentration of the relief in services remain in place throughout, even where the size of the response diminishes.

7. CONCLUSIONS

Several sources have been identified to explain inflation persistence following large disturbances. The framework developed here introduces an additional source of that persistence by relocating size-dependence into the monetary-policy reaction itself, embedding a shock-size-dependent inflation objective within a two-sector economy and allowing its effect to transmit through the signal-extraction problem that agents face. When a sufficiently large disturbance shifts the persistent component of the inflation target, agents must infer whether the movement represents a durable reorientation of the nominal anchor or a transitory deviation. Their interpretation of that movement shapes expected inflation, the ex-ante real interest rate, and ultimately the output cost and persistence of the adjustment. The empirical evidence provides reduced-form motivation for this mechanism by showing that the implicit target gap increased broadly across economies during the inflation episode and was systematically associated with

large supply-like shocks.

The quantitative analysis establishes that permitting the target to shift contingently improves the output–inflation trade-off, reducing cumulative output losses while tolerating a more sustained inflation path rather than eliminating persistence, with the gains largest for cost-push disturbances. The advantage is entirely conditional on credibility, and its interaction with the policy horizon makes the conditionality sharpest. Under a credible anchor, a mild extension of the operational horizon shortens the inflation-convergence horizon and lowers output losses at once. Under an impaired anchor, the same extension reverses both margins, lengthening convergence and deepening output losses, since the intended accommodation does not pass into expected inflation, real rates tighten, and output absorbs the adjustment the mechanism was designed to spare it. Horizon flexibility is therefore no substitute for credibility, the operative object being the pass-through from the announced target to beliefs rather than the extension of the target path itself.

Three features bound the interpretation by design. The empirical exercise is reduced-form, associating large supply-like shocks with a higher implied short-run target path without identifying a target revision or a structural policy innovation, and bundling the accommodation and horizon margins that only the model separates. The trigger is conditioned on the inflationary tail, leaving sign-contingent responses aside. Credibility enters as an exogenous learning speed rather than an equilibrium outcome, so its consequences are characterized without its formation being explained.

Each boundary indicates a natural extension. To endogenize the credibility of monetary policy would close the loop between policy conduct and the beliefs on which it relies. A sign-contingent formulation would extend the mechanism to favorable tail events and persistent undershooting. Most consequentially, separating the accommodation and horizon margins in the data, not in the model alone, would turn the recovered gap from a motivating regularity into a testable structural claim.

The last-mile problem after large shocks therefore reflects not only the duration of the originating disturbance but a temporary, belief-mediated displacement of the nominal anchor, the stabilizing value of which is inseparable from the credibility that transmits it.

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Appendix A. Empirical Appendix

A.1. *Dynamics of Inflation and its Sectoral Components*

The empirical exercise for subsection 2.1 uses a monthly country-level panel of all-items, goods, and services price indices, in addition to inflation targets. The working sample runs from January 2000 to December 2025 and excludes regional aggregates. After applying these exclusions, the panel contains 42 country-level entries and 13,104 country-month observations. The number of valid year-on-year observations is 12,556 for all-items inflation, 12,178 for goods inflation, 12,178 for services inflation, and 9,885 for the target observations.

Table A.1 reports the raw sources used to build the panel. The source-specific price indices are harmonized into three common variables: all-items prices, goods prices, and services prices. For European economies, the data are based on Eurostat HICP indices. For the remaining economies, national and international sources are used.

Table A.1. Data Sources and Raw Inflation Series

Source	Coverage	All-items index	Goods index	Services index
FRED (BEA)	United States	PCEPI	DGDSRG3M086SBEA	DSERRG3M086SBEA
BCCh (INE)	Chile	G073.IPC.IND.2023.M	F074.IPCB.IND.Z.EP23.Z.M	F074.IPCS.IND.Z.EP23.Z.M
Eurostat	European economies	HICP CP00	HICP GD	HICP SERV
e-Stat	Japan	Category 0201	Category 0202	Category 0220
OECD	Korea, United Kingdom, Costa Rica, Australia	CPI total	CPI goods	CPI services
Banxico	Mexico	SP1	SP74626	SP74628
Statistics Canada	Canada	Vector 41690973	Vector 41691222	Vector 41691230
BCRP	Peru	PN01301PM	PN38705PM	PN01296PM

Note: The table reports the raw source identifiers used to construct the monthly panel. All series are harmonized into all-items, goods, and services price indices.

For each country c, month t, and price category $x \in \{AI, G, S\}$, year-on-year inflation is computed as

$$\pi^x_{c,t} = 100 \times \left(\frac{P^x_{c,t}}{P^x_{c,t-12}} - 1 \right),$$

where AI, G, and S denote all-items, goods, and services prices, respectively.

Table A.2 reports the country composition of the panel. The sample is unbalanced because the availability of sectoral indices differs across countries and sources. The coverage columns report the first and last valid all-items year-on-year observation, while the remaining columns report the number of valid observations for each year-on-year inflation series.

Table A.2: Panel Composition and Series Availability

Country	Code	Source	All-items coverage	N^{AI}	N^G	N^S	Target coverage	N^T
Albania	ALB	Eurostat HICP	2016-12 – 2025-12	109	109	109	2009-01 – 2025-12	204
Austria	AUT	Eurostat HICP	2000-01 – 2025-12	312	301	301	2000-01 – 2025-12	312
Belgium	BEL	Eurostat HICP	2000-01 – 2025-12	312	312	312	2000-01 – 2025-12	312
Bulgaria	BGR	Eurostat HICP	2000-01 – 2025-12	312	312	312	–	0
Canada	CAN	Statistics Canada CPI	2000-01 – 2025-12	312	312	312	2000-01 – 2025-12	312
Chile	CHL	BCCh/INE CPI	2000-01 – 2025-12	312	181	181	2001-01 – 2025-12	300
Croatia	HRV	Eurostat HICP	2000-01 – 2025-12	312	312	312	2023-01 – 2025-12	36
Cyprus	CYP	Eurostat HICP	2000-01 – 2025-12	312	312	312	2008-01 – 2025-12	216
Czechia	CZE	Eurostat HICP	2000-01 – 2025-12	312	301	301	2010-01 – 2025-12	192
Denmark	DNK	Eurostat HICP	2000-01 – 2025-12	312	301	301	–	0
Estonia	EST	Eurostat HICP	2000-01 – 2025-12	312	289	289	2011-01 – 2025-12	180
Finland	FIN	Eurostat HICP	2000-01 – 2025-12	312	301	301	2000-01 – 2025-12	312
France	FRA	Eurostat HICP	2000-01 – 2025-12	312	312	312	2000-01 – 2025-12	312
Germany	DEU	Eurostat HICP	2000-01 – 2025-12	312	312	312	2000-01 – 2025-12	312
Greece	GRC	Eurostat HICP	2000-01 – 2025-12	312	312	312	2001-01 – 2025-12	300
Hungary	HUN	Eurostat HICP	2000-01 – 2025-12	312	289	289	2007-01 – 2025-12	228
Iceland	ISL	Eurostat HICP	2000-01 – 2025-12	312	301	301	2001-03 – 2025-12	298
Ireland	IRL	Eurostat HICP	2000-01 – 2025-12	312	301	301	2000-01 – 2025-12	312

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Table A.2: Panel Composition and Series Availability (continued)

Country	Code	Source	All-items coverage	N^{AI}	N^G	N^S	Target coverage	N^T
Italy	ITA	Eurostat HICP	2000-01 – 2025-12	312	301	301	2000-01 – 2025-12	312
Japan	JPN	e-Stat CPI	2000-01 – 2025-12	312	312	312	2013-01 – 2025-12	156
Korea	KOR	OECD CPI	2000-01 – 2025-12	312	312	312	2010-01 – 2025-12	192
Latvia	LVA	Eurostat HICP	2000-01 – 2025-12	312	312	312	2014-01 – 2025-12	144
Lithuania	LTU	Eurostat HICP	2000-01 – 2025-12	312	312	312	2015-01 – 2025-12	132
Luxembourg	LUX	Eurostat HICP	2000-01 – 2025-12	312	301	301	2000-01 – 2025-12	312
Malta	MLT	Eurostat HICP	2000-01 – 2025-12	312	312	312	2008-01 – 2025-12	216
Mexico	MEX	Banxico CPI	2000-01 – 2025-12	312	312	312	2003-01 – 2025-12	276
Montenegro	MNE	Eurostat HICP	2015-12 – 2025-12	121	121	121	–	0
Netherlands	NLD	Eurostat HICP	2000-01 – 2025-12	312	301	301	2000-01 – 2025-12	312
Norway	NOR	Eurostat HICP	2000-01 – 2025-12	312	301	301	2001-03 – 2025-12	298
Peru	PER	BCRP CPI	2000-01 – 2025-12	312	312	312	2002-01 – 2025-12	288
Poland	POL	Eurostat HICP	2000-01 – 2025-12	312	312	312	2004-01 – 2025-12	264
Portugal	PRT	Eurostat HICP	2000-01 – 2025-12	312	301	301	2000-01 – 2025-12	312
Romania	ROU	Eurostat HICP	2000-01 – 2025-12	312	289	289	2013-12 – 2025-12	145
Serbia	SRB	Eurostat HICP	2006-12 – 2025-12	229	229	229	2009-01 – 2025-12	204
Slovakia	SVK	Eurostat HICP	2000-01 – 2025-12	312	312	312	2005-01 – 2025-12	252
Slovenia	SVN	Eurostat HICP	2000-01 – 2025-12	312	289	289	2007-01 – 2025-12	228
Spain	ESP	Eurostat HICP	2000-01 – 2025-12	312	289	289	2000-01 – 2025-12	312
Sweden	SWE	Eurostat HICP	2000-01 – 2025-12	312	301	301	2000-01 – 2025-12	312
Switzerland	CHE	Eurostat HICP	2005-12 – 2025-12	241	241	241	2000-01 – 2025-12	312
Türkiye	TUR	Eurostat HICP	2000-01 – 2025-12	312	312	312	2002-01 – 2025-12	288
UK	GBR	OECD CPI	2000-01 – 2025-12	312	312	312	2000-01 – 2025-12	312
USA	USA	FRED/BEA PCE	2000-01 – 2025-12	312	312	312	2012-01 – 2025-12	168

Note: N^{AI}, N^G, and N^S denote the number of valid monthly year-on-year observations for all-items, goods, and services inflation, respectively. N^T denotes the number of valid monthly observations for the country-specific inflation target. The all-items coverage column reports the first and last valid all-items year-on-year observation in the Jan-2000–Dec-2025 sample. The target coverage column reports the first and last valid target observation over the same sample.

Table A.3 reports country-level descriptive statistics for the three year-on-year inflation series. The statistics are computed over the valid observations available for each country between January 2000 and December 2025.

Table A.3: Country-Level Summary Statistics for Year-on-Year Inflation

		Mean				Median				Maximum			
Country	Code	N^{AI}	N^G	N^S	N^T	N^{AI}	N^G	N^S	N^T	N^{AI}	N^G	N^S	N^T
Albania	ALB	3.09	3.47	2.29	0.09	2.44	2.62	1.87	-0.56	8.04	10.12	5.75	5.04
Austria	AUT	2.50	2.15	2.91	0.58	1.99	1.62	2.52	0.09	11.57	15.78	8.54	9.57
Belgium	BEL	2.43	2.40	2.44	0.51	2.05	1.93	2.14	0.15	13.09	18.94	6.93	11.09
Bulgaria	BGR	4.27	4.11	4.84	–	3.37	3.16	3.69	–	15.63	17.76	17.02	–
Canada	CAN	2.23	1.76	2.65	0.23	2.08	1.40	2.38	0.08	8.13	11.16	5.75	6.13
Chile	CHL	3.75	3.78	4.73	0.75	3.30	2.95	4.61	0.22	14.08	17.21	11.43	11.08
Croatia	HRV	2.88	2.66	3.42	3.63	2.29	2.12	2.58	2.68	12.95	14.53	11.41	10.49
Cyprus	CYP	1.98	1.77	2.22	-0.29	2.00	1.69	2.26	-0.55	10.57	13.03	8.10	8.57
Czechia	CZE	3.12	2.70	3.92	1.44	2.37	1.83	3.16	0.36	19.06	21.28	14.38	17.06
Denmark	DNK	1.86	1.33	2.52	–	1.65	0.71	2.47	–	11.36	16.05	7.20	–
Estonia	EST	4.27	4.10	4.42	2.37	3.83	3.52	3.73	1.65	25.16	30.47	13.67	23.16
Finland	FIN	1.92	1.43	2.53	0.01	1.48	0.78	2.58	-0.46	9.06	12.71	4.79	7.06
France	FRA	1.88	1.80	1.98	-0.04	1.75	1.62	1.90	-0.17	7.29	9.89	4.26	5.29
Germany	DEU	2.07	2.20	1.88	0.15	1.75	1.69	1.50	-0.16	11.56	17.64	6.77	9.56
Greece	GRC	2.27	2.30	2.19	0.33	2.83	2.21	2.90	0.87	12.12	15.81	7.38	10.12
Hungary	HUN	5.23	4.61	5.35	1.88	4.22	3.69	4.42	0.85	26.20	30.52	19.15	23.20
Iceland	ISL	4.26	4.03	4.61	1.76	3.49	2.89	4.76	0.93	21.86	28.43	12.18	19.36
Ireland	IRL	2.00	0.72	3.12	0.09	1.75	0.80	3.00	-0.17	9.60	14.32	8.66	7.60
Italy	ITA	2.12	2.12	2.06	0.21	1.97	1.69	2.03	0.05	12.58	17.88	5.12	10.58
Japan	JPN	0.52	0.84	0.19	-0.68	0.15	0.30	0.10	-1.19	4.39	7.28	2.34	2.39
Korea	KOR	2.48	2.61	2.40	-0.29	2.41	2.25	2.27	-0.47	6.33	9.29	5.23	4.33
Latvia	LVA	4.16	4.15	4.18	1.76	2.93	2.86	3.03	0.32	22.05	26.35	16.59	20.05
Lithuania	LTU	3.34	3.01	4.45	2.26	2.42	1.85	4.08	0.61	22.52	25.70	14.62	20.52
Luxembourg	LUX	2.48	2.42	2.49	0.57	2.46	2.25	2.58	0.55	10.33	13.46	5.39	8.33
Malta	MLT	2.28	2.08	2.54	0.34	2.11	1.71	2.22	-0.16	7.42	7.88	7.45	5.42

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Table A.3: Country-Level Summary Statistics for Year-on-Year Inflation (continued)

		Mean				Median				Maximum			
Country	Code	N^{AI}	N^{G}	N^{S}	N^{T}	N^{AI}	N^{G}	N^{S}	N^{T}	N^{AI}	N^{G}	N^{S}	N^{T}
Mexico	MEX	4.73	4.61	4.19	1.43	4.33	3.98	3.67	1.21	11.02	12.23	14.30	5.70
Montenegro	MNE	3.62	3.44	4.02	–	2.83	2.69	3.02	–	15.78	17.48	12.60	–
Netherlands	NLD	2.45	2.20	2.76	0.54	1.87	1.66	2.45	-0.03	17.06	24.75	7.49	15.06
Norway	NOR	2.43	2.03	3.09	0.06	2.18	1.63	2.88	-0.30	8.45	10.26	6.62	6.45
Peru	PER	3.00	2.51	3.26	0.90	2.75	2.27	3.09	0.62	8.81	7.29	6.56	6.81
Poland	POL	3.51	3.14	4.42	0.82	2.87	2.22	3.06	0.35	17.19	18.74	14.48	14.69
Portugal	PRT	2.20	1.77	2.76	0.29	2.22	1.53	2.61	0.32	10.57	13.59	7.99	8.57
Romania	ROU	8.98	6.44	6.43	1.63	5.44	5.04	4.93	1.25	56.72	29.26	36.78	12.09
Serbia	SRB	5.76	6.00	4.90	1.92	4.42	3.98	4.56	0.77	15.48	17.27	11.06	12.48
Slovakia	SVK	4.00	3.46	5.27	1.06	3.15	2.61	3.85	0.48	16.80	17.04	15.98	13.40
Slovenia	SVN	3.31	2.51	3.52	0.53	2.37	2.11	3.04	0.11	11.70	14.59	12.49	9.70
Spain	ESP	2.39	2.26	2.43	0.48	2.56	2.31	2.18	0.66	10.70	14.63	4.92	8.70
Sweden	SWE	2.03	1.82	2.37	0.03	1.64	1.36	2.03	-0.36	10.84	14.52	7.13	8.84
Switzerland	CHE	0.49	0.14	0.73	-1.41	0.33	-0.17	0.68	-1.57	3.35	6.10	2.39	1.45
Türkiye	TUR	22.68	22.35	23.63	12.58	10.76	11.44	11.75	4.93	85.45	94.81	96.69	80.45
UK	GBR	2.52	1.70	3.11	0.44	2.19	1.25	2.86	0.19	9.61	14.79	6.43	7.61
USA	USA	2.21	0.90	2.88	0.27	2.11	0.39	2.67	-0.24	7.24	10.71	5.99	5.24

Note: N^{AI}, N^{G}, and N^{S} denote all-items, goods, and services year-on-year inflation, respectively. N^{T} denotes the all-items inflation gap relative to the country-specific inflation target, computed as $\pi^{AI} - \pi^{*}$. Statistics are expressed in percentage points and computed over valid observations between Jan-2000 and Dec-2025.

Table A.4 summarizes the distribution of all-items, goods, and services inflation across selected periods, previous crisis are included as pre-pandemic comparison windows.

Table A.4. Period-Level Summary Statistics for Year-on-Year Inflation

	All Items				Inflation Gap				Goods				Services			
Period	N	Mean	Median	[p10–p90]	N	Mean	Median	[p10–p90]	N	Mean	Median	[p10–p90]	N	Mean	Median	[p10–p90]
Pre-COVID	9532	2.93	2.16	[0.10–5.60]	6947	0.09	-0.13	[-1.95–2.04]	9154	2.46	1.84	[-0.89–5.59]	9154	3.24	2.51	[0.71–5.91]
GFC	1440	3.60	2.81	[0.00–8.42]	996	0.91	0.54	[-1.92–3.80]	1404	3.26	2.56	[-1.17–8.09]	1404	4.06	3.24	[1.46–8.37]
Euro-Area Debt Crisis	1920	2.66	2.47	[-0.07–5.07]	1500	0.47	0.38	[-1.84–2.35]	1897	2.71	2.72	[-0.98–5.77]	1897	2.56	2.28	[0.50–5.03]
COVID 2020	504	1.13	0.81	[-0.91–3.25]	456	-0.97	-1.16	[-2.81–0.87]	504	0.70	0.07	[-1.59–3.45]	504	1.81	1.53	[-0.39–3.65]
Inflation 2021–22	1008	7.16	5.48	[1.17–13.59]	912	4.97	3.19	[-0.90–10.82]	1008	8.64	6.64	[1.37–16.33]	1008	4.52	3.23	[0.75–9.47]
Disinflation 2023–24	1008	5.98	3.65	[1.55–9.87]	936	3.84	1.44	[-0.48–7.74]	1008	5.13	2.96	[-0.42–11.04]	1008	7.47	5.30	[2.89–10.22]
Last-mile 2024–25	1008	3.94	2.79	[1.34–4.69]	936	1.78	0.70	[-0.72–2.40]	1008	2.62	1.76	[-0.54–4.03]	1008	6.12	4.41	[2.45–7.50]

Note: The table reports pooled country-month descriptive statistics for the country-level panel. N^{AI}, N^{G}, and N^{S} denote all-items, goods, and services year-on-year inflation, respectively. The inflation gap is defined as all-items year-on-year inflation minus the country-specific inflation target, $T = \pi^{AI} - \pi^{*}$. Periods are defined as follows: Pre-COVID = 2000M1–2019M12; Global Financial Crisis (GFC) = 2007M1–2009M12; Euro-area debt crisis = 2009M1–2012M12; COVID 2020 = 2020M1–2020M12; Inflation episode = 2021M1–2022M12; Disinflation = 2023M1–2024M12; Last-mile = 2024M1–2025M12. Statistics are expressed in percentage points. The entry [p10–p90] identifies the bounded interval spanning the 10th to 90th percentiles.

To summarize the sectoral asymmetry behind the recent inflation episode, define the services-goods inflation gap as $gap^{S-G}_{i,t} = \pi^{S}_{i,t} - \pi^{G}_{i,t}$. Table A.5 reports the distribution of this gap by period.

The country-specific target series is constructed as a monthly institutional series that records the numerical inflation objective relevant for each economy at each date. For inflation-targeting economies, the series follows the announced point target, or the midpoint of the announced target band when the framework is expressed as a range. For euro-area economies, the ECB price-stability objective is assigned from the date at which the country enters the monetary union, with the July 2021 strategy revision coded as a common shift to a symmetric 2 percent target. Countries without a comparable numerical target in the sample are left with missing target observations. Table A.6 reports the target spells used to construct the inflation-gap variable.

TABLE A.5. Services-Goods Inflation Gap by Period

Period	N	Mean gap	Median gap	$p10$	$p90$
Pre-COVID	8729	0.75	0.81	-1.88	3.45
Global Financial Crisis	1440	0.82	0.88	-2.32	4.16
Euro-area debt crisis	1945	-0.05	-0.08	-3.09	3.51
COVID 2020	516	1.15	1.18	-1.32	3.45
Inflation episode 2021–2022	1020	-4.11	-3.34	-9.00	0.39
Disinflation 2023–2024	1008	2.34	2.26	-3.10	6.08
Last-mile 2024–2025	1017	3.48	2.82	0.54	5.96

Note: The services-goods inflation gap is defined as services year-on-year inflation minus goods year-on-year inflation. Periods are defined as follows: Pre-COVID = 2000M1–2019M12; Global Financial Crisis = 2007M1–2009M12; Euro-area debt crisis = 2009M1–2012M12; COVID 2020 = 2020M1–2020M12; Inflation episode = 2021M1–2022M12; Disinflation = 2023M1–2024M12. Statistics are expressed in percentage points and are computed over observations for which both goods and services inflation are available.

Table A.6: Inflation Target Coverage and Target Changes

Country	Date	Target	Comment
Albania	2009-01	3.0	Domestic numerical target.
Austria	2000-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Belgium	2000-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Bulgaria	–	–	–
Canada	2000-01	2.0	Domestic numerical target.
Chile	2001-01	3.0	Domestic numerical target.
Croatia	2023-01	2.0	ECB price-stability objective below 2%.
Cyprus	2008-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Czechia	2010-01	2.0	Domestic numerical target.
Denmark	–	–	–
Estonia	2011-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Finland	2000-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
France	2000-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Germany	2000-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Greece	2001-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Hungary	2007-01	3.0	Domestic numerical target.
Iceland	2001-03	2.5	Domestic numerical target.
—	2000-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Italy	2000-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Japan	2013-01	2.0	Domestic numerical target.
Korea	2010-01	3.0	Domestic numerical target.
—	2016-01	2.0	Domestic target change.
Latvia	2014-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Lithuania	2015-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Luxembourg	2000-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Malta	2008-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Mexico	2003-01	3.0	Domestic numerical target.
Montenegro	–	–	No comparable numerical target.
Netherlands	2000-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Norway	2001-03	2.5	Domestic numerical target.
—	2018-03	2.0	Domestic target change.

Continued on next page

Table A.6: Inflation Target Coverage and Target Changes (continued)

Country	Date	Target	Comment
Peru	2002-01	2.5	Domestic numerical target.
—	2007-01	2.0	Domestic target change.
Poland	2004-01	2.5	Domestic numerical target.
Portugal	2000-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Romania	2013-12	2.5	Domestic numerical target.
Serbia	2009-01	4.0	Domestic numerical target.
—	2017-01	3.0	Domestic target change.
Slovakia	2005-01	3.5	Domestic numerical target.
—	2006-01	2.5	Domestic target change.
—	2007-01	2.0	Domestic target change.
—	2009-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
—	2007-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Spain	2000-01	1.9	ECB price-stability objective below 2%.
—	2021-07	2.0	ECB strategy revision with symmetric 2% target.
Sweden	2000-01	2.0	Domestic numerical target.
Switzerland	2000-01	1.9	Domestic numerical target.
Türkiye	2002-01	35.0	Domestic numerical target.
—	2003-01	20.0	Domestic target change.
—	2004-01	12.0	Domestic target change.
—	2005-01	8.0	Domestic target change.
—	2006-01	5.0	Domestic target change.
—	2007-01	4.0	Domestic target change.
—	2009-01	7.5	Domestic target change.
—	2010-01	6.5	Domestic target change.
—	2011-01	5.5	Domestic target change.
—	2012-01	5.0	Domestic target change.
UK	2000-01	2.5	Domestic numerical target.
—	2003-12	2.0	Domestic target change.
USA	2012-01	2.0	Domestic numerical target.

Note: The table reports target spells used to construct the country-specific inflation target series, $\pi^*_{c,t}$, in the January 2000–December 2025 sample. A row is reported when the target first becomes available in the panel and whenever the coded target changes. The target column reports the numerical point target used in the inflation-gap construction, expressed in percentage points. For target ranges or upper-bound formulations, the table reports the numerical value coded in the panel. Countries without a comparable numerical target are left with missing target observations.

A.2. *Implicit Inflation Target and State-Dependent Accommodation under Large Shocks*

A.2.1. *Data sources and variables*

Table A.7 reports the non-inflation variables used to construct the empirical panel. All series are harmonized at quarterly frequency. Real GDP is used only to construct output growth and trend growth, while the interest-rate variables define alternative measures of the monetary-policy stance. The baseline policy stance comes from IMF Monetary and Financial Statistics and national central bank sources. For euro-area countries, the policy stance uses the ECB main refinancing operations rate (ECBMRRFR) outside the floor-binding period and the deposit facility rate (ECBDFR) from June 2014 to July 2022. The shadow rate policy stance follows Krippner (2013) when available, and the NFC measures, detailed below, are used for robustness exercise.

A.2.2. *Country coverage and sample construction*

The panel is constructed at quarterly frequency. Monthly interest-rate and lending-rate series are converted to quarterly averages, while policy rates are measured at quarter end. The country sample is restricted by the availability of announced inflation targets mainly, however, data availability is still a limiting factor. For euro-area countries, the baseline policy stance is common across member states, while the NFC robustness uses country-specific lending rates to non-financial corporations. Table A.8 presents country coverage statistics. Sample

TABLE A.7. Data Sources and Variable Definitions

Raw variable	Symbol	Series	Source	Transformation
Announced inflation target	$\pi^*_{i,t}$	Country-specific target schedule	Central-bank target coding	Applied from each stated quarter until the next target change.
Real GDP	$Y_{i,t}$	B1GQ, constant prices, seasonally adjusted	IMF National Economic Accounts	Used to compute $\Delta y_{i,t}$ and $g^y_{i,t}$
Output growth	$\Delta y_{i,t}$	Constructed from B1GQ	Constructed	$100[\log Y_{i,t} - \log Y_{i,t-4fixed}]$.
Trend output growth	$g^y_{i,t}$	Constructed from B1GQ	Constructed	One-sided HP trend growth, frozen after 2019Q4.
Policy-rate stance	$i^{\text{policy}}_{c,t}$	MFS166_RT_PT_A_PT; national central-bank policy-rate series	IMF MFS Interest Rates; national central banks	Quarter-end value
Shadow-rate stance	$i^{shadow}_{i,t}$	–	Krippner (2013)	Quarterly average; policy rate if unavailable.
NFC lending rate	$i^{NFC,raw}_{i,t}$	MIR.M.{Code}.B.A2A.A.R.A.2240.EUR.N	ECB Data Portal	Quarterly average; euro-area countries only.
Ten-year sovereign yield	$r^{10y}_{i,t}$	M.MCBY.{Code}; FRED long-rate series	Eurostat; FRED	Quarterly average
German Bund yield	$r^{10y}_{DE,t}$	IRLTLT01DEM156N	FRED	Quarterly average; euro-area reference yield
US Treasury yield	$r^{10y}_{US,t}$	GS10	FRED	Quarterly average; non-euro-area reference yield
Sovereign spread	$spread^{10y}_{i,t}$	Constructed from ten-year yields	Constructed	Country yield minus Bund for EA, minus UST for non-EA.
NFC net stance	$i^{NFC,net}_{i,t}$	Constructed from NFC rate and sovereign spread	Constructed	$i^{NFC,net}_{i,t} - spread^{10y}_{i,t}$

Notes: Inflation variables are excluded because they are documented in the preceding data section. NFC denotes non-financial corporations. EA denotes euro area. For non-euro-area countries, the NFC stance variables are set equal to the baseline policy-rate stance.

is limited up to 2024Q4. Coverage counts report the number of non-missing country-quarter observations available for each series.

A.2.3. *Recovery of the implicit target*

The recovered implicit target is constructed by inverting a calibrated Taylor-rule that represents the monetary stance. For each country c, quarter t, and stance measure s, the rule is written as

$$i^{(s)}_{c,t} = a^{(s)}_c + \gamma i^{(s)}_{c,t-1} + (1-\gamma)\left[\phi_\pi\left(\pi_{c,t} - \pi^{T,s)}_{c,t}\right) + \phi_y\left(\Delta y_{c,t} - g^y_c\right)\right].$$

The stance $i^{(s)}_{c,t}$ corresponds either to the policy-rate or to the shadow-rate stance. The term $gap_{c,t}$ denotes the output-growth gap, defined as the deviation of realized output growth from its pre-COVID trend-growth component recovered from a Hodrick-Prescott filter. The parameters are fixed at the values used in the model calibration, $\gamma = 0.50$, $\phi_\pi = 2.00$, and $\phi_y = 0.50$.

The exercise does not estimate a Taylor rule. Instead, it imposes a rule and asks which short-run target would rationalize the observed policy stance ($i^{(s)}_{c,t}$), conditional on inflation ($\pi_{c,t}$), output-growth conditions ($\Delta y_{c,t} - g^y_c$), policy inertia ($i^{(s)}_{c,t-1}$), and the country-specific intercept ($a^{(s)}_c$).

Solving the rule for the unobserved target gives

$$\widehat{\pi}^{T,(s)}_{c,t} = \pi_{c,t} - \frac{1}{\phi_\pi}\left[\frac{i^{(s)}_{c,t} - a^{(s)}_c - \gamma i^{(s)}_{c,t-1}}{1-\gamma} - \phi_y\left(\Delta y_{c,t} - g^y_c\right)\right].$$

The recovered implicit target gap is then $g^{T,(s)}_{c,t} = \widehat{\pi}^{T,(s)}_{c,t} - \pi^*_{c,t}$, where $\pi^*_{c,t}$ is the announced numerical inflation target.

The country-specific intercept $a^{(s)}_c$ is computed separately for each stance measure. It is chosen so that the average recovered implicit target equals the average announced target in the pre-COVID anchoring window,

$$\frac{1}{T^0_c}\sum_{t\in T^0_c}\left(\widehat{\pi}^{T,(s)}_{c,t} - \pi^*_{c,t}\right) = 0.$$

TABLE A.8. Country Coverage and Sample Availability

			Data Coverage			
Country	ISO3	Periods	Policy gap	Shadow gap	NFC	Shock
Austria	AUT	1999Q1 – 2024Q4	105	105	101	106
Belgium	BEL	1999Q1 – 2024Q4	105	105	89	106
Canada	CAN	1999Q1 – 2024Q4	101	101	101	106
Chile	CHL	2001Q1 – 2024Q4	98	98	98	103
Czech Republic	CZE	2010Q1 – 2024Q4	62	62	62	106
Finland	FIN	1999Q1 – 2024Q4	105	105	99	106
France	FRA	1999Q1 – 2024Q4	105	105	101	106
Germany	DEU	1999Q1 – 2024Q4	105	105	101	106
Hungary	HUN	2007Q1 – 2024Q4	74	74	74	106
Iceland	ISL	2001Q1 – 2024Q4	94	94	94	106
Ireland	IRL	1999Q1 – 2024Q4	105	105	101	106
Italy	ITA	1999Q1 – 2024Q4	105	105	101	106
Japan	JPN	2013Q1 – 2024Q4	50	50	50	106
Korea	KOR	2010Q1 – 2024Q4	62	62	62	106
Mexico	MEX	2003Q1 – 2024Q4	90	90	90	106
Netherlands	NLD	1999Q1 – 2024Q4	105	105	101	106
Norway	NOR	2001Q1 – 2024Q4	98	98	98	106
Poland	POL	2004Q1 – 2024Q4	86	86	86	106
Portugal	PRT	1999Q1 – 2024Q4	105	105	101	106
Spain	ESP	1999Q1 – 2024Q4	105	105	101	106
Sweden	SWE	1999Q1 – 2024Q4	105	105	105	106
Switzerland	CHE	2005Q4 – 2024Q4	79	79	79	79
United Kingdom	GBR	1999Q1 – 2024Q4	105	105	105	106
United States	USA	2012Q1 – 2024Q4	54	54	54	106

Notes: Coverage columns report non-missing country-quarter observations in the attached panel. Periods column corresponds to the first and last quarters defined over the period in which the announced target is observed. Policy and shadow gap coverage require the corresponding stance variable, its lag, the announced target, inflation, and the output-growth gap. NFC coverage requires the raw and net NFC stance variables. For non-euro-area countries, the NFC stance variables coincide with the baseline policy-rate stance. Shock coverage refers to the availability of the inflation and output-growth inputs used in the pseudo-real-time shock construction, before losses from forecasting lags.

Equivalently, the intercept satisfies

$$a_c^{(s)} = \bar{i}_c^{(s)} - \gamma \bar{i}_{c,-1}^{(s)} - (1-\gamma)\left[\phi_\pi\left(\bar{\pi}_c - \bar{\pi}_c^*\right) + \phi_y \overline{(\Delta y_{c,t} - g_c^y)}\right],$$

where bars denote averages over the pre-COVID anchoring window T_c^0. This normalization centers the average pre-COVID target gap around zero. The recovered series is therefore interpreted as a conditional short-run target path, not as a structural estimate of the central bank's objective.

Table A.9 presents diagnostics for different procedure components under the two policy stances.

A.2.4. *Robustness of the recovered target gap*

The first robustness check varies the Taylor-rule inflation coefficient, ϕ_π. This addresses the concern that the recovered target gap may be mechanically driven by the strength of the assumed policy response to inflation. The inversion is recomputed for alternative values of ϕ_π, and the country-specific intercept is re-estimated under the same pre-COVID normalization.

The second robustness check modifies the anchoring rule. This addresses the possibility that the baseline pre-COVID mean anchor may mechanically shape the level of the recovered gap after 2020. The exercise compares the baseline anchor with a steady-state target-anchor specification and a no-anchor diagnostic. In the target-anchor case, the intercept is set to

TABLE A.9. Recovery Diagnostics

Stance	Country	$a_c^{(s)}$ Stats				$\widehat{\pi}^T$ Observations				Target Gap	
		Min	Med.	Mean	Max	Non-miss	Miss	> 10	> 20	Min	Max
Policy	24	0.48	1.02	1.18	2.07	2193	15	29	0	-8.41	16.31
Shadow	24	-0.40	0.57	0.85	2.07	2193	15	24	0	-7.99	16.31

Notes: The table reports diagnostics for the HP-gap inversions using the policy-rate and shadow-rate stance measures. The intercept $a_c^{(s)}$ is computed separately by country and stance measure using the pre-COVID normalization. The columns > 10 and > 20 report the number of recovered implicit-target observations whose absolute value exceeds 10 and 20 percentage points, respectively. The final two columns report the minimum and maximum recovered implicit target gap, $y_{c,t}^{(s)} = \widehat{\pi}_{c,t}^{T,(s)} - \pi_{c,t}^*$.

$a_c^{(s)} = (1-\gamma)(r_c^* + \pi_c^*)$, where r_c^* is proxied by the country-specific pre-COVID average of the real policy rate, $i_{c,t}^{(s)} - \pi_{c,t}$, and π_c^* is the country's announced numerical inflation target, averaged over the same anchoring window when it varies over time. The no-anchor diagnostic sets $a_c^{(s)} = 0$.

The third robustness check replaces contemporaneous inflation with a forward-looking inflation measure. This addresses the concern that the baseline inversion may overstate the recovered target gap if central banks were responding to expected inflation rather than current inflation. Under this interpretation, policy rates may appear too low relative to realized inflation because the relevant policy variable is expected future inflation. If this explanation were driving the baseline result, replacing $\pi_{c,t}$ with a forecast $E_t\pi_{c,t+4}$ should reduce the recovered gap. The exercise implements this test by constructing an AR(2)-based four-quarter-ahead inflation forecast and using it in the Taylor-rule inversion in place of contemporaneous inflation.

The fourth robustness check modifies the treatment of output-growth trends. The baseline specification uses a pre-COVID HP trend that is frozen after 2019Q4, avoidint the COVID collapse and recovery to contaminate the trend estimate. The robustness exercise instead uses a rolling real-time HP trend. For each country and quarter t, the trend is recomputed using only observations available up to t. Operationally, the procedure applies the HP filter to the expanding history of log real GDP and retains only the terminal trend estimate for that date; trend growth is then computed as the year-over-year change in this real-time trend. This produces an output-growth gap based on information available contemporaneously rather than on a frozen pre-pandemic benchmark.

Table A.10 presents the summary of results and their respective interpretation. Overall, these exercises show that the positive recovered target gap during the 2021–2022 inflation episode is not specific to the baseline recovery choices. The gap remains positive under alternative Taylor-rule coefficients, anchoring conventions, inflation timing assumptions, and output-trend measures. The evidence should still be read descriptively.

A.2.5. *Pseudo real-time forecast shock construction*

The shock variables used in the following regressions are constructed as pseudo-real-time forecast-error proxies. They are not structural shocks from the theoretical model. Instead, they are empirical innovations in inflation and output growth, built to distinguish between supply-like and demand-like surprises. The identifying logic, in the spirit of Shapiro (2026), is that a supply-like disturbance is associated with inflation above forecast and output growth below forecast, whereas a demand-like disturbance is associated with both inflation and output growth above forecast.

TABLE A.10. Robustness of the Recovered Implicit Target Gap

Exercise	Modification	Main result	Interpretation
Taylor-rule inflation coefficient	Recompute the inversion with $\phi_\pi = 1.5$, re-estimating $a_c^{(s)}$ under the pre-COVID mean anchor	Mean 2021–2022 gap: 4.19 pp; 22/24 countries positive	The recovered gap remains positive under a lower inflation-response coefficient
Taylor-rule inflation coefficient	Recompute the inversion with $\phi_\pi = 2.0$, re-estimating $a_c^{(s)}$ under the pre-COVID mean anchor	Mean 2021–2022 gap: 3.96 pp; 22/24 countries positive	The result is not mechanically tied to a single implementation of the baseline calibration
Taylor-rule inflation coefficient	Recompute the inversion with $\phi_\pi = 2.5$, re-estimating $a_c^{(s)}$ under the pre-COVID mean anchor	Mean 2021–2022 gap: 3.82 pp; 22/24 countries positive	The magnitude declines mildly, but the sign and breadth of the result are preserved
Taylor-rule inflation coefficient	Recompute the inversion with $\phi_\pi = 3.0$, re-estimating $a_c^{(s)}$ under the pre-COVID mean anchor	Mean 2021–2022 gap: 3.73 pp; 22/24 countries positive	A stronger inflation-response coefficient does not eliminate the recovered gap
Anchoring rule	Replace the pre-COVID mean anchor with the target-anchor condition $a_c^{(s)} = (1-\gamma)(r_c^* + \bar{\pi}_c^*)$, where $r_c^* = \overline{i_{c,t}^{(s)} - \pi_{c,t}}$	Mean 2021–2022 gap: 4.38 pp; 24/24 countries positive	The result is not generated by the pre-COVID mean anchor
Anchoring rule	Remove the intercept normalization by setting $a_c^{(s)} = 0$	Mean 2021–2022 gap: 3.19 pp; 23/24 countries positive	The recovered gap survives under the most restrictive anchoring diagnostic
Forward-looking inflation	Replace contemporaneous inflation $\pi_{c,t}$ with an AR(2)-based four-quarter-ahead forecast, $E_t\pi_{c,t+4}$	Mean 2021–2022 gap: 9.12 pp; 24/24 countries positive	Forward-looking policy behavior does not explain away the recovered gap
Real-time trend	Replace the pre-COVID HP trend with a rolling real-time HP trend computed from expanding country-level samples	Mean 2021–2022 gap: 4.47 pp; 24/24 countries positive	The result is not driven by the frozen-trend assumption

Notes: The table reports robustness checks for the recovered implicit target gap during the 2021–2022 inflation episode. The baseline specification uses contemporaneous inflation, the policy-rate stance, the frozen pre-COVID HP trend for output growth, and the pre-COVID mean anchoring normalization.

For each country c, inflation and output growth are forecast using only lagged inflation and lagged output growth. The baseline forecasting equations are

$$\pi_{c,t} = \alpha_{\pi,c} + \sum_{\ell=1}^{p_{\pi\pi,c}} b_{\pi\pi,c,\ell}\pi_{c,t-\ell} + \sum_{\ell=1}^{q_{\pi y,c}} b_{\pi y,c,\ell}\Delta y_{c,t-\ell} + u_{c,t}^{\pi},$$

$$\Delta y_{c,t} = \alpha_{y,c} + \sum_{\ell=1}^{p_{y\pi,c}} b_{y\pi,c,\ell}\pi_{c,t-\ell} + \sum_{\ell=1}^{q_{yy,c}} b_{yy,c,\ell}\Delta y_{c,t-\ell} + u_{c,t}^{y}.$$

The lag orders, limited from 1 up to 4, are selected separately by country and by forecasting equation using the Hannan and Quinn information criterion.[32] The notation $p_{\pi\pi,c}$ denotes the number of inflation lags in the inflation equation, $q_{\pi y,c}$ the number of output-growth lags in the inflation equation, $p_{y\pi,c}$ the number of inflation lags in the output-growth equation, and $q_{yy,c}$ the number of output-growth lags in the output-growth equation.

Two pseudo-real-time forecasting procedures are implemented. In the first, lag orders are selected using only pre-2020 information and are then held fixed throughout the post-2020 period. Conditional on these selected lag orders, the forecasting coefficients are updated recursively with an expanding window, so that the forecast for quarter t uses only information available through quarter $t-1$.

The second procedure allows lag orders to update annually after 2020. At the end of each year, the lag structure is re-selected using only the information available up to that point, and the selected lag orders are then used for forecasts in the following year. Coefficients continue to be estimated recursively with expanding samples. This option preserves the pseudo-real-time discipline while allowing the forecasting model to adapt to post-pandemic changes in inflation and output growth dynamics.

[32]For a candidate model m, the criterion is computed as

$$HQC(m) = \log\left(\widehat{\sigma}_m^2\right) + \frac{2k_m \log(\log T_m)}{T_m},$$

where $\widehat{\sigma}_m^2$ is the residual variance, k_m is the number of estimated parameters, and T_m is the effective sample size. The selected specification is the one that minimizes $HQC(m)$ over the admissible lag grid.

The forecast errors from the two equations are standardized by country using pre-2020 moments,

$$\widetilde{u}^{\pi}_{c,t} = \frac{u^{\pi}_{c,t} - \bar{u}^{\pi,pre}_{c}}{\sigma^{\pi,pre}_{c}}, \qquad \widetilde{u}^{y}_{c,t} = \frac{u^{y}_{c,t} - \bar{u}^{y,pre}_{c}}{\sigma^{y,pre}_{c}}.$$

the use of pre-2020 moments fixes the normalization before the inflation episode and makes post-2020 realizations interpretable relative to each country's pre-pandemic forecast-error distribution.

The standardized forecast errors are then rotated into supply-like and demand-like components,

$$z^{S,raw}_{c,t} = \frac{\widetilde{u}^{\pi}_{c,t} - \widetilde{u}^{y}_{c,t}}{\sqrt{2}}, \qquad z^{D,raw}_{c,t} = \frac{\widetilde{u}^{\pi}_{c,t} + \widetilde{u}^{y}_{c,t}}{\sqrt{2}}.$$

this rotation maps high-inflation and low-output-growth surprises into the supply-like component, and high-inflation and high-output-growth surprises into the demand-like component. The rotated series are standardized again using pre-2020 country-specific moments to obtain the final shock proxies, $z^{S}_{c,t}$ and $z^{D}_{c,t}$. Under this normalization, the thresholds used in the regressions, $\tau \in \{1.0, 1.5, 2.0\}$, are expressed in country-specific standard-deviation units.

The selected forecasting lag orders are reported separately for the two procedures. Table A.11 presents the selected lag orders obtained from the fixed pre-2020 selection window and held constant afterward. Table A.12 presents the lag orders used under the annual lag-update procedure. For both tables, each cell reports $(p_{\pi\pi}, q_{\pi y}; p_{y\pi}, q_{yy})$, with the first pair corresponding to the inflation equation and the second to the output-growth equation. The distribution of the final standardized shocks is reported in Table A.13.

Table A.11. Forecasting Lag Orders under the Fixed Pre-2020 Procedure

Country	ISO3	$p^{fixed}_{\pi\pi}$	$q^{fixed}_{\pi y}$	$p^{fixed}_{y\pi}$	q^{fixed}_{yy}
Austria	AUT	2	2	2	2
Belgium	BEL	3	1	1	2
Canada	CAN	2	1	2	2
Chile	CHL	2	1	2	1
Czech Republic	CZE	2	1	2	2
Finland	FIN	2	1	1	4
France	FRA	2	1	1	3
Germany	DEU	1	1	2	4
Hungary	HUN	2	3	1	2
Iceland	ISL	2	1	1	4
Ireland	IRL	2	1	1	1
Italy	ITA	1	1	4	2
Japan	JPN	1	1	1	3
Korea	KOR	1	1	2	2
Mexico	MEX	1	3	3	2
Netherlands	NLD	1	1	1	2
Norway	NOR	4	1	1	4
Poland	POL	2	1	1	4
Portugal	PRT	2	1	1	1
Spain	ESP	2	1	3	2
Sweden	SWE	1	1	1	4
Switzerland	CHE	2	1	1	2
United Kingdom	GBR	2	1	1	4
United States	USA	2	1	3	2

Notes: The subscripts indicate the forecasting equation and the lagged regressor. $p_{\pi\pi}$ is the number of inflation lags in the inflation equation, $q_{\pi y}$ is the number of output-growth lags in the inflation equation, $p_{y\pi}$ is the number of inflation lags in the output-growth equation, and q_{yy} is the number of output-growth lags in the output-growth equation.

A.2.6. *Empirical methodology*

The empirical methodology aims to estimate whether large forecast-error shocks are associated with the recovered implicit target gap after controlling for the linear effects of supply and demand-like shocks. The dependent variable is the recovered implicit target gap, under two different policy stance measures. The regressions are estimated separately for the policy-rate and shadow-rate gaps, and separately for the two shock-construction panels described above.

TABLE A.12. Forecasting Lag Orders under the Annual Lag-Update Procedure

Country	ISO3	2020	2021	2022	2023	2024
Austria	AUT	(2, 2; 2, 2)	(2, 1; 1, 2)	(2, 3; 1, 4)	(2, 1; 1, 4)	(2, 1; 1, 4)
Belgium	BEL	(3, 1; 1, 2)	(3, 1; 1, 1)	(3, 4; 1, 4)	(3, 4; 1, 4)	(4, 2; 1, 4)
Canada	CAN	(2, 1; 2, 2)	(2, 1; 1, 1)	(2, 1; 1, 4)	(2, 1; 1, 4)	(2, 1; 1, 4)
Chile	CHL	(2, 1; 2, 1)	(2, 1; 1, 1)	(2, 1; 1, 4)	(2, 1; 1, 4)	(2, 1; 1, 4)
Czech Republic	CZE	(2, 1; 2, 2)	(2, 1; 3, 1)	(2, 1; 1, 4)	(2, 1; 1, 4)	(2, 1; 1, 4)
Finland	FIN	(2, 1; 1, 4)	(1, 1; 1, 4)	(2, 3; 1, 4)	(2, 3; 2, 4)	(2, 3; 1, 4)
France	FRA	(2, 1; 1, 3)	(2, 1; 1, 1)	(2, 3; 1, 4)	(2, 1; 1, 4)	(3, 1; 1, 4)
Germany	DEU	(1, 1; 2, 4)	(1, 1; 4, 1)	(2, 4; 2, 1)	(1, 3; 2, 1)	(4, 4; 2, 4)
Hungary	HUN	(2, 3; 1, 2)	(2, 1; 1, 1)	(2, 2; 1, 4)	(2, 2; 1, 4)	(2, 2; 1, 4)
Iceland	ISL	(2, 1; 1, 4)	(2, 1; 1, 4)	(2, 1; 1, 4)	(2, 1; 1, 4)	(2, 1; 1, 4)
Ireland	IRL	(2, 1; 1, 1)	(2, 1; 1, 1)	(2, 1; 1, 4)	(3, 1; 1, 4)	(3, 1; 1, 4)
Italy	ITA	(1, 1; 4, 2)	(3, 1; 1, 1)	(3, 3; 1, 4)	(3, 1; 2, 4)	(4, 1; 2, 4)
Japan	JPN	(1, 1; 1, 3)	(1, 1; 1, 3)	(1, 1; 1, 3)	(4, 1; 1, 3)	(4, 1; 1, 3)
Korea	KOR	(1, 1; 2, 2)	(1, 2; 2, 2)	(1, 1; 2, 2)	(1, 1; 2, 2)	(1, 1; 2, 2)
Mexico	MEX	(1, 3; 3, 2)	(1, 1; 1, 1)	(1, 4; 2, 4)	(2, 3; 1, 4)	(2, 3; 1, 4)
Netherlands	NLD	(1, 1; 1, 2)	(1, 1; 2, 1)	(1, 3; 2, 4)	(3, 4; 2, 4)	(4, 1; 2, 4)
Norway	NOR	(4, 1; 1, 4)	(4, 1; 1, 4)	(4, 1; 1, 4)	(1, 2; 1, 4)	(1, 2; 1, 4)
Poland	POL	(2, 1; 1, 4)	(4, 1; 1, 4)	(4, 3; 1, 4)	(4, 1; 1, 4)	(2, 1; 1, 4)
Portugal	PRT	(2, 1; 1, 1)	(2, 1; 1, 1)	(2, 1; 1, 4)	(4, 1; 1, 4)	(4, 1; 1, 4)
Spain	ESP	(2, 1; 3, 2)	(2, 1; 1, 1)	(2, 3; 1, 4)	(3, 1; 1, 4)	(3, 1; 1, 4)
Sweden	SWE	(1, 1; 1, 4)	(1, 1; 1, 4)	(1, 3; 2, 4)	(2, 2; 2, 4)	(3, 2; 2, 4)
Switzerland	CHE	(2, 1; 1, 2)	(2, 1; 2, 1)	(2, 1; 1, 4)	(2, 1; 1, 4)	(2, 1; 1, 4)
United Kingdom	GBR	(2, 1; 1, 4)	(2, 1; 1, 1)	(2, 4; 1, 4)	(3, 1; 1, 4)	(3, 1; 1, 4)
United States	USA	(2, 1; 3, 2)	(2, 1; 1, 1)	(2, 1; 1, 4)	(2, 1; 1, 4)	(2, 1; 1, 4)

Notes: Each cell reports $(p_{\pi\pi}, q_{\pi y}; p_{y\pi}, q_{yy})$. The first pair gives the inflation and output-growth lag orders in the inflation forecasting equation. The second pair gives the inflation and output-growth lag orders in the output-growth forecasting equation. The entry for year Y reports the lag orders used to forecast observations in year Y. The 2020 entries are based on the pre-2020 selection window. From 2021 onward, lag orders are re-selected annually using information available up to the end of the previous year.

The common empirical specification is

$$y_{c,t}^{(s)} = \alpha_c + \lambda_t + \rho y_{c,t-1}^{(s)} + \theta_S z_{c,t}^S + \theta_D z_{c,t}^D + \beta_S \mathcal{N}_{c,t}^S(\tau) + \beta_D \mathcal{N}_{c,t}^D(\tau) + \varepsilon_{c,t}.$$

The terms α_c and λ_t denote country and year fixed effects, respectively. The lagged dependent variable captures persistence in the recovered implicit target gap. Standard errors are clustered at the country level.

The variables $z_{c,t}^S$ and $z_{c,t}^D$ are the standardized supply-like and demand-like shocks.[33] The coefficients θ_S and θ_D measure the linear association between each shock and the recovered gap. The nonlinear terms $\mathcal{N}_{c,t}^S(\tau)$ and $\mathcal{N}_{c,t}^D(\tau)$ capture the additional association that appears when shocks cross a size threshold. Since the shocks are standardized by country, the thresholds $\tau \in \{1.0, 1.5, 2.0\}$ are interpreted in standard-deviation units.

The definition of the nonlinear terms $\mathcal{N}_{c,t}^S(\tau)$ and $\mathcal{N}_{c,t}^D(\tau)$ has two alternatives. In the one-sided specification, these terms are defined using positive upper-tail shocks and are designed to capture the empirical relevance of large inflationary disturbances. In the symmetric specification, $\mathcal{N}_{c,t}^S(\tau)$ and $\mathcal{N}_{c,t}^D(\tau)$ are defined using absolute-value triggers, which is closer to the theoretical shock-size function because activation depends on shock magnitude rather than shock sign.

One-sided trigger specification. The one-sided version of the main empirical specification defines the nonlinear terms as positive upper-tail transformations of the supply-like and demand-like shocks,

$$N_{c,t}^S(\tau) = \left(z_{c,t}^S - \tau\right)_+, \qquad N_{c,t}^D(\tau) = \left(z_{c,t}^D - \tau\right)_+.$$

Since the shocks are standardized, the thresholds $\tau \in \{1.0, 1.5, 2.0\}$ are measured in country-specific standard-deviation units. The coefficients β_S and β_D capture the additional association between the recovered implicit target gap and the corresponding shock once the threshold is exceeded.

Table A.14 reports the one-sided estimates for the policy-rate and shadow-rate recovered gaps. The table includes the linear coefficients θ_S and θ_D, the nonlinear coefficients β_S and

[33] That is, the variables $z_{c,t}^{S,raw}$ and $z_{c,t}^{D,raw}$ standardized using the pre-2020 moments.

TABLE A.13. Shock Coverage and Distribution

Panel A. Fixed pre-2020 procedure

Period	N	Mean	Std. dev.	p10	p50	p90
Supply-like shock						
Pre-2020	1339	0.000	0.991	−1.195	−0.052	1.260
COVID 2020	96	−0.978	13.235	−10.823	1.030	8.080
Inflation 2021–2022	192	−0.023	4.903	−4.255	0.444	4.191
Disinflation 2023–2024	192	−0.746	2.294	−3.562	−0.267	1.162
Demand-like shock						
Pre-2020	1339	0.000	0.991	−1.129	0.029	1.194
COVID 2020	96	1.313	14.090	−8.270	−0.599	11.289
Inflation 2021–2022	192	2.389	4.721	−2.082	2.024	6.311
Disinflation 2023–2024	192	0.095	1.753	−1.616	0.201	1.760

Panel B. Annual update procedure

Period	N	Mean	Std. dev.	p10	p50	p90
Supply-like shock						
Pre-2020	1339	0.000	0.991	−1.195	−0.052	1.260
COVID 2020	96	−0.978	13.235	−10.823	1.030	8.080
Inflation 2021–2022	192	−0.255	5.089	−5.816	0.437	4.736
Disinflation 2023–2024	192	−0.379	2.360	−3.473	0.094	1.817
Demand-like shock						
Pre-2020	1339	0.000	0.991	−1.129	0.029	1.194
COVID 2020	96	1.313	14.090	−8.270	−0.599	11.289
Inflation 2021–2022	192	2.699	4.927	−1.686	1.797	8.715
Disinflation 2023–2024	192	0.259	1.916	−1.351	0.288	2.065

Notes: The fixed pre-2020 procedure selects lag orders using pre-2020 information and holds them fixed afterward. The annual update procedure re-selects lag orders annually after 2020 while preserving the pseudo-real-time forecasting discipline. Pre-2020 is the normalization sample. COVID 2020 = 2020Q1–2020Q4; Inflation 2021–2022 = 2021Q1–2022Q2; Disinflation 2023–2024 = 2023Q1–2024Q4. Statistics are computed over non-missing country-quarter observations.

β_D, and the upper-tail slopes $\theta_S + \beta_S$ and $\theta_D + \beta_D$. Standard errors for the upper-tail slopes are computed from the corresponding linear-combination tests.

Symmetric trigger specification. The symmetric version of main empirical specification defines the nonlinear terms as absolute-value transformations of the supply-like and demand-like shocks,

$$N^S_{c,t}(\tau) = \left(|z^S_{c,t}| - \tau\right)_+, \qquad N^D_{c,t}(\tau) = \left(|z^D_{c,t}| - \tau\right)_+.$$

This specification keeps the signed shocks $z^S_{c,t}$ and $z^D_{c,t}$ as linear controls, while allowing the nonlinear component to activate when the magnitude of each shock exceeds the threshold τ.

Table A.15 reports the symmetric-trigger estimates for the policy-rate and shadow-rate recovered gaps. The coefficients θ^{sym}_S and θ^{sym}_D denote the linear coefficients estimated within the symmetric-trigger specification. The coefficients β^{abs}_S and β^{abs}_D denote the coefficients attached to the absolute-value trigger terms. The table also reports the positive-tail slopes, $\theta^{sym} + \beta^{abs}$, and the negative-tail slopes, $\theta^{sym} - \beta^{abs}$. Standard errors for all tail slopes are computed using Wald tests for the corresponding linear restrictions.

TABLE A.14. One-Sided Coefficients and Tail Slopes

Panel A. Fixed lag-order specification

	Supply side coeff.			Demand side coeff.		
τ	θ_S	β_S	$\theta_S + \beta_S$	θ_D	β_D	$\theta_D + \beta_D$
Policy Rate gap						
1.0	0.109***	0.024	0.134**	0.471***	−0.295***	0.176***
	(0.029)	(0.068)	(0.049)	(0.041)	(0.057)	(0.056)
1.5	0.097***	0.046*	0.142**	0.469***	−0.309***	0.159***
	(0.029)	(0.023)	(0.054)	(0.038)	(0.056)	(0.054)
2.0	0.092***	0.051*	0.144**	0.463***	−0.312***	0.151***
	(0.029)	(0.028)	(0.059)	(0.036)	(0.054)	(0.052)
Shadow Rate gap						
1.0	0.080**	0.077	0.157***	0.458***	−0.304***	0.153**
	(0.032)	(0.067)	(0.047)	(0.040)	(0.055)	(0.057)
1.5	0.069**	0.103	0.171***	0.456***	−0.319***	0.137**
	(0.032)	(0.074)	(0.053)	(0.037)	(0.055)	(0.055)
2.0	0.066**	0.109*	0.175***	0.450***	−0.320***	0.130**
	(0.032)	(0.062)	(0.059)	(0.034)	(0.054)	(0.054)

Panel B. Annual lag-update specification

	One-sided supply side			One-sided demand side		
τ	θ_S	β_S	$\theta_S + \beta_S$	θ_D	β_D	$\theta_D + \beta_D$
Policy Rate gap						
1.0	0.071***	0.091*	0.162***	0.495***	−0.356***	0.139***
	(0.023)	(0.050)	(0.042)	(0.044)	(0.040)	(0.046)
1.5	0.060**	0.116**	0.176***	0.491***	−0.368***	0.123***
	(0.023)	(0.054)	(0.045)	(0.041)	(0.040)	(0.043)
2.0	0.057**	0.127**	0.183***	0.483***	−0.369***	0.114***
	(0.022)	(0.057)	(0.049)	(0.039)	(0.039)	(0.041)
Shadow Rate gap						
1.0	0.045*	0.127**	0.173***	0.473***	−0.355***	0.119**
	(0.026)	(0.053)	(0.044)	(0.045)	(0.039)	(0.046)
1.5	0.035	0.158**	0.193***	0.471***	−0.369***	0.103**
	(0.026)	(0.057)	(0.047)	(0.041)	(0.039)	(0.044)
2.0	0.033	0.170**	0.203***	0.464***	−0.369***	0.096**
	(0.026)	(0.062)	(0.052)	(0.039)	(0.039)	(0.042)

Note: The dependent variable is the recovered implicit target gap, computed using either the policy-rate or shadow-rate stance. The nonlinear terms are $N^S_{c,t}(\tau) = (z^S_{c,t} - \tau)_+$ and $N^D_{c,t}(\tau) = (z^D_{c,t} - \tau)_+$. All regressions include country fixed effects, year fixed effects, and the lagged dependent variable. Standard errors are clustered at the country level and reported in parentheses. The columns $\theta_S + \beta_S$ and $\theta_D + \beta_D$ report upper-tail slopes, with standard errors and significance levels computed from the corresponding linear-combination tests. Panel A holds forecasting lag orders fixed at their pre-2020 selection. Panel B presents forecasts updating lag orders annually from 2020 onwards. Sample in quarters with $N = 1{,}719$ observations across all specifications. Significance levels are denoted by $^*p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

A.2.7. *Distributed-lag specification*

The distributed-lag specifications extend the nonlinear component of the main empirical strategy by allowing the trigger variables to enter with lags, keeping the previous controls. For a maximum lag length J, econometric specification is given by

$$y^{(s)}_{c,t} = \alpha_c + \lambda_t + \rho y^{(s)}_{c,t-1} + \theta_S z^S_{c,t} + \theta_D z^D_{c,t} + \sum_{j=0}^{J} \beta_{j,S} N^S_{c,t-j}(\tau) + \sum_{j=0}^{J} \beta_{j,D} N^D_{c,t-j}(\tau) + \varepsilon_{c,t},$$

TABLE A.15. Symmetric-Trigger Coefficients and Tail Slopes

Panel A. Fixed lag-order specification

	Supply side coeff.				Demand side coeff.			
τ	θ_S^{sym}	β_S^{abs}	$\theta_S^{\text{sym}}+\beta_S^{\text{abs}}$	$\theta_S^{\text{sym}}-\beta_S^{\text{abs}}$	θ_D^{sym}	β_D^{abs}	$\theta_D^{\text{sym}}+\beta_D^{\text{abs}}$	$\theta_D^{\text{sym}}-\beta_D^{\text{abs}}$
Policy Rate gap								
1.0	0.136***	−0.009	0.126***	0.145***	0.353***	−0.137***	0.216***	0.490***
	(0.019)	(0.026)	(0.038)	(0.026)	(0.046)	(0.026)	(0.056)	(0.050)
1.5	0.135***	0.005	0.140***	0.130***	0.357***	−0.157***	0.200***	0.513***
	(0.019)	(0.031)	(0.043)	(0.028)	(0.046)	(0.029)	(0.055)	(0.055)
2.0	0.135***	0.012*	0.147***	0.122***	0.360***	−0.170***	0.190***	0.529***
	(0.019)	(0.007)	(0.047)	(0.031)	(0.046)	(0.031)	(0.055)	(0.057)
Shadow Rate gap								
1.0	0.124***	0.001	0.126***	0.123***	0.330***	−0.130***	0.200***	0.460***
	(0.021)	(0.025)	(0.038)	(0.027)	(0.047)	(0.025)	(0.054)	(0.053)
1.5	0.124***	0.016*	0.140***	0.109***	0.334***	−0.149***	0.185***	0.482***
	(0.021)	(0.01)	(0.042)	(0.029)	(0.047)	(0.028)	(0.053)	(0.057)
2.0	0.124***	0.023**	0.147***	0.101***	0.336***	−0.162***	0.175***	0.498***
	(0.021)	(0.013)	(0.046)	(0.032)	(0.047)	(0.031)	(0.054)	(0.058)

Panel B. Annual lag-update specification

	Supply side coeff.				Demand side coeff.			
τ	θ_S^{sym}	β_S^{abs}	$\theta_S^{\text{sym}}+\beta_S^{\text{abs}}$	$\theta_S^{\text{sym}}-\beta_S^{\text{abs}}$	θ_D^{sym}	β_D^{abs}	$\theta_D^{\text{sym}}+\beta_D^{\text{abs}}$	$\theta_D^{\text{sym}}-\beta_D^{\text{abs}}$
Policy Rate gap								
1.0	0.125***	0.032	0.157***	0.093***	0.350***	−0.182***	0.168***	0.531***
	(0.021)	(0.022)	(0.037)	(0.021)	(0.048)	(0.021)	(0.044)	(0.060)
1.5	0.125***	0.054**	0.179***	0.071***	0.355***	−0.209***	0.146***	0.564***
	(0.021)	(0.024)	(0.041)	(0.020)	(0.049)	(0.025)	(0.039)	(0.067)
2.0	0.126***	0.070**	0.196***	0.056***	0.359***	−0.230***	0.129***	0.589***
	(0.022)	(0.026)	(0.044)	(0.019)	(0.049)	(0.029)	(0.035)	(0.073)
Shadow Rate gap								
1.0	0.112***	0.039*	0.151***	0.073***	0.325***	−0.172***	0.153***	0.497***
	(0.024)	(0.022)	(0.040)	(0.023)	(0.050)	(0.022)	(0.043)	(0.064)
1.5	0.112***	0.061**	0.173***	0.052**	0.330***	−0.198***	0.131***	0.528***
	(0.024)	(0.025)	(0.044)	(0.022)	(0.051)	(0.026)	(0.039)	(0.070)
2.0	0.113***	0.077***	0.190***	0.036*	0.334***	−0.219***	0.115***	0.553***
	(0.024)	(0.027)	(0.046)	(0.021)	(0.051)	(0.029)	(0.036)	(0.075)

Note: The dependent variable is the recovered implicit target gap, computed using either the policy-rate stance or the shadow-rate stance. The nonlinear terms are $N_{c,t}^S(\tau) = (|z_{c,t}^S| - \tau)_+$ and $N_{c,t}^D(\tau) = (|z_{c,t}^D| - \tau)_+$. All regressions include country fixed effects, year fixed effects, and the lagged dependent variable. Standard errors are clustered at the country level and reported in parentheses. The superscript *sym* indicates that the linear coefficient is estimated within the symmetric-trigger specification. The superscript *abs* denotes the coefficient on the absolute-value trigger. Standard errors and significance levels for the reported positive-tail and negative-tail slopes are computed using Wald tests for the corresponding linear restrictions. Panel A holds forecasting lag orders fixed at their pre-2020 selection. Panel B presents forecasts updating lag orders annually from 2020 onwards. Sample in quarters with $N = 1{,}719$ observations across all specifications. Significance levels are denoted by $^*p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

where $N_{c,t}^S(\tau) = (z_{c,t}^S - \tau)_+$ and $N_{c,t}^D(\tau) = (z_{c,t}^D - \tau)_+$ for the one-sided specification., and $N_{c,t}^S(\tau) = (|z_{c,t}^S| - \tau)_+$ and $N_{c,t}^D(\tau) = (|z_{c,t}^D| - \tau)_+$ for the symmetric specification.

Table A.16 report the cumulative nonlinear coefficients for $J \in \{0, 1, 2\}$. The reported p-values correspond to joint Wald tests for the null that all nonlinear distributed-lag coefficients for the corresponding shock are jointly equal to zero.

TABLE A.16. Distributed-Lag Models: One-Sided vs. Symmetric Trigger Specifications

		ONE-SIDED TRIGGER SPECIFICATION				SYMMETRIC TRIGGER SPECIFICATION			
		Panel A: Fixed Lags		Panel B: Annual Update		Panel A: Fixed Lags		Panel B: Annual Update	
τ	J	$\sum \beta_{j,S}$	$\sum \beta_{j,D}$	$\sum \beta_{j,S}$	$\sum \beta_{j,D}$	$\sum \beta_{j,S}$	$\sum \beta_{j,D}$	$\sum \beta_{j,S}$	$\sum \beta_{j,D}$
Policy Rate gap									
		0.024	-0.295^{***}	0.091^{*}	-0.356^{***}	0.136^{***}	0.353^{***}	0.125^{***}	0.350^{***}
1.0	0	(0.723)	(< 0.001)	(0.083)	(< 0.001)	(0.019)	(0.046)	(0.021)	(0.048)
		0.045	-0.309^{***}	0.094	-0.378^{***}	0.135^{***}	0.357^{***}	0.125^{***}	0.355^{***}
1.0	1	(0.876)	(< 0.001)	(0.199)	(< 0.001)	(0.019)	(0.046)	(0.021)	(0.049)
		0.198^{***}	-0.350^{***}	0.230^{***}	-0.399^{***}	0.135^{***}	0.360^{***}	0.126^{***}	0.359^{***}
1.0	2	(< 0.001)	(< 0.001)	(0.004)	(< 0.001)	(0.019)	(0.046)	(0.022)	(0.049)
		0.046	-0.309^{***}	0.116^{**}	-0.368^{***}	0.124^{***}	0.330^{***}	0.112^{***}	0.325^{***}
1.5	0	(0.546)	(< 0.001)	(0.041)	(< 0.001)	(0.021)	(0.047)	(0.024)	(0.050)
		0.059	-0.323^{***}	0.103	-0.387^{***}	0.124^{***}	0.334^{***}	0.112^{***}	0.330^{***}
1.5	1	(0.806)	(< 0.001)	(0.116)	(< 0.001)	(0.021)	(0.047)	(0.024)	(0.051)
		0.212^{***}	-0.359^{***}	0.249^{***}	-0.411^{***}	0.124^{***}	0.336^{***}	0.113^{***}	0.334^{***}
1.5	2	(< 0.001)	(< 0.001)	(0.003)	(< 0.001)	(0.021)	(0.047)	(0.024)	(0.051)
		0.051	-0.312^{***}	0.127^{**}	-0.369^{***}	0.125^{***}	0.350^{***}	0.112^{***}	0.325^{***}
2.0	0	(0.522)	(< 0.001)	(0.038)	(< 0.001)	(0.021)	(0.048)	(0.024)	(0.050)
		0.059	-0.325^{***}	0.098	-0.386^{***}	0.125^{***}	0.355^{***}	0.112^{***}	0.330^{***}
2.0	1	(0.770)	(< 0.001)	(0.110)	(< 0.001)	(0.021)	(0.049)	(0.024)	(0.051)
		0.203^{***}	-0.356^{***}	0.252^{***}	-0.412^{***}	0.126^{***}	0.359^{***}	0.113^{***}	0.334^{***}
2.0	2	(0.002)	(< 0.001)	(0.002)	(< 0.001)	(0.022)	(0.049)	(0.024)	(0.051)
Shadow Rate gap									
		0.077	-0.304^{***}	0.127^{**}	-0.355^{***}	0.124^{***}	0.330^{***}	0.112^{***}	0.325^{***}
1.0	0	(0.266)	(< 0.001)	(0.025)	(< 0.001)	(0.021)	(0.047)	(0.024)	(0.050)
		0.082	-0.311^{***}	0.116^{*}	-0.365^{***}	0.124^{***}	0.334^{***}	0.112^{***}	0.330^{***}
1.0	1	(0.460)	(< 0.001)	(0.073)	(< 0.001)	(0.021)	(0.047)	(0.024)	(0.051)
		0.233^{***}	-0.356^{***}	0.277^{***}	-0.397^{***}	0.124^{***}	0.336^{***}	0.113^{***}	0.334^{***}
1.0	2	(< 0.001)	(< 0.001)	(< 0.001)	(< 0.001)	(0.021)	(0.047)	(0.024)	(0.051)
		0.103	-0.319^{***}	0.158^{**}	-0.369^{***}	0.124^{***}	0.330^{***}	0.112^{***}	0.325^{***}
1.5	0	(0.180)	(< 0.001)	(0.011)	(< 0.001)	(0.021)	(0.047)	(0.024)	(0.050)
		0.099	-0.326^{***}	0.130^{**}	-0.377^{***}	0.124^{***}	0.334^{***}	0.112^{***}	0.330^{***}
1.5	1	(0.251)	(< 0.001)	(0.032)	(< 0.001)	(0.021)	(0.047)	(0.024)	(0.051)
		0.247^{***}	-0.366^{***}	0.303^{***}	-0.412^{***}	0.124^{***}	0.336^{***}	0.113^{***}	0.334^{***}
1.5	2	(< 0.001)	(< 0.001)	(< 0.001)	(< 0.001)	(0.021)	(0.047)	(0.024)	(0.051)
		0.109	-0.320^{***}	0.170^{**}	-0.369^{***}	0.124^{***}	0.330^{***}	0.112^{***}	0.325^{***}
2.0	0	(0.191)	(< 0.001)	(0.012)	(< 0.001)	(0.021)	(0.047)	(0.024)	(0.050)
		0.096	-0.327^{***}	0.124^{**}	-0.376^{***}	0.124^{***}	0.334^{***}	0.112^{***}	0.330^{***}
2.0	1	(0.185)	(< 0.001)	(0.028)	(< 0.001)	(0.021)	(0.047)	(0.024)	(0.051)
		0.237^{***}	-0.363^{***}	0.308^{***}	-0.415^{***}	0.124^{***}	0.336^{***}	0.113^{***}	0.334^{***}
2.0	2	(0.002)	(< 0.001)	(< 0.001)	(< 0.001)	(0.021)	(0.047)	(0.024)	(0.051)

Note: The dependent variable is the recovered implicit target gap under the policy-rate or shadow-rate stances. One-sided shocks follow $N^i_{c,t}(\tau) = (z^i_{c,t} - \tau)_+$, while symmetric shocks follow $N^i_{c,t}(\tau) = (|z^i_{c,t}| - \tau)_+$. The parameter J represents the maximum lag included in the distributed lag. The values $\sum \beta_{j,S}$ and $\sum \beta_{j,D}$ indicate the cumulative nonlinear coefficients for supply and demand sides, respectively. Values inside parentheses report joint Wald test p-values. Panel A preserves lag selections fixed at their pre-2020 layout, whereas Panel B updates lags annually from 2020 onward. Regressions incorporate country and year fixed effects with standard errors clustered at the country level. Sample in quarters, for $J = 0$ the $N = 1{,}719$, for $J = 1$ $N = 1{,}700$, for $J = 2$ the $N = 1{,}681$. Significance levels are denoted by $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

A.2.8. *Non-Financial Corporations Lending as Policy Stance Proxy*

The baseline recovery uses the policy-rate and shadow-rate stance measures. For euro-area countries, however, this assigns a common policy-rate stance to economies that may face different country-level financing conditions. This subsection reconstructs the recovered implicit

target gap using lending rates to non-financial corporations, NFC, as an alternative monetary-stance measure.

Two NFC-based stance measures are used. The first is the raw NFC lending rate, $i_{c,t}^{NFC,raw}$. The second subtracts the ten-year sovereign spread from the raw lending rate, yielding $i_{c,t}^{NFC,net}$. The raw measure captures country-level firm borrowing conditions, while the net measure removes part of the sovereign-risk component embedded in domestic lending rates. For non-euro-area countries, the NFC-based stance variables coincide with the baseline policy-rate stance, so the robustness exercise modifies the euro-area monetary-stance measure while preserving the rest of the panel construction.

The same specifications as in Appendix A.2.6 are estimated using the NFC raw and NFC net recovered gaps as dependent variables. Table A.17 reports the one-sided trigger estimates, while Table A.18 reports the symmetric-trigger estimates.

TABLE A.17. One-Sided Coefficients and Upper-Tail Slopes

Panel A. Fixed lag-order specification

	Supply side coeff.			Demand side coeff.		
τ	θ_S	β_S	$\theta_S+\beta_S$	θ_D	β_D	$\theta_D+\beta_D$
NFC Raw Rate						
1.0	0.114***	0.029	0.144***	0.478***	−0.299***	0.179***
	(0.030)	(0.071)	(0.050)	(0.042)	(0.059)	(0.056)
1.5	0.104***	0.041	0.146**	0.471***	−0.306***	0.165***
	(0.031)	(0.078)	(0.056)	(0.039)	(0.059)	(0.055)
2.0	0.101***	0.044	0.145**	0.464***	−0.307***	0.157***
	(0.030)	(0.082)	(0.060)	(0.036)	(0.056)	(0.052)
NFC Net Rate						
1.0	0.122***	0.018	0.140***	0.469***	−0.286***	0.183***
	(0.030)	(0.069)	(0.048)	(0.040)	(0.057)	(0.055)
1.5	0.111***	0.033	0.144**	0.464***	−0.296***	0.168***
	(0.030)	(0.076)	(0.054)	(0.038)	(0.056)	(0.053)
2.0	0.107***	0.036	0.143**	0.458***	−0.298***	0.160***
	(0.029)	(0.080)	(0.059)	(0.035)	(0.054)	(0.050)

Panel B. Annual lag-update specification

	Supply side coeff.			Demand side coeff.		
τ	θ_S	β_S	$\theta_S+\beta_S$	θ_D	β_D	$\theta_D+\beta_D$
NFC Raw Rate						
1.0	0.077***	0.088	0.164***	0.498***	−0.356***	0.142***
	(0.024)	(0.056)	(0.046)	(0.046)	(0.045)	(0.045)
1.5	0.068***	0.105*	0.173***	0.490***	−0.361***	0.129***
	(0.024)	(0.059)	(0.049)	(0.043)	(0.044)	(0.043)
2.0	0.066***	0.110*	0.176***	0.481***	−0.360***	0.122***
	(0.023)	(0.061)	(0.052)	(0.041)	(0.042)	(0.041)
NFC Net Rate						
1.0	0.083***	0.080	0.163***	0.489***	−0.344***	0.145***
	(0.024)	(0.054)	(0.044)	(0.044)	(0.042)	(0.044)
1.5	0.073***	0.100*	0.173***	0.484***	−0.353***	0.131***
	(0.024)	(0.057)	(0.047)	(0.041)	(0.041)	(0.041)
2.0	0.071***	0.107*	0.177***	0.476***	−0.352***	0.124***
	(0.023)	(0.059)	(0.050)	(0.039)	(0.040)	(0.039)

Note: The nonlinear terms are $N^S_{c,t}(\tau)=(z^S_{c,t}-\tau)_+$ and $N^D_{c,t}(\tau)=(z^D_{c,t}-\tau)_+$. All regressions include country fixed effects, year fixed effects, and the lagged dependent variable. Standard errors are clustered at the country level and reported in parentheses. The columns $\theta_S+\beta_S$ and $\theta_D+\beta_D$ report upper-tail slopes. Standard errors and significance levels for these slopes are computed using Wald tests for the corresponding linear restrictions. Significance levels are denoted by $^*p<0.10$, $^{**}p<0.05$, and $^{***}p<0.01$.

APPENDIX B. RESULTS

B.1. *Sacrifice-ratio statistics*

Let y_{t+h} and π_{t+h} denote the GIRFs of output and inflation at horizon h, and let H denote the horizon over which the statistic is computed.

The recession-based sacrifice ratio is defined as

$$SR^H_{rec}=\frac{\sum_{h=0}^{H}\min\{y_{t+h},0\}}{\sum_{h=0}^{H}\pi_{t+h}}.$$

TABLE A.18. Symmetric-Trigger Coefficients and Tail Slopes

Panel A. Fixed lag-order specification

	Supply side coeff.				Demand side coeff.			
τ	θ_S^{sym}	β_S^{abs}	$\theta_S^{sym}+\beta_S^{abs}$	$\theta_S^{sym}-\beta_S^{abs}$	θ_D^{sym}	β_D^{abs}	$\theta_D^{sym}+\beta_D^{abs}$	$\theta_D^{sym}-\beta_D^{abs}$
NFC Raw Rate								
1.0	0.143^{***} (0.019)	-0.004 (0.028)	0.139^{***} (0.039)	0.147^{***} (0.027)	0.358^{***} (0.046)	-0.142^{***} (0.028)	0.216^{***} (0.055)	0.500^{***} (0.051)
1.5	0.142^{***} (0.019)	0.007 (0.034)	0.150^{***} (0.046)	0.135^{***} (0.031)	0.361^{***} (0.046)	-0.158^{***} (0.032)	0.203^{***} (0.055)	0.520^{***} (0.056)
2.0	0.142^{***} (0.019)	0.013 (0.039)	0.154^{***} (0.050)	0.129^{***} (0.034)	0.364^{***} (0.045)	-0.169^{***} (0.035)	0.195^{***} (0.056)	0.533^{***} (0.058)
NFC Net Rate								
1.0	0.146^{***} (0.019)	-0.011 (0.027)	0.135^{***} (0.037)	0.156^{***} (0.027)	0.355^{***} (0.044)	-0.133^{***} (0.026)	0.222^{***} (0.054)	0.488^{***} (0.048)
1.5	0.145^{***} (0.019)	0.001 (0.032)	0.146^{***} (0.043)	0.144^{***} (0.030)	0.358^{***} (0.044)	-0.150^{***} (0.029)	0.208^{***} (0.053)	0.508^{***} (0.052)
2.0	0.144^{***} (0.019)	0.008 (0.036)	0.152^{***} (0.047)	0.137^{***} (0.032)	0.361^{***} (0.044)	-0.162^{***} (0.032)	0.199^{***} (0.054)	0.523^{***} (0.054)

Panel B. Annual lag-update specification

	Supply side coeff.				Demand side coeff.			
τ	θ_S^{sym}	β_S^{abs}	$\theta_S^{sym}+\beta_S^{abs}$	$\theta_S^{sym}-\beta_S^{abs}$	θ_D^{sym}	β_D^{abs}	$\theta_D^{sym}+\beta_D^{abs}$	$\theta_D^{sym}-\beta_D^{abs}$
NFC Raw Rate								
1.0	0.130^{***} (0.022)	0.034 (0.025)	0.164^{***} (0.042)	0.096^{***} (0.021)	0.353^{***} (0.048)	-0.184^{***} (0.024)	0.169^{***} (0.043)	0.538^{***} (0.064)
1.5	0.130^{***} (0.022)	0.053^{*} (0.028)	0.183^{***} (0.046)	0.077^{***} (0.021)	0.358^{***} (0.049)	-0.208^{***} (0.029)	0.150^{***} (0.039)	0.566^{***} (0.070)
2.0	0.131^{***} (0.022)	0.067^{**} (0.030)	0.198^{***} (0.049)	0.063^{***} (0.020)	0.362^{***} (0.049)	-0.227^{***} (0.032)	0.135^{***} (0.036)	0.588^{***} (0.075)
NFC Net Rate								
1.0	0.133^{***} (0.022)	0.029 (0.023)	0.162^{***} (0.040)	0.104^{***} (0.021)	0.350^{***} (0.047)	-0.176^{***} (0.022)	0.173^{***} (0.042)	0.526^{***} (0.060)
1.5	0.133^{***} (0.022)	0.049^{*} (0.026)	0.182^{***} (0.044)	0.084^{***} (0.020)	0.354^{***} (0.047)	-0.201^{***} (0.026)	0.153^{***} (0.038)	0.555^{***} (0.066)
2.0	0.133^{***} (0.022)	0.064^{**} (0.028)	0.197^{***} (0.046)	0.069^{***} (0.020)	0.358^{***} (0.048)	-0.221^{***} (0.029)	0.138^{***} (0.034)	0.579^{***} (0.071)

Note: The nonlinear terms are $N_{c,t}^S(\tau) = (|z_{c,t}^S| - \tau)_+$ and $N_{c,t}^D(\tau) = (|z_{c,t}^D| - \tau)_+$. All regressions include country fixed effects, year fixed effects, and the lagged dependent variable. Standard errors are clustered at the country level and reported in parentheses. The superscript *sym* indicates that the linear coefficient is estimated within the symmetric-trigger specification. The superscript *abs* denotes the coefficient on the absolute-value trigger. The columns $\theta_j^{sym}+\beta_j^{abs}$ and $\theta_j^{sym}-\beta_j^{abs}$, for $j \in \{S, D\}$, report positive-tail and negative-tail slopes, respectively. Standard errors and significance levels for these slopes are computed using Wald tests for the corresponding linear restrictions. Significance levels are denoted by $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

The numerator accumulates only negative output responses, while the denominator accumulates the inflation response generated by the shock. The objective of this statistic is to measure the recessionary output cost associated with each unit of cumulative inflation pressure.

The net sacrifice ratio is defined as

$$SR_{net}^H = \frac{\sum_{h=0}^{H} y_{t+h}}{\sum_{h=0}^{H} \pi_{t+h}}.$$

The numerator accumulates the full output response, allowing positive output responses to offset negative ones. The denominator is the same cumulative inflation response used in SR_{rec}^H.

The objective of this statistic is to measure the net output cost associated with each unit of cumulative inflation pressure. If the output GIRF does not cross the zero line over the relevant horizon, then SR^H_{rec} and SR^H_{net} coincide.

Finally, the volatility ratio is defined as

$$SR^H_{vol} = \frac{\sum_{h=0}^{H} y^2_{t+h}}{\sum_{h=0}^{H} \pi^2_{t+h}}.$$

The numerator accumulates squared output responses, while the denominator accumulates squared inflation responses. The objective of this statistic is to summarize the relative magnitude of output fluctuations compared with inflation fluctuations over the same horizon.

B.2. *The Shock-Size Mechanism and the Output-Inflation Trade-Off*

TABLE B.1. Shock-Size Mechanism: Complete Summary Statistics

Period 40	–	$\sum^{40} \pi_t$	$\sum^{40} y_t$	SR^{40}_{rec}	SR^{40}_{net}	SR^{40}_{vol}	
Productivity Shock							
Baseline		1.350	-2.424	-1.796 $(-,+)$	-1.796 $(-,+)$	2.221	
Active Mechanism		1.443	-2.282	-1.582 $(-,+)$	-1.582 $(-,+)$	1.670	
Cost-Push Shock							
Baseline		0.372	-0.716	-1.924 $(-,+)$	-1.924 $(-,+)$	2.364	
Active Mechanism		0.464	-0.576	-1.239 $(-,+)$	-1.239 $(-,+)$	0.854	
Endogenous Convergence	$H^*_{5\%}$	$H^{\pi-\pi_T}_{5\%}$	$\sum^{H} \pi_t$	$\sum^{H} y_t$	SR^H_{rec}	SR^H_{net}	SR^H_{vol}
Productivity Shock							
Baseline	26	26	1.229	-2.210	-1.798 $(-,+)$	-1.798 $(-,+)$	2.214
Active Mechanism	24	27	1.306	-2.036	-1.560 $(-,+)$	-1.560 $(-,+)$	1.659
Cost-Push Shock							
Baseline	32	32	0.350	-0.677	-1.934 $(-,+)$	-1.934 $(-,+)$	2.360
Active Mechanism	25	42	0.422	-0.500	-1.184 $(-,+)$	-1.184 $(-,+)$	0.837

Note: The table reports summary statistics by shock and policy scenario. The variables π_t and y_t denote inflation and output, respectively. The upper panel reports statistics computed over a fixed 40-period horizon. The lower panel reports statistics computed up to the endogenous convergence horizon $H^*_{5\%}$, defined as the first horizon at which inflation falls to at most 5% of its initial response and remains there for four consecutive periods, while $H^{\pi-\pi_T}_{5\%}$ denotes the analogous relative convergence horizon for the inflation gap $(\pi_t - \pi_{T,t})$. The signs in parentheses report the signs of the numerator and denominator of each sacrifice ratio, respectively.

TABLE B.2. Shock-Size Mechanism: Complete Policy-Stance Summary Statistics

Period 40	–	$\sum^{40} i_t$	$\sum^{40} r_t$	$\sum^{40} \mathbb{E}_t \pi_{t+1}$	$\sum^{40} \pi_{T,t}$
Productivity Shock					
Baseline		1.479	0.342	1.137	–
Active Mechanism		1.516	0.309	1.207	0.232
Cost-Push Shock					
Baseline		0.383	0.073	0.310	–
Active Mechanism		0.420	0.040	0.380	0.230
Endogenous Convergence	$H^*_{5\%}$	$\sum^{H^*_{5\%}} i_t$	$\sum^{H^*_{5\%}} r_t$	$\sum^{H^*_{5\%}} \mathbb{E}_t \pi_{t+1}$	$\sum^{H^*_{5\%}} \pi_{T,t}$
Productivity Shock					
Baseline	26	1.342	0.322	1.020	–
Active Mechanism	24	1.361	0.287	1.074	0.232
Cost-Push Shock					
Baseline	32	0.358	0.069	0.289	–
Active Mechanism	25	0.373	0.033	0.340	0.230

Note: The table reports cumulative policy-stance statistics by shock and policy scenario. The variables i_t, r_t, $\mathbb{E}_t \pi_{t+1}$, and $\pi_{T,t}$ denote the nominal interest rate, real interest rate, expected inflation rate, and inflation target, respectively. The upper panel reports statistics computed over a fixed 40-period horizon. The lower panel reports statistics computed up to the endogenous convergence horizon $H^*_{5\%}$, defined as the first horizon at which inflation falls to at most 5% of its initial response and remains there for four consecutive periods.

TABLE B.3. Shock-Size Mechanism: Sectoral Summary Statistics

Period 40	–	$\sum^{40} \pi_{S,t}$	$\sum^{40} \pi_{G,t}$	$\sum^{40} y_{S,t}$	$\sum^{40} y_{G,t}$	$\sum^{40} \pi^W_{S,t}$	$\sum^{40} \pi^W_{G,t}$
Productivity Shock							
Baseline		1.334	1.378	-1.563	-4.434	1.323	1.309
Active Mechanism		1.427	1.472	-1.385	-4.375	1.418	1.403
Cost-Push Shock							
Baseline		0.368	0.380	-0.456	-1.324	0.362	0.358
Active Mechanism		0.460	0.473	-0.280	-1.265	0.456	0.451
Endogenous Convergence	$H^*_{5\%}$	$\sum^{H^*_{5\%}} \pi_{S,t}$	$\sum^{H^*_{5\%}} \pi_{G,t}$	$\sum^{H^*_{5\%}} y_{S,t}$	$\sum^{H^*_{5\%}} y_{G,t}$	$\sum^{H^*_{5\%}} \pi^W_{S,t}$	$\sum^{H^*_{5\%}} \pi^W_{G,t}$
Productivity Shock							
Baseline	26	1.211	1.263	-1.474	-3.926	1.187	1.175
Active Mechanism	24	1.286	1.341	-1.289	-3.780	1.262	1.253
Cost-Push Shock							
Baseline	32	0.345	0.358	-0.436	-1.238	0.337	0.332
Active Mechanism	25	0.417	0.433	-0.245	-1.095	0.408	0.403

Note: The table reports sectoral summary statistics by shock and policy scenario. The subscripts S and G denote services and goods, respectively. The variables $\pi_{s,t}$, $y_{s,t}$, and $\pi^W_{s,t}$ denote sectoral price inflation, sectoral output, and sectoral wage inflation, respectively. The upper panel reports statistics computed over a fixed 40-period horizon. The lower panel reports statistics computed up to the endogenous convergence horizon $H^*_{5\%}$, defined from aggregate inflation as the first horizon at which inflation falls to at most 5% of its initial response and remains there for four consecutive periods.

B.3. *Policy-Horizon Flexibility under High Credibility*

TABLE B.4. Policy-Horizon Flexibility: Complete Summary Statistics

Period 40	–	–	$\sum^{40}\pi_t$	$\sum^{40} y_t$	SR^{40}_{rec}	SR^{40}_{net}	SR^{40}_{vol}
Productivity Shock							
$\rho_p = 0.65$			1.443	-2.282	-1.582 $(-,+)$	-1.582 $(-,+)$	1.670
$\rho_p = 0.70$			1.481	-2.242	-1.514 $(-,+)$	-1.514 $(-,+)$	1.545
$\rho_p = 0.75$			1.542	-2.189	-1.419 $(-,+)$	-1.419 $(-,+)$	1.395
$\rho_p = 0.80$			1.644	-2.114	-1.286 $(-,+)$	-1.286 $(-,+)$	1.218
$\rho_p = 0.85$			1.834	-2.007	-1.094 $(-,+)$	-1.094 $(-,+)$	1.012
Cost-Push Shock							
$\rho_p = 0.65$			0.464	-0.576	-1.239 $(-,+)$	-1.239 $(-,+)$	0.854
$\rho_p = 0.70$			0.503	-0.536	-1.066 $(-,+)$	-1.066 $(-,+)$	0.643
$\rho_p = 0.75$			0.563	-0.483	-0.858 $(-,+)$	-0.858 $(-,+)$	0.443
$\rho_p = 0.80$			0.664	-0.409	-0.616 $(-,+)$	-0.616 $(-,+)$	0.272
$\rho_p = 0.85$			0.852	-0.303	-0.355 $(-,+)$	-0.355 $(-,+)$	0.146
Endogenous Convergence	$H^*_{5\%}$	$H^{\pi-\pi_T}_{5\%}$	$\sum^{H}\pi_t$	$\sum^{H} y_t$	SR^{H}_{rec}	SR^{H}_{net}	SR^{H}_{vol}
Productivity Shock							
$\rho_p = 0.65$	24	27	1.306	-2.036	-1.560 $(-,+)$	-1.560 $(-,+)$	1.659
$\rho_p = 0.70$	23	27	1.335	-1.979	-1.483 $(-,+)$	-1.483 $(-,+)$	1.532
$\rho_p = 0.75$	23	26	1.398	-1.932	-1.382 $(-,+)$	-1.382 $(-,+)$	1.383
$\rho_p = 0.80$	23	24	1.502	-1.869	-1.244 $(-,+)$	-1.244 $(-,+)$	1.208
$\rho_p = 0.85$	25	23	1.709	-1.817	-1.063 $(-,+)$	-1.063 $(-,+)$	1.006
Cost-Push Shock							
$\rho_p = 0.65$	25	42	0.422	-0.500	-1.184 $(-,+)$	-1.184 $(-,+)$	0.837
$\rho_p = 0.70$	23	39	0.455	-0.450	-0.989 $(-,+)$	-0.989 $(-,+)$	0.625
$\rho_p = 0.75$	23	35	0.518	-0.403	-0.779 $(-,+)$	-0.779 $(-,+)$	0.429
$\rho_p = 0.80$	23	28	0.621	-0.341	-0.549 $(-,+)$	-0.549 $(-,+)$	0.264
$\rho_p = 0.85$	27	23	0.820	-0.268	-0.327 $(-,+)$	-0.327 $(-,+)$	0.144

Note: The table reports summary statistics by shock and policy scenario. The variables π_t and y_t denote inflation and output, respectively. The upper panel reports statistics computed over a fixed 40-period horizon. The lower panel reports statistics computed up to the endogenous convergence horizon $H^*_{5\%}$, defined as the first horizon at which inflation falls to at most 5% of its initial response and remains there for four consecutive periods, while $H^{\pi-\pi_T}_{5\%}$ denotes the analogous relative convergence horizon for the inflation gap $(\pi_t - \pi_{T,t})$. The signs in parentheses report the signs of the numerator and denominator of each sacrifice ratio, respectively.

TABLE B.5. Policy-Horizon Flexibility: Complete Policy-Stance Summary Statistics

Period 40	–	$\sum^{40} i_t$	$\sum^{40} r_t$	$\sum^{40} \mathbb{E}_t \pi_{t+1}$	$\sum^{40} \pi_{T,t}$
Productivity Shock					
$\rho_p = 0.65$		1.516	0.309	1.207	0.232
$\rho_p = 0.70$		1.546	0.305	1.241	0.270
$\rho_p = 0.75$		1.597	0.302	1.295	0.324
$\rho_p = 0.80$		1.688	0.299	1.389	0.405
$\rho_p = 0.85$		1.865	0.297	1.568	0.540
Cost-Push Shock					
$\rho_p = 0.65$		0.420	0.040	0.380	0.230
$\rho_p = 0.70$		0.450	0.037	0.413	0.268
$\rho_p = 0.75$		0.500	0.033	0.467	0.321
$\rho_p = 0.80$		0.591	0.030	0.561	0.402
$\rho_p = 0.85$		0.766	0.028	0.738	0.534
Endogenous Convergence	$H^*_{5\%}$	$\sum^{H^*_{5\%}} i_t$	$\sum^{H^*_{5\%}} r_t$	$\sum^{H^*_{5\%}} \mathbb{E}_t \pi_{t+1}$	$\sum^{H^*_{5\%}} \pi_{T,t}$
Productivity Shock					
$\rho_p = 0.65$	24	1.361	0.287	1.074	0.232
$\rho_p = 0.70$	23	1.381	0.283	1.098	0.270
$\rho_p = 0.75$	23	1.435	0.280	1.155	0.323
$\rho_p = 0.80$	23	1.529	0.277	1.252	0.401
$\rho_p = 0.85$	25	1.726	0.278	1.448	0.527
Cost-Push Shock					
$\rho_p = 0.65$	25	0.373	0.033	0.340	0.230
$\rho_p = 0.70$	23	0.397	0.030	0.367	0.268
$\rho_p = 0.75$	23	0.450	0.027	0.423	0.320
$\rho_p = 0.80$	23	0.543	0.024	0.519	0.398
$\rho_p = 0.85$	27	0.731	0.024	0.707	0.526

Note: The table reports cumulative policy-stance statistics by shock and policy scenario. The variables i_t, r_t, $\mathbb{E}_t \pi_{t+1}$, and $\pi_{T,t}$ denote the nominal interest rate, real interest rate, expected inflation rate, and inflation target, respectively. The upper panel reports statistics computed over a fixed 40-period horizon. The lower panel reports statistics computed up to the endogenous convergence horizon $H^*_{5\%}$, defined as the first horizon at which inflation falls to at most 5% of its initial response and remains there for four consecutive periods.

B.4. *The Cost of Low Credibility*

TABLE B.6. Cost of Low Credibility: Complete Summary Statistics

Period 40	–	–	$\sum^{40} \pi_t$	$\sum^{40} y_t$	SR^{40}_{rec}	SR^{40}_{net}	SR^{40}_{vol}
Productivity Shock							
$k_g = 0.15$			1.397	-2.368	-1.695 $(-,+)$	-1.695 $(-,+)$	1.985
$k_g = 0.25$			1.412	-2.344	-1.660 $(-,+)$	-1.660 $(-,+)$	1.893
$k_g = 0.50$			1.431	-2.310	-1.614 $(-,+)$	-1.614 $(-,+)$	1.764
$k_g = 0.75$			1.439	-2.291	-1.592 $(-,+)$	-1.592 $(-,+)$	1.701
$k_g = 0.95$			1.443	-2.282	-1.582 $(-,+)$	-1.582 $(-,+)$	1.670
Cost-Push Shock							
$k_g = 0.15$			0.419	-0.660	-1.575 $(-,+)$	-1.575 $(-,+)$	1.592
$k_g = 0.25$			0.434	-0.637	-1.466 $(-,+)$	-1.466 $(-,+)$	1.346
$k_g = 0.50$			0.453	-0.603	-1.330 $(-,+)$	-1.330 $(-,+)$	1.046
$k_g = 0.75$			0.461	-0.585	-1.268 $(-,+)$	-1.268 $(-,+)$	0.914
$k_g = 0.95$			0.464	-0.576	-1.239 $(-,+)$	-1.239 $(-,+)$	0.854
Endogenous Convergence	$H^*_{5\%}$	$H^{\pi-\pi_T}_{5\%}$	$\sum^{H} \pi_t$	$\sum^{H} y_t$	SR^{H}_{rec}	SR^{H}_{net}	SR^{H}_{vol}
Productivity Shock							
$k_g = 0.15$	25	29	1.268	-2.137	-1.686 $(-,+)$	-1.686 $(-,+)$	1.976
$k_g = 0.25$	25	29	1.284	-2.115	-1.647 $(-,+)$	-1.647 $(-,+)$	1.884
$k_g = 0.50$	24	28	1.293	-2.062	-1.595 $(-,+)$	-1.595 $(-,+)$	1.753
$k_g = 0.75$	24	28	1.302	-2.045	-1.571 $(-,+)$	-1.571 $(-,+)$	1.689
$k_g = 0.95$	24	27	1.306	-2.036	-1.560 $(-,+)$	-1.560 $(-,+)$	1.659
Cost-Push Shock							
$k_g = 0.15$	29	48	0.388	-0.606	-1.559 $(-,+)$	-1.559 $(-,+)$	1.581
$k_g = 0.25$	28	47	0.401	-0.577	-1.440 $(-,+)$	-1.440 $(-,+)$	1.334
$k_g = 0.50$	26	44	0.414	-0.532	-1.287 $(-,+)$	-1.287 $(-,+)$	1.030
$k_g = 0.75$	25	43	0.418	-0.508	-1.215 $(-,+)$	-1.215 $(-,+)$	0.897
$k_g = 0.95$	25	42	0.422	-0.500	-1.184 $(-,+)$	-1.184 $(-,+)$	0.837

Note: The table reports summary statistics by shock and credibility scenario. The variables π_t and y_t denote inflation and output, respectively. The upper panel reports statistics computed over a fixed 40-period horizon. The lower panel reports statistics computed up to the endogenous convergence horizon $H^*_{5\%}$, defined as the first horizon at which inflation falls to at most 5% of its initial response and remains there for four consecutive periods, while $H^{\pi-\pi_T}_{5\%}$ denotes the analogous relative convergence horizon for the inflation gap $(\pi_t - \pi_{T,t})$. The signs in parentheses report the signs of the numerator and denominator of each sacrifice ratio, respectively. Complete statistics are reported in Table B.6 in Appendix B.

TABLE B.7. Cost of Low Credibility: Complete Policy-Stance Summary Statistics

Period 40	–	$\sum^{40} i_t$	$\sum^{40} r_t$	$\sum^{40} \mathbb{E}_t \pi_{t+1}$	$\sum^{40} \pi_{T,t}$
Productivity Shock					
$k_g = 0.15$		1.512	0.336	1.176	0.232
$k_g = 0.25$		1.519	0.331	1.188	0.232
$k_g = 0.50$		1.522	0.322	1.200	0.232
$k_g = 0.75$		1.519	0.314	1.205	0.232
$k_g = 0.95$		1.516	0.309	1.207	0.232
Cost-Push Shock					
$k_g = 0.15$		0.416	0.066	0.350	0.230
$k_g = 0.25$		0.423	0.062	0.361	0.230
$k_g = 0.50$		0.426	0.053	0.373	0.230
$k_g = 0.75$		0.423	0.045	0.378	0.230
$k_g = 0.95$		0.420	0.040	0.380	0.230
Endogenous Convergence	$H^*_{5\%}$	$\sum^{H^*_{5\%}} i_t$	$\sum^{H^*_{5\%}} r_t$	$\sum^{H^*_{5\%}} \mathbb{E}_t \pi_{t+1}$	$\sum^{H^*_{5\%}} \pi_{T,t}$
Productivity Shock					
$k_g = 0.15$	25	1.365	0.314	1.051	0.232
$k_g = 0.25$	25	1.373	0.310	1.063	0.232
$k_g = 0.50$	24	1.366	0.299	1.067	0.232
$k_g = 0.75$	24	1.364	0.292	1.072	0.232
$k_g = 0.95$	24	1.361	0.287	1.074	0.232
Cost-Push Shock					
$k_g = 0.15$	29	0.381	0.061	0.320	0.230
$k_g = 0.25$	28	0.385	0.056	0.329	0.230
$k_g = 0.50$	26	0.382	0.046	0.336	0.230
$k_g = 0.75$	25	0.376	0.038	0.338	0.230
$k_g = 0.95$	25	0.373	0.033	0.340	0.230

Note: The table reports cumulative policy-stance statistics by shock and credibility scenario. The variables i_t, r_t, $\mathbb{E}_t \pi_{t+1}$, and $\pi_{T,t}$ denote the nominal interest rate, real interest rate, expected inflation rate, and inflation target, respectively. The upper panel reports statistics computed over a fixed 40-period horizon. The lower panel reports statistics computed up to the endogenous convergence horizon $H^*_{5\%}$, defined as the first horizon at which inflation falls to at most 5% of its initial response and remains there for four consecutive periods.

B.5. *Horizon Flexibility under Credibility Stress*

TABLE B.8. Horizon Flexibility and Credibility Stress: Complete Summary Statistics

Period 40	–	–	$\sum^{40}\pi_t$	$\sum^{40} y_t$	SR^{40}_{rec}	SR^{40}_{net}	SR^{40}_{vol}
Productivity Shock							
High Credibility and Short Horizon			1.443	-2.282	-1.582 $(-,+)$	-1.582 $(-,+)$	1.670
Low Credibility and Extended Horizon			1.421	-2.349	-1.653 $(-,+)$	-1.653 $(-,+)$	1.914
Cost-Push Shock							
High Credibility and Short Horizon			0.464	-0.576	-1.239 $(-,+)$	-1.239 $(-,+)$	0.854
Low Credibility and Extended Horizon			0.443	-0.641	-1.449 $(-,+)$	-1.449 $(-,+)$	1.400
Endogenous Convergence	$H^*_{5\%}$	$H^{\pi-\pi_T}_{5\%}$	$\sum^{H}\pi_t$	$\sum^{H} y_t$	SR^{H}_{rec}	SR^{H}_{net}	SR^{H}_{vol}
Productivity Shock							
High Credibility and Short Horizon	24	27	1.306	-2.036	-1.560 $(-,+)$	-1.560 $(-,+)$	1.659
Low Credibility and Extended Horizon	25	29	1.293	-2.120	-1.640 $(-,+)$	-1.640 $(-,+)$	1.905
Cost-Push Shock							
High Credibility and Short Horizon	25	42	0.422	-0.500	-1.184 $(-,+)$	-1.184 $(-,+)$	0.837
Low Credibility and Extended Horizon	28	47	0.409	-0.582	-1.422 $(-,+)$	-1.422 $(-,+)$	1.388

Note: The table reports summary statistics by shock and credibility-horizon scenario. The variables π_t and y_t denote inflation and output, respectively. The upper panel reports statistics computed over a fixed 40-period horizon. The lower panel reports statistics computed up to the endogenous convergence horizon $H^*_{5\%}$, defined as the first horizon at which inflation falls to at most 5% of its initial response and remains there for four consecutive periods, while $H^{\pi-\pi_T}_{5\%}$ denotes the analogous relative convergence horizon for the inflation gap $(\pi_t - \pi_{T,t})$. The signs in parentheses report the signs of the numerator and denominator of each sacrifice ratio, respectively. Complete statistics are reported in Table B.8 in Appendix B.

TABLE B.9. Horizon Flexibility and Credibility Stress: Complete Policy-Stance Summary Statistics

Period 40	–	$\sum^{40} i_t$	$\sum^{40} r_t$	$\sum^{40}\mathbb{E}_t\pi_{t+1}$	$\sum^{40}\pi_{T,t}$
Productivity Shock					
$k_g = 0.95,\ \rho_p = 0.65$		1.516	0.309	1.207	0.232
$k_g = 0.15,\ \rho_p = 0.72$		1.533	0.335	1.198	0.270
Cost-Push Shock					
$k_g = 0.95,\ \rho_p = 0.65$		0.420	0.040	0.380	0.230
$k_g = 0.15,\ \rho_p = 0.72$		0.437	0.066	0.371	0.268
Endogenous Convergence	$H^*_{5\%}$	$\sum^{H^*_{5\%}} i_t$	$\sum^{H^*_{5\%}} r_t$	$\sum^{H^*_{5\%}}\mathbb{E}_t\pi_{t+1}$	$\sum^{H^*_{5\%}}\pi_{T,t}$
Productivity Shock					
$k_g = 0.95,\ \rho_p = 0.65$	24	1.361	0.287	1.074	0.232
$k_g = 0.15,\ \rho_p = 0.72$	25	1.387	0.314	1.073	0.270
Cost-Push Shock					
$k_g = 0.95,\ \rho_p = 0.65$	25	0.373	0.033	0.340	0.230
$k_g = 0.15,\ \rho_p = 0.72$	28	0.399	0.060	0.339	0.268

Note: The table reports cumulative policy-stance statistics by shock and credibility-horizon scenario. The variables i_t, r_t, $\mathbb{E}_t\pi_{t+1}$, and $\pi_{T,t}$ denote the nominal interest rate, real interest rate, expected inflation rate, and inflation target, respectively. The upper panel reports statistics computed over a fixed 40-period horizon. The lower panel reports statistics computed up to the endogenous convergence horizon $H^*_{5\%}$, defined as the first horizon at which inflation falls to at most 5% of its initial response and remains there for four consecutive periods.

Appendix C. Model

C.1. *Problem of the Firm*

C.1.1. *Sectoral price aggregator*

The sectoral price aggregator is given by

$$P_{s,t}^{1-\epsilon_s} = \int_0^1 P_{s,t}(j)^{1-\epsilon_s}\,\mathrm{d}\,j. \tag{C.1}$$

Given that the sectoral probability of not adjusting prices α_s can be interpreted as the proportion of firms that cannot change prices in a given period, the integer can be partitioned

$$\begin{aligned}
P_{s,t}^{1-\epsilon_s} &= \int_0^1 P_{s,t}(j)^{1-\epsilon_s}\,\mathrm{d}\,j,\\
&= \int_0^{\alpha_s} P_{s,t-1}(j)^{1-\epsilon_s}\,\mathrm{d}\,j \;+\; \int_{\alpha_s}^1 P_{s,t}^C(j)^{1-\epsilon_s}\,\mathrm{d}\,j,\\
&= \int_0^{\alpha_s} P_{s,t-1}(j)^{1-\epsilon_s}\,\mathrm{d}\,j \;+\; (1-\alpha_s)\,(P_{s,t}^C)^{1-\epsilon_s},\\
&= \alpha_s \cdot P_{s,t-1}^{1-\epsilon_s} + (1-\alpha_s)\cdot (P_{s,t}^C)^{1-\epsilon_s},\\
\left(\frac{P_{s,t}}{P_{s,t-1}}\right)^{1-\epsilon_s} &= \alpha_s \cdot \left(\frac{P_{s,t-1}}{P_{s,t-1}}\right)^{1-\epsilon_s} + (1-\alpha_s)\cdot\left(\frac{P_{s,t}^C}{P_{s,t-1}}\right)^{1-\epsilon_s},\\
\Pi_{s,t}^{1-\epsilon_s} &= \alpha_s \;+\; (1-\alpha_s)\cdot\left(\frac{P_{s,t}^C}{P_{s,t-1}}\right)^{1-\epsilon_s}. \qquad \text{(C.2)}
\end{aligned}$$

Log-linearizing the RHS of equation (C.2)

$$\begin{aligned}
\Pi_{s,t}^{1-\epsilon_s} &\approx \Pi^{1-\epsilon_s} + (1-\epsilon_s)\,\Pi^{-\epsilon_s}\cdot(\Pi_{s,t}-\Pi)\,,\\
&\approx 1 + (1-\epsilon_s)\cdot(\Pi_{s,t}-1),\\
&\approx 1 + (1-\epsilon_s)\cdot(1+\pi_{s,t}-1),\\
&\approx 1 + (1-\epsilon_s)\cdot\pi_{s,t}. \qquad \text{(C.3)}
\end{aligned}$$

Note that $\Pi_{s,t} = 1+\pi_{s,t}$, where $\Pi_{s,t}$ is the gross inflation rate and $\pi_{s,t}$ is the net inflation rate. Additionally, we have used that the steady state value of the gross inflation rate is one. Log-linearizing the LHS of equation (C.2),

$$\begin{aligned}
\alpha_s \;+\; (1-\alpha_s)\cdot\left(\frac{P_{s,t}^C}{P_{s,t-1}}\right)^{1-\epsilon_s} &\approx \left(\alpha_s + (1-\alpha_s)\left(\frac{P_s}{P_s}\right)^{1-\epsilon_s}\right)\\
&\quad + \left((1-\alpha_s)\,(1-\epsilon_s)\,\left(\frac{P_s}{P_s}\right)^{-\epsilon_s}\,\frac{1}{P_s}\cdot(P_{s,t}^C - P_s)\right)\\
&\quad + \left((1-\alpha_s)\,(1-\epsilon_s)\,\left(\frac{P_s}{P_s}\right)^{-\epsilon_s}\,P_s\,(-1)\,\frac{1}{P_s^2}\cdot(P_{s,t-1}-P_s)\right),\\
&\approx (\alpha_s + (1-\alpha_s)\,) + (1-\alpha_s)\,(1-\epsilon_s)\cdot\left[\left(\frac{P_{s,t}^C - P_s}{P_s}\right) - \left(\frac{P_{s,t-1}-P_s}{P_s}\right)\right],\\
&\approx 1 + (1-\alpha_s)\,(1-\epsilon_s)\cdot(p_{s,t}^C - p_{s,t-1}). \qquad \text{(C.4)}
\end{aligned}$$

Combining both results[34],

$$\begin{aligned}
1 + (1-\epsilon_s)\cdot\pi_{s,t} &= 1 + (1-\alpha_s)\,(1-\epsilon_s)\cdot(p_{s,t}^C - p_{s,t-1}),\\
(1-\epsilon_s)\,\pi_{s,t} &= (1-\alpha_s)\,(1-\epsilon_s)\cdot(p_{s,t}^C - p_{s,t-1}),\\
\pi_{s,t} &= (1-\alpha_s)\cdot(p_{s,t}^C - p_{s,t-1}). \qquad \text{(C.5)}
\end{aligned}$$

with the later as the sectoral inflation, resulted by the fact that there exists a proportion of firms that changes prices, to their optimum onesc and with an indexed rule, and others that do not.

[34]Lower case variables represent the first-order expansion around the steady state of the upper case variable.

C.1.2. *Forward-looking firms*

The Lagrangian that resumes the optimization problem of the forward-looking firm j of sector s at period t is given by

$$\mathcal{L}: \sum_{l=0}^{\infty} Q_{t,\,t+l}\,\alpha_s^l \left[\Big(P_{s,\,t}(j) - MC_{s,\,t+l}(j)\Big)\cdot\left(\left[\frac{P_{s,\,t}(j)}{P_{t+l}}\right]^{-\frac{1+\theta_s}{\theta_s}}\cdot Y_{s\,t+l}\right)\right]. \tag{C.6}$$

This problem yields the following First Order Condition (the firm chooses the price that maximizes equation (C.6)),

$$\begin{aligned}
&\mathbb{E}_t \sum_{l=0}^{\infty} Q_{t,\,t+l}\,\alpha_s^l\Bigg[\left(\frac{1}{P_{t+l}}\right)^{-\frac{1+\theta}{\theta}}\cdot Y_{s,\,t+l}\cdot\left(-\frac{1}{\theta_s}\right)\cdot(P^*_{s,\,t}(j))^{-\frac{1}{\theta_s}-1}\\
&\qquad -MC_{s,\,t+l}(j)\cdot\left(\frac{1}{P_{t+l}}\right)^{-\frac{1+\theta_s}{\theta_s}}\cdot Y_{s,\,t+l}\cdot\left(-\frac{1+\theta_s}{\theta_s}\right)\cdot\frac{P^*_{s,\,t}(j)^{-\frac{1+\theta_s}{\theta_s}}}{P^*_{s,\,t}(j)}\Bigg]=0,\\
&\mathbb{E}_t \sum_{l=0}^{\infty} Q_{t,\,t+l}\,\alpha_s^l\Bigg[\left(-\frac{1}{\theta_s}\right)-\left(-\frac{1+\theta_s}{\theta_s}\right)\cdot MC_{s,\,t+l}(j)\cdot\frac{1}{P^*_{s,\,t}(j)}\Bigg]Y_{s,\,t+l}(j)=0,\\
&\mathbb{E}_t \sum_{l=0}^{\infty} Q_{t,\,t+l}\,\alpha_s^l\Bigg[\left(\frac{1}{1+\theta_s}\right)-MC_{s,\,t+l}(j)\cdot\frac{1}{P^*_{s,\,t}(j)}\Bigg]Y_{s,\,t+l}(j)=0,\\
&\mathbb{E}_t \sum_{l=0}^{\infty} Q_{t,\,t+l}\,\alpha_s^l\Bigg[\left(\frac{P^*_{s,\,t}(j)}{1+\theta_s}\right)-MC_{s,\,t+l}(j)\Bigg]\cdot Y_{s,\,t+l}(j)=0.
\end{aligned} \tag{C.7}$$

Solving for prices,

$$\begin{aligned}
&\mathbb{E}_t \sum_{l=0}^{\infty} Q_{t,\,t+l}\,\alpha_s^l\Bigg[\left(\frac{P^*_{s,\,t}(j)}{1+\theta_s}\right)-MC_{s,\,t+l}(j)\Bigg]Y_{s,\,t+l}(j)=0,\\
&\mathbb{E}_t \sum_{l=0}^{\infty} Q_{t,\,t+l}\,\alpha_s^l\Bigg[\left(\frac{P^*_{s,\,t}(j)}{1+\theta_s}\right)\cdot Y_{s,\,t+l}(j)\Bigg]=\mathbb{E}_t\sum_{l=0}^{\infty} Q_{t,\,t+l}\,\alpha_s^l\Big[MC_{s,\,t+l}(j)\Big]Y_{s,\,t+l}(j),\\
&\frac{P_{s,\,t}(j)^*}{1+\theta_s}\mathbb{E}_t \sum_{l=0}^{\infty} Q_{t,\,t+l}\,\alpha_s^l\Big[Y_{s,\,t+l}(j)\Big]=\mathbb{E}_t \sum_{l=0}^{\infty} Q_{t,\,t+l}\,\alpha_s^l\Big[MC_{s,\,t+l}(j)\Big]Y_{s,\,t+l}(j),\\
&P^*_{s,\,t}(j)=(1+\theta_s)\cdot\frac{\mathbb{E}_t\sum_{l=0}^{\infty} Q_{t,\,t+l}\cdot\alpha_s^l\cdot MC_{s,\,t+l}(j)\cdot Y_{s,\,t+l}(j)}{\mathbb{E}_t\sum_{l=0}^{\infty} Q_{t,\,t+l}\cdot\alpha_s^l\cdot Y_{s,\,t+l}(j)},\\
&P^*_{s,\,t}(j)=(1+\theta_s)\cdot\frac{\mathbb{E}_t\sum_{l=0}^{\infty}(\beta\,\alpha_s)^l\cdot\frac{U_{c,\,t+l}}{U_{c,\,t}}\cdot\frac{P_t}{P_{t+l}}\cdot MC_{s,\,t+l}(j)\cdot Y_{s,\,t+l}(j)}{\mathbb{E}_t\sum_{l=0}^{\infty}(\beta\,\alpha_s)^l\cdot\frac{U_{c,\,t+l}}{U_{c,\,t}}\cdot\frac{P_t}{P_{t+l}}\cdot Y_{s,\,t+l}(j)}.
\end{aligned} \tag{C.8}$$

In which we have used that the Stochastic Discount Factor, $Q_{t,t+l}$, is given by

$$Q_{t,t+l}=\beta^l\,\mathbb{E}_t\left[\frac{U_{c,\,t+l}}{U_{c,\,t}}\,\frac{P_t}{P_{t+l}}\right]$$

Defining the following auxiliary expression

$$P^*_{s,\,t}(j)=(1+\theta_s)\cdot\frac{X_{1,\,t}}{X_{2,\,t}}. \tag{C.9}$$

with

$$X_{1,\,t}=\mathbb{E}_t\sum_{l=0}^{\infty}(\beta\,\alpha_s)^l\cdot U_{c,\,t+l}\cdot P_{t+l}^{-1}\cdot MC_{s,\,t+l}(j)\cdot Y_{s,\,t+l}(j). \tag{C.10}$$

$$X_{2,\,t}=\mathbb{E}_t\sum_{l=0}^{\infty}(\beta\,\alpha_s)^l\cdot U_{c,\,t+l}\cdot P_{t+l}^{-1}\cdot Y_{s,\,t+l}(j). \tag{C.11}$$

The log-linearized form of (C.9) is given by

$$\frac{\mathrm{d}\,P^*_{s,\,t}(j)}{P^*_s(j)}=\frac{\mathrm{d}\,X_{1,\,t}}{X_1}-\frac{\mathrm{d}\,X_{2,\,t}}{X_2}\Longleftrightarrow p^*_{s,\,t}(j)=x_{1,\,t}-x_{2,\,t}. \tag{C.12}$$

Note that the variable x_t represents the log-differenced variable X_t. Differentiating the first term in equation (C.12)

$$\mathrm{d}\,X_{1,\,t}=\mathbb{E}_t\sum_{l=0}^{\infty}(\beta\,\alpha_s)^l\Big\{U_c\,P^{-1}\,MC_s(j)\,Y_s(j)\times \tag{C.13}$$

$$\left(\frac{\mathrm{d}\ MC_{s,\,t+l}(j)}{MC_s(j)} + \frac{\mathrm{d}\ U_{c,\,t+l}\, P_{t+l}^{-1}\, Y_{s,\,t+l}(j)}{U_c\, P^{-1}\, Y_s(j)}\right)\Bigg\}.$$

Note that in equation (C.13) we have isolated the term $\frac{\mathrm{d}\ U_{c,\,t+l}\, P_{t+l}^{-1}\, MC_{s,\,t+l}(j)\, Y_{s,\,t+l}(j)}{U_c\, P^{-1}\, MC_s(j)\, Y_s(j)}$), procedure that will be useful when we form the complete log-linearized expression. The steady state value for $X_{1,\,t}$

$$\begin{aligned} X_1 &= \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l \cdot U_c \cdot P^{-1} \cdot MC_s(j) \cdot Y_s(j), \\ &= \frac{1}{1-\beta\,\alpha_s} \cdot U_c \cdot P^{-1} \cdot MC_s(j) \cdot Y_s(j). \end{aligned}$$

Thus,

$$\begin{aligned} \frac{\mathrm{d}\ X_{1,\,t}}{X_1} &= \frac{1}{X_1} \cdot \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l \left\{ U_c\, P^{-1}\, MC_s(j)\, Y_s(j) \cdot \left(\frac{\mathrm{d}\ MC_{s,\,t+l}(j)}{MC_s(j)} + \frac{\mathrm{d}\ U_{c,\,t+l}\, P_{t+l}^{-1}\, Y_{s,\,t+l}(j)}{U_c\, P^{-1}\, Y_s(j)}\right)\right\}, \\ &= \frac{U_c\, P^{-1}\, MC_s(j)\, Y_s(j)}{(1-\beta\alpha_s)^{-1} U_c\, P^{-1}\, MC_s(j)\, Y_s(j)} \cdot \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l \left(\frac{\mathrm{d}\ MC_{s,\,t+l}(j)}{MC_s(j)} + \frac{\mathrm{d}\, U_{c,\,t+l}\, P_{t+l}^{-1}\, Y_{s,\,t+l}(j)}{U_c\, P^{-1}\, Y_s(j)}\right), \\ &= (1-\beta\,\alpha_s) \cdot \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l \left(\frac{\mathrm{d}\ MC_{s,\,t+l}(j)}{MC_s(j)} + \frac{\mathrm{d}\, U_{c,\,t+l}\, P_{t+l}^{-1}\, Y_{s,\,t+l}(j)}{U_c\, P^{-1}\, Y_s(j)}\right), \\ &= (1-\beta\,\alpha_s) \cdot \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l \left(mc_{s,\,t+l}(j) + \frac{\mathrm{d}\ U_{c,\,t+l}\, P_{t+l}^{-1}\, Y_{s,\,t+l}(j)}{U_c\, P^{-1}\, Y_s(j)}\right), \end{aligned}$$

$$= (1-\beta\,\alpha_s) \cdot \left(\mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l\, mc_{s,\,t+l}(j) + \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l\, \frac{\mathrm{d}\ U_{c,\,t+l}\, P_{t+l}^{-1}\, Y_{s,\,t+l}(j)}{U_c\, P^{-1}\, Y_s(j)}\right). \tag{C.14}$$

If we now turn to the second term of the equation (C.12)

$$\mathrm{d}\ X_{2,\,t} = \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l \left\{ U_c\, P^{-1}\, Y_s(j) \cdot \left(\frac{\mathrm{d}\ U_{c,\,t+l}\, P_{t+l}^{-1}\, Y_{s,\,t+l}(j)}{U_c\, P^{-1}\, Y_s(j)}\right)\right\}. \tag{C.15}$$

with the following steady state for $X_{2\,t}$

$$\begin{aligned} X_2 &= \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l \cdot U_c \cdot P^{-1} \cdot Y(j), \\ &= \frac{1}{1-\beta\,\alpha_s} \cdot U_c \cdot P^{-1} \cdot Y(j). \end{aligned}$$

which implies that

$$\begin{aligned} \frac{\mathrm{d}\ X_{2,\,t}}{X_2} &= \frac{1}{X_2} \cdot \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l \left\{ U_c\, P^{-1}\, Y_s(j) \cdot \left(\frac{\mathrm{d}\ U_{c,\,t+l}\, P_{t+l}^{-1}\, Y_{s,\,t+l}(j)}{U_c\, P^{-1}\, Y_s(j)}\right)\right\}, \\ &= \frac{U_c\, P^{-1}\, Y_s(j)}{(1-\beta\alpha_s)^{-1} U_c\, P^{-1}\, Y_s(j)} \cdot \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l \left(\frac{\mathrm{d}\ U_{c,\,t+l}\, P_{t+l}^{-1}\, Y_{s,\,t+l}(j)}{U_c\, P^{-1}\, Y_s(j)}\right), \\ &= (1-\beta\,\alpha_s) \cdot \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l \left(\frac{\mathrm{d}\ U_{c,\,t+l}\, P_{t+l}^{-1}\, Y_{s,\,t+l}(j)}{U_c\, P^{-1}\, Y_s(j)}\right). \end{aligned} \tag{C.16}$$

Note that (C.16) can be replaced in equation (C.14).

$$\begin{aligned} \frac{\mathrm{d}\ X_{1,\,t}}{X_1} &= (1-\beta\,\alpha_s) \cdot \Bigg(\mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l\, mc_{s,\,t+l}(j) \\ &\qquad + \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l\, \frac{\mathrm{d}\ U_{c,\,t+l}\, P_{t+l}^{-1}\, Y_{s,\,t+l}(j)}{U_c\, P^{-1}\, Y_s(j)}\Bigg), \\ &= (1-\beta\,\alpha_s) \cdot \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l\, mc_{s,\,t+l}(j) \\ &\qquad + (1-\beta\,\alpha_s) \cdot \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l\, \frac{\mathrm{d}\ U_{c,\,t+l}\, P_{t+l}^{-1}\, Y_{s,\,t+l}(j)}{U_c\, P^{-1}\, Y_s(j)}, \\ &= (1-\beta\,\alpha_s) \cdot \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s)^l\, mc_{s,\,t+l}(j) + \frac{\mathrm{d}\, X_{2,\,t}}{X_2}, \end{aligned}$$

$$x_{1,t} = (1-\beta\,\alpha_s)\cdot\mathbb{E}_t\sum_{l=0}^{\infty}(\beta\,\alpha_s)^l\,mc_{s,\,t+l}(j)+x_{2,\,t}. \tag{C.17}$$

Combining the previous result, equation (C.17), with equation (C.12), we obtain

$$p^*_{s,\,t}(j) = (1-\beta\,\alpha_s)\cdot\mathbb{E}_t\sum_{l=0}^{\infty}(\beta\,\alpha_s)^l\,mc_{s,\,t+l}(j). \tag{C.18}$$

This implies that the steady-state change of optimal prices is a weighted average of the expected variations of the present and future costs. Working with (C.18)

$$\begin{aligned}
p^*_{s,\,t}(j) &= (1-\beta\,\alpha_s)\cdot\mathbb{E}_t\sum_{l=0}^{\infty}(\beta\,\alpha_s)^l\,mc_{s,\,t+l}(j),\\
&= (1-\beta\,\alpha_s)\cdot\left(mc_{s,t+l}+\mathbb{E}_t\sum_{l=1}^{\infty}(\beta\,\alpha_s)^l\,mc_{s,\,t+l}(j)\right),\\
&= (1-\beta\,\alpha_s)\cdot\left(mc_{s,t+l}+\left((\beta\,\alpha_s)\,mc_{s,\,t+1}(j)+(\beta\,\alpha_s)^2\,mc_{s,\,t+2}(j)+\ldots\right)\right),\\
&= (1-\beta\,\alpha_s)\cdot\left(mc_{s,t+l}+(\beta\,\alpha_s)\,\left(mc_{s,\,t+1}(j)+(\beta\,\alpha_s)\,mc_{s,\,t+2}(j)+(\beta\,\alpha_s)^2\,mc_{s,\,t+3}(j)+\ldots\right)\right),\\
&= (1-\beta\,\alpha_s)\cdot\left(mc_{s,t+l}+(\beta\,\alpha_s)\,\mathbb{E}_t\sum_{l=0}(\beta\,\alpha_s)^l\,mc_{s,\,t+l+1}\right),\\
&= (1-\beta\,\alpha_s)\cdot mc_{s,t+l}+(1-\beta\,\alpha_s)\cdot(\beta\,\alpha_s)\cdot\mathbb{E}_t\sum_{l=0}(\beta\,\alpha_s)^l\,mc_{s,\,t+l+1},\\
&= (1-\beta\,\alpha_s)\cdot mc_{s,t+l}+(\beta\,\alpha_s)\cdot\left((1-\beta\,\alpha_s)\cdot\mathbb{E}_t\sum_{l=0}(\beta\,\alpha_s)^l\,mc_{s,\,t+l+1}\right),\\
&= (1-\beta\,\alpha_s)\cdot mc_{s,t+l}+(\beta\,\alpha_s)\cdot\mathbb{E}_t\,p^*_{s,\,t+1}(j),
\end{aligned}$$

$$p^*_{s,\,t} = (1-\beta\,\alpha_s)\cdot mc_{s,t+l}+(\beta\,\alpha_s)\cdot\mathbb{E}_t\,p^*_{s,\,t+1}. \tag{C.19}$$

In which we have used for the final expression symmetric results for the forward-looking firm.

C.1.3. *Hybrid sectoral Phillips curve*

We know that the system of equations that determines the problem of the firm is given by

$$\pi_{s,\,t} = (1-\alpha_s)\cdot\left(p^C_{s,\,t}-p_{s,\,t-1}\right), \tag{C.20}$$

$$p^C_{s,\,t} = (1-\omega_s)\cdot p^*_{s,\,t}+\omega_s\cdot p^b_{s,t}, \tag{C.21}$$

$$p^b_{s,\,t} = p^C_{s,\,t-1}+l_s\cdot\pi_{t-1}, \tag{C.22}$$

$$p^*_{s,\,t} = (1-\beta\cdot\alpha_s)\cdot mc_{s,\,t}+(\beta\cdot\alpha_s)\cdot\mathbb{E}_t p^*_{s,\,t+1}. \tag{C.23}$$

Subtracting $p_{s,\,t-1}$ equation (C.21)

$$\begin{aligned}
p^C_{s,\,t} &= (1-\omega_s)\cdot p^*_{s,\,t}+(\omega_s)\cdot p^b_{s,t},\\
p^C_{s,\,t}-p_{s,\,t-1} &= (1-\omega_s)\cdot p^*_{s,\,t}+(\omega_s)\cdot p^b_{s,t}-(1-\omega_s)\cdot p_{s,\,t-1}-(\omega_s)\cdot p_{s,\,t-1},\\
p^C_{s,\,t}-p_{s,\,t-1} &= (1-\omega_s)\cdot\left(p^*_{s,\,t}-p_{s,\,t-1}\right)+(\omega_s)\cdot\left(p^b_{s,t}-p_{s,\,t-1}\right).
\end{aligned} \tag{C.24}$$

Substituting equation (C.22) in the previous result

$$\begin{aligned}
p^C_{s,\,t}-p_{s,\,t-1} &= (1-\omega_s)\cdot\left(p^*_{s,\,t}-p_{s,\,t-1}\right)+(\omega_s)\cdot\left(p^b_{s,t}-p_{s,\,t-1}\right),\\
p^C_{s,\,t}-p_{s,\,t-1} &= (1-\omega_s)\cdot\left(p^*_{s,\,t}-p_{s,\,t-1}\right)+(\omega_s)\cdot\left(p^C_{s,\,t-1}+(l_s)\cdot\pi_{t-1}-p_{s,\,t-1}\right),\\
p^C_{s,\,t}-p_{s,\,t-1} &= (1-\omega_s)\cdot\left(p^*_{s,\,t}-p_{s,\,t-1}\right)+(\omega_s)\cdot\left(p^C_{s,\,t-1}-p_{s,\,t-1}\right)+(\omega_s\cdot l_s)\cdot\pi_{t-1}.
\end{aligned} \tag{C.25}$$

Defining the following auxiliary term $x_{s,\,t}=p^*_{s,\,t}-p_{s,\,t}$, thus

$$\begin{aligned}
p^*_{s,\,t}-p_{s,\,t-1} &= p^*_{s,\,t}-p_{s,\,t}-p_{s,\,t-1}+p_{s,\,t},\\
p^*_{s,\,t}-p_{s,\,t-1} &= x_{s,\,t}+\pi_{s,\,t}.
\end{aligned} \tag{C.26}$$

Then, equation (C.25) turns into

$$p^C_{s,\,t} - p_{s,\,t-1} = (1-\omega_s)\cdot(x_{s,\,t}+\pi_{s,\,t}) + (\omega_s)\cdot\left(p^C_{s,\,t-1}-p_{s,\,t-1}\right)+(\omega_s\cdot l_s)\cdot\pi_{t-1}. \tag{C.27}$$

To find an expression for $p^C_{s,\,t-1}-p_{s,\,t-1}$ for the previous expression, we lad equation (C.20).

$$\begin{aligned}
\pi_{s,\,t-1} &= (1-\alpha_s)\cdot\left(p^C_{s,\,t-1}-p_{s,\,t-2}\right),\\
\pi_{s,\,t-1} &= (1-\alpha_s)\cdot\left(p^C_{s,\,t-1}-p_{s,\,t-2}+p_{s,\,t-1}-p_{s,\,t-1}\right),\\
\pi_{s,\,t-1} &= (1-\alpha_s)\cdot\left(\,(p^C_{s,\,t-1}-p_{s,\,t-1})\;+\;(p_{s,\,t-1}-p_{s,\,t-2})\,\right),\\
\pi_{s,\,t-1} &= (1-\alpha_s)\cdot\left(\,(p^C_{s,\,t-1}-p_{s,\,t-1})\;+\;\pi_{s,\,t-1}\,\right),\\
\pi_{s,\,t-1}-(1-\alpha_s)\cdot\pi_{s,\,t-1} &= (1-\alpha_s)\cdot\left(p^C_{s,\,t-1}-p_{s,\,t-1}\right),\\
p^C_{s,\,t-1}-p_{s,\,t-1} &= \left[\frac{\alpha_s}{1-\alpha_s}\right]\cdot\pi_{s,\,t-1}.
\end{aligned} \tag{C.28}$$

Replacing (C.28) in equation (C.27)

$$\begin{aligned}
p^C_{s,\,t}-p_{s,\,t-1} &= (1-\omega_s)\cdot(x_{s,\,t}+\pi_{s,\,t})+(\omega_s)\cdot\left(p^C_{s,\,t-1}-p_{s,\,t-1}\right)+(\omega_s\cdot l_s)\cdot\pi_{t-1},\\
p^C_{s,\,t}-p_{s,\,t-1} &= (1-\omega_s)\cdot(x_{s,\,t}+\pi_{s,\,t})+(\omega_s)\cdot\left(\left[\frac{\alpha_s}{1-\alpha_s}\right]\cdot\pi_{s,\,t-1}\right)+(\omega_s\cdot l_s)\cdot\pi_{t-1}.
\end{aligned} \tag{C.29}$$

Replacing the previous result in equation (C.20)

$$\begin{aligned}
\pi_{s,\,t} &= (1-\alpha_s)\cdot\left(p^C_{s,\,t}-p_{s,\,t-1}\right),\\
\pi_{s,\,t} &= (1-\alpha_s)\cdot\left((1-\omega_s)\cdot(x_{s,\,t}+\pi_{s,\,t})+(\omega_s)\cdot\left(\left[\frac{\alpha_s}{1-\alpha_s}\right]\cdot\pi_{s,\,t-1}\right)+(\omega_s\cdot l_s)\cdot\pi_{t-1}\right),
\end{aligned}$$

$$\pi_{s,\,t}-(1-\alpha_s)(1-\omega_s)\pi_{s,\,t} = (1-\alpha_s)\left((1-\omega_s)x_{s,\,t}+(\omega_s)\left(\left[\frac{\alpha_s}{1-\alpha_s}\right]\pi_{s,\,t-1}\right)+(\omega_s\,l_s)\pi_{t-1}\right),$$

$$\pi_{s,\,t} = F_s\,x_{s,\,t}+G_s\,\pi_{s,\,t-1}+H_s\,\pi_{t-1}. \tag{C.30}$$

where

$$\begin{aligned}
D_s &= \alpha_s+(1-\alpha_s)\,\omega_s,\\
F_s &= \frac{(1-\alpha_s)\,(1-\omega_s)}{D_s},\\
G_S &= \frac{\alpha_s\,\omega_s}{D_s},\\
H_s &= \frac{(1-\alpha_s)\,\omega_s\,l_s}{D_s}.
\end{aligned}$$

Let's now turn into the $x_{s,\,t}$ term. Subtracting $p_{s,\,t}$ from equation (C.23),

$$\begin{aligned}
p^*_{s,\,t} &= (1-\beta\cdot\alpha_s)\cdot mc_{s,\,t}+(\beta\cdot\alpha_s)\cdot\mathbb{E}_t p^*_{s,\,t+1},\\
p^*_{s,\,t}-p_{s,\,t} &= (1-\beta\cdot\alpha_s)\cdot mc_{s,\,t}+(\beta\cdot\alpha_s)\cdot\mathbb{E}_t p^*_{s,\,t+1}-p_{s,\,t},\\
x_{s,\,t} &= (1-\beta\cdot\alpha_s)\cdot mc_{s,\,t}+(\beta\cdot\alpha_s)\cdot\mathbb{E}_t p^*_{s,\,t+1}-p_{s,\,t},\\
x_{s,\,t} &= (1-\beta\cdot\alpha_s)\cdot mc_{s,\,t}+(\beta\cdot\alpha_s)\cdot\mathbb{E}_t p^*_{s,\,t+1}-(1-\beta\cdot\alpha_s)\cdot p_{s,\,t}-(\beta\cdot\alpha_s)\cdot p_{s,\,t},\\
x_{s,\,t} &= (1-\beta\cdot\alpha_s)\cdot(mc_{s,\,t}-p_{s,\,t})+(\beta\cdot\alpha_s)\cdot\mathbb{E}_t(p^*_{s,\,t+1}-p_{s,\,t}).
\end{aligned} \tag{C.31}$$

Noting that

$$\begin{aligned}
&\mathbb{E}_t(p^*_{s,\,t+1}-p_{s,\,t}),\\
&\mathbb{E}_t(p^*_{s,\,t+1}+p_{s,\,t+1}-p_{s,\,t+1}-p_{s,\,t}),\\
&\mathbb{E}_t\left((p^*_{s,\,t+1}-p_{s,\,t+1})+(p_{s,\,t+1}-p_{s,\,t})\right),\\
&\mathbb{E}_t\left(x_{s,\,t+1}+\pi_{s,\,t+1}\right)
\end{aligned}$$

Then, equation (C.31) turns into

$$x_{s,\,t} = (1-\beta\cdot\alpha_s)\cdot(mc_{s,\,t}-p_{s,\,t})+(\beta\cdot\alpha_s)\cdot\mathbb{E}_t\left(x_{s,\,t+1}+\pi_{s,\,t+1}\right). \tag{C.32}$$

Multiplying the previous by F_s,

$$F_s x_{s,\,t}-F_s(\beta\cdot\alpha_s)E_t x_{s,\,t+1} = F_s(1-\beta\,\alpha_s)(mc_{s,\,t}-p_{s,\,t})+F_s(\beta\,\alpha_s)\mathbb{E}_t\left(\pi_{s,\,t+1}\right). \tag{C.33}$$

Solving for $F_s x_{s,t}$ in equation (C.30)

$$F_s \cdot x_{s,t} = \pi_{s,t} - G_s\,\pi_{s,t-1} - H_s\,\pi_{t-1} \tag{C.34}$$

and for $t+1$,

$$F_s \cdot x_{s,t+1} = \pi_{s,t+1} - G_s\,\pi_{s,t} - H_s\,\pi_t \tag{C.35}$$

Replacing both equations (C.34) and (C.35) in equation (C.33) leads to

$$\begin{aligned}\pi_{s,t} - G_s\,\pi_{s,t-1} - H_s\,\pi_{t-1} - (\beta\,\alpha_s)E_t\left(\pi_{s,t+1} - G_s\,\pi_{s,t} - H_s\,\pi_t\right) = \\ F_s(1-\beta\,\alpha_s)(mc_{s,t} - p_{s,t}) + F_s(\beta\,\alpha_s)\,\mathbb{E}_t\,\left(\pi_{s,t+1}\right).\end{aligned} \tag{C.36}$$

Solving for $\pi_{s,t}$, then,

$$\begin{aligned}(1+\beta\,\alpha_s\,G_s)\cdot\pi_{s,t} = \beta\,\alpha_s(1+F_s)\,E_t(\pi_{s,t+1}) \\ + G_s\,(\pi_{s,t-1}) + H_s\,(\pi_{t-1}) \\ - \beta\,\alpha_s\,H_s\,E_t(\pi_t) \\ + F_s\,(1-\beta\,\alpha_s)\,(mc_{s,t} - p_{s,t}).\end{aligned} \tag{C.37}$$

Assuming that $\mathbb{E}_t(\pi_t) = \pi_t$, the hybrid sectoral Phillips Curve is given by

$$\pi_{s,t} = \gamma_{f,s}\mathbb{E}_t\,\pi_{s,t+1} + \gamma_{b,s}\,\pi_{s,t-1} + \gamma_{\text{lag agg},s}\,\pi_{t-1} - \gamma_{\text{agg},s}\,\pi_t + \gamma_{mc,s}\,(mc_{s,t} - p_{s,t}). \tag{C.38}$$

with the following coefficients, knowing that $Q_s = 1 + \beta\,\alpha_s\,G_s$,

$$\begin{aligned}\gamma_{f,s} &= \frac{\beta\,\alpha_s\,(1+F_s)}{Q_s}, \\ \gamma_{b,s} &= \frac{G_s}{Q_s}, \\ \gamma_{\text{lag agg}} &= \frac{H_s}{Q_s}, \\ \gamma_{\text{agg}} &= \frac{H_s\,(\beta\,\alpha_s)}{Q_s}, \\ \gamma_{mc,s} &= \frac{F_s\,(1-\beta\,\alpha_s)}{Q_s}.\end{aligned}$$

C.2. *Problem of the Household*

C.2.1. *Labor demand problem*

The problem of the sector s labor packer is given by the cost minimization given a labor aggregator of the household's supply,

$$\begin{aligned}&\min_{N_{s,t}(h)} \int_0^1 W_{s,t}(h)\cdot N_{s,t}(h)\,\mathrm{d}\,h \\ &S.t.\ N_{s,t} = \left[\int_0^1 N_{s,t}(h)^{\frac{1}{1+\theta_n}}\,\mathrm{d}\,h\right]^{1+\theta_n}.\end{aligned} \tag{C.39}$$

with the following First Order Condition

$$\begin{aligned}N_{s,t}(h)\ :&W_{s,t}(h) - \lambda_t\left[(1+\theta_n)\left[\int_0^1 N_{s,t}(h)^{\frac{1}{1+\theta_n}}\,\mathrm{d}\,h\right]^{\theta_n}\frac{1}{1+\theta_n}\,N_{s,t}(h)^{\frac{1}{1+\theta_n}-1}\right] = 0 \\ &W_{s,t}(h) - \lambda_t\left[\left[\int_0^1 N_{s,t}(h)^{\frac{1}{1+\theta_n}}\,\mathrm{d}\,h\right]^{\theta_n} N_{s,t}(h)^{\frac{-\theta_n}{1+\theta_n}}\right] = 0 \\ &W_{s,t}(h) - W_{s,t}\left[\left[\int_0^1 N_{s,t}(h)^{\frac{1}{1+\theta_n}}\,\mathrm{d}\,h\right]^{\theta_n} N_{s,t}(h)^{\frac{-\theta_n}{1+\theta_n}}\right] = 0 \\ &\frac{W_{s,t}(h)}{W_{s,t}} = \left[\int_0^1 N_{s,t}(h)^{\frac{1}{1+\theta_n}}\,\mathrm{d}\,h\right]^{\theta_n} N_{s,t}(h)^{\frac{-\theta_n}{1+\theta_n}} \\ &\left[\frac{W_{s,t}(h)}{W_{s,t}}\right]^{-\frac{1+\theta_n}{\theta_n}} = \left[\int_0^1 N_{s,t}(h)^{\frac{1}{1+\theta_n}}\,\mathrm{d}\,h\right]^{-1-\theta_n} N_{s,t}(h)\end{aligned}$$

$$\left[\frac{W_{s,\,t}(h)}{W_{s,\,t}}\right]^{-\frac{1+\theta_n}{\theta_n}} = N_{s,\,t}^{-1}\, N_{s,\,t}(h) \tag{C.40}$$

from which the household's labor demand is yielded as

$$N_{s,\,t}(h) \;=\; \left[\frac{W_{s,\,t}(h)}{W_{s,\,t}}\right]^{-\frac{1+\theta_n}{\theta_n}} N_{s,\,t}, \tag{C.41}$$

C.2.2. *Utility maximization problem and optimal equations*

The optimization problem of the Household h at time t is given by

$$\begin{aligned}
\max\ \mathbb{E}_t \sum_{l=0}^{\infty}\beta^l\Bigg[&\frac{C_{t+l}(h)^{1-\sigma}}{1-\sigma} - \frac{\chi_0}{\chi}\, N_{t+l}(h)^{\chi} + \frac{\mu_0}{1-\mu}\left(\frac{M_{t+l}(h)}{P_{t+l}}\right)^{1-\mu}\Bigg], \\
S.t.\ &\sum_{s=S,\,G} P_{s,\,t+l}\, C_{s,\,t+l}(h) + \sum_{s=S,\,G} P_{s,\,t+l}\, I_{s,\,t+l}(h) \\
&\quad + M_{t+l+1}(h) - M_{t+l}(h) + \int \xi_{t+l+1}\, B_{t+l+1}(h) - B_{t+l}(h) \\
&= \sum_{s=S,\,G} W_{s,\,t+l}(h)\, N_{s,\,t+l}(h) \\
&\quad + \sum_{s=S,\,G} R^s_{K,t+l}(h)\, K_{s,\,t+l}(h) \\
&\quad - 0.5\,\phi_I^s\, P_{s,\,t+l}\, I_{s,\,t+l}(h)\left(\frac{I_{s,\,t+l}(h)}{I_{s,\,t+l-1}(h)} - 1\right)^2, \\
&N_{S,\,t+l}(h) = \left[\frac{W_{S,\,t+l}(h)}{W_{S,\,t+l}}\right]^{-\frac{1+\theta_n}{\theta_n}} N_{S,\,t+l}, \\
&N_{G,\,t+l}(h) = \left[\frac{W_{G,\,t+l}(h)}{W_{G,\,t+l}}\right]^{-\frac{1+\theta_n}{\theta_n}} N_{G,\,t+l}, \\
&N_{t+l}(h) = \left[\omega_{N,\,S}^{\frac{1}{\eta_n}}\cdot\left(N_{S,\,t+l}(h)\right)^{\frac{\eta_n-1}{\eta_n}} + \omega_{N,\,G}^{\frac{1}{\eta_n}}\cdot\left(N_{G,\,t+l}(h)\right)^{\frac{\eta_n-1}{\eta_n}}\right]^{\frac{\eta_n}{\eta_n-1}}, \\
&C_{t+l}(h) = \left[\omega_{C,\,S}^{\frac{1}{\eta_c}}\cdot\left(C_{S,\,t+l}(h)\right)^{\frac{\eta_c-1}{\eta}} + \omega_{C,\,G}^{\frac{1}{\eta_c}}\cdot\left(C_{G,\,t+l}(h)\right)^{\frac{\eta_c-1}{\eta}}\right]^{\frac{\eta_c}{\eta_c-1}}, \\
&K_{S,\,t+l+1}(h) = (1-\delta_S)\, K_{S,\,t+l}(h) + I_{S,\,t+l}(h), \\
&K_{G,\,t+l+1}(h) = (1-\delta_G)\, K_{G,\,t+l}(h) + I_{G,\,t+l}(h).
\end{aligned} \tag{C.42}$$

in the previous equation the term $\Gamma_{t+l}(h)$ is suppressed, given that it is irrelevant for the optimization procedure. Reducing the amount of restrictions,

$$\begin{aligned}
\max\ \mathbb{E}_t \sum_{l=0}^{\infty}\beta^l\Bigg[&\frac{1}{1-\sigma}\left[\omega_{C,\,S}^{\frac{1}{\eta_c}}\cdot\left(C_{S,\,t+l}(h)\right)^{\frac{\eta_c-1}{\eta}} + \omega_{C,\,G}^{\frac{1}{\eta_c}}\cdot\left(C_{G,\,t+l}(h)\right)^{\frac{\eta_c-1}{\eta}}\right]^{\frac{\eta_c\,(1-\sigma)}{\eta_c-1}} \\
&- \frac{\chi_0}{\chi}\left[\omega_{N,\,S}^{\frac{1}{\eta_n}}\cdot\left(N_{S,\,t+l}(h)\right)^{\frac{\eta_n-1}{\eta_n}} + \omega_{N,\,G}^{\frac{1}{\eta_n}}\cdot\left(N_{G,\,t+l}(h)\right)^{\frac{\eta_n-1}{\eta_n}}\right]^{\frac{\eta_n\,(\chi)}{\eta_n-1}} \\
&+ \frac{\mu_0}{1-\mu}\left(\frac{M_{t+l}(h)}{P_{t+l}}\right)^{1-\mu}\Bigg], \\
S.t.\ &\sum_{s=S,\,G} P_{s,\,t+l}\, C_{s,\,t+l}(h) + \sum_{s=S,\,G} P_{s,\,t+l}\, I_{s,\,t+l}(h) \\
&\quad + M_{t+l+1}(h) - M_{t+l}(h) + \int \xi_{t+l+1}\, B_{t+l+1}(h) - B_{t+l}(h) \\
&= \sum_{s=S,\,G} W_{s,\,t+l}(h)\, N_{s,\,t+l}(h) \\
&\quad + \sum_{s=S,\,G} R^s_{K,t+l}(h)\, K_{s,\,t+l}(h) \\
&\quad - \sum_{s=S,\,G} 0.5\,\phi_I^s\, P_{s,\,t+l}\, I_{s,\,t+l}(h)\left(\frac{I_{s,\,t+l}(h)}{I_{s,\,t+l-1}(h)} - 1\right)^2,
\end{aligned} \tag{C.43}$$

$$
\begin{aligned}
K_{S,\,t+l+1}(h) &= (1-\delta_S)\,K_{S,\,t+l}(h) + I_{S,\,t+l}(h),\\
K_{G,\,t+l+1}(h) &= (1-\delta_G)\,K_{G,\,t+l}(h) + I_{G,\,t+l}(h),\\
N_{S,\,t+l}(h) &= \left[\frac{W_{S,\,t+l}(h)}{W_{S,\,t+l}}\right]^{-\frac{1+\theta_n}{\theta_n}} N_{S,\,t+l},\\
N_{G,\,t+l}(h) &= \left[\frac{W_{G,\,t+l}(h)}{W_{G,\,t+l}}\right]^{-\frac{1+\theta_n}{\theta_n}} N_{G,\,t+l}.
\end{aligned}
$$

Household solves the previous problem taking as given aggregate and sectoral prices ($P_{S,\,t}(h)$, $P_{G,\,t}(h)$ and $P_t(h)$), wages ($W_t(h)$), contingent returns (ξ_t), capital rent costs ($R_{K,\,t}$), and initial conditions for contingent claims, sectoral capital, money holdings, and nominal wages ($B_{t-1}(h)$, $K_{s,\,t-1}(h)$, $M_{t-1}(h)$ and $W_{t-1}(h)$). The utility maximization problem yields the following First Order Conditions[35]

$$
C_{S,t+l}(h) : (C_{t+l}^{1/\eta_c-\sigma})\,\omega_{C,\,S}^{\frac{1}{\eta_c}}\,C_{S,\,t+l}(h)^{\frac{\eta_c-1}{\eta_c}-1} - \lambda_{t+l}^1(P_{S,\,t+l}) = 0, \tag{C.44}
$$

$$
C_{G,t+l}(h) : (C_{t+l}^{1/\eta_c-\sigma})\,\omega_{C,\,G}^{\frac{1}{\eta_c}}\,C_{G,\,t+l}(h)^{\frac{\eta_c-1}{\eta_c}-1} - \lambda_{t+l}^1(P_{G,\,t+l}) = 0, \tag{C.45}
$$

$$
N_{S,t+l}(h) : -\chi_o\,N_{t+l}(h)^{\chi-1+1/\eta_n}(\omega_{N,\,S}^{\frac{1}{\eta_n}})\left(N_{S,\,t+l}(h)^{\frac{\eta_n-1}{\eta_n}-1}\right) + \lambda_t^1(W_{S,\,t+l}(h)) = 0, \tag{C.46}
$$

$$
N_{G,t+l}(h) : -\chi_o\,N_{t+l}(h)^{\chi-1+1/\eta_n}(\omega_{N,\,G}^{\frac{1}{\eta_n}})\left(N_{G,\,t+l}(h)^{\frac{\eta_n-1}{\eta_n}-1}\right) + \lambda_t^1(W_{G,\,t+l}(h)) = 0, \tag{C.47}
$$

$$
\begin{aligned}
M_{t+l+1}(h) :& \beta^l\left(-\lambda_{t+l}^1\right)\\
&+\beta^{l+1}\mathbb{E}_t\left[\frac{\mu_0}{1-\mu}\left(\frac{1}{P_{t+l}}\right)^{1-\mu}(1-\mu)\,M_{t+l+1}^{-\mu} + \lambda_{t+l+1}^1\right] = 0,
\end{aligned} \tag{C.48}
$$

$$
B_{t+l+1}(h) : \beta^l\left(-\xi_{t+l+1}\,\lambda_{t+l}^1\right) + \beta^{l+1}\mathbb{E}_t\left[\lambda_{t+l+1}^1\right] = 0, \tag{C.49}
$$

$$
K_{S,t+l+1}(h) : (-\lambda_{t+l}^2) + \beta\left[\lambda_{t+l+1}^1\,R_{K,\,t+l}^S + \lambda_{t+l+1}^2(1-\delta_S)\right] = 0, \tag{C.50}
$$

$$
K_{G,t+l+1}(h) : (-\lambda_{t+l}^3) + \beta\left[\lambda_{t+l+1}^1\,R_{K,\,t+l}^G + \lambda_{t+l+1}^3(1-\delta_G)\right] = 0. \tag{C.51}
$$

$$
\begin{aligned}
I_{S,t+l}(h) :& (-\lambda_{t+l}^1)P_{S,\,t+l}\left(1 + \frac{\phi_I^S}{2}\left(\frac{I_{S,\,t+l}}{I_{S,\,t+l-1}} - 1\right)^2 + \phi_I^S\left(\frac{I_{S,\,t+l}}{I_{S,\,t+l-1}}\right)\left(\frac{I_{S,\,t+l}}{I_{S,\,t+l-1}} - 1\right)\right)\\
&+\lambda_{t+l}^2 + \beta\left[(-\lambda_{t+l+1}^1)\,P_{S,\,t+l+1}\,I_{S,\,t+l+1}\,\phi_I^S\left(\frac{I_{S,\,t+l+1}}{I_{S,\,t+l}}\right)^2\left(\frac{I_{S,\,t+l+1}}{I_{S,\,t+l}} - 1\right)\right] = 0,
\end{aligned} \tag{C.52}
$$

$$
\begin{aligned}
I_{G,t+l}(h) :& (-\lambda_{t+l}^1)P_{G,\,t+l}\left(1 + \frac{\phi_I^G}{2}\left(\frac{I_{G,\,t+l}}{I_{G,\,t+l-1}} - 1\right)^2 + \phi_I^G\left(\frac{I_{G,\,t+l}}{I_{G,\,t+l-1}}\right)\left(\frac{I_{G,\,t+l}}{I_{G,\,t+l-1}} - 1\right)\right)\\
&+\lambda_{t+l}^3 + \beta\left[(-\lambda_{t+l+1}^1)\,P_{G,\,t+l+1}\,I_{G,\,t+l+1}\,\phi_I^G\left(\frac{I_{G,\,t+l+1}}{I_{G,\,t+l}}\right)^2\left(\frac{I_{G,\,t+l+1}}{I_{G,\,t+l}} - 1\right)\right] = 0.
\end{aligned} \tag{C.53}
$$

C.2.3. *Euler equation for consumption*

Working for the usual Euler's Equation, we first use the First Order Condition for aggregate consumption

$$
\begin{aligned}
C_{t+l}(h) \,:& \beta^l\left(C_{t+l}^{-\sigma} - \lambda_{t+l}^1(P_{t+l})\right) = 0,\\
& C_{t+l}^{-\sigma} = \lambda_{t+l}^1\,(P_{t+l}),\\
& \frac{1}{P_{t+l}}\,C_{t+l}^{-\sigma} = \lambda_{t+l}^1.
\end{aligned} \tag{C.54}
$$

Combining this result with the State Contingent Claim Condition,

$$
\begin{aligned}
B_{t+l+1}(h) \,:& \beta^l\left(-\xi_{t+l+1}\,\lambda_{t+l}^1\right) + \beta^{l+1}\mathbb{E}_t\left[\lambda_{t+l+1}^1\right] = 0,\\
& \beta^l\left(\xi_{t+l+1}\,\lambda_{t+l}^1\right) = \beta^{l+1}\mathbb{E}_t\left[\lambda_{t+l+1}^1\right],\\
& \left(\xi_{t+l+1}\,\lambda_{t+l}^1\right) = \beta\mathbb{E}_t\left[\lambda_{t+l+1}^1\right],\\
& \lambda_{t+l}^1 = \beta\,\mathbb{E}_t\frac{1}{\xi_{t+l+1}}\left[\lambda_{t+l+1}^1\right]
\end{aligned} \tag{C.55}
$$

[35] λ_t^1, λ_t^2, λ_t^3, λ_t^4 and λ_t^5 are the Lagrange multipliers of the first, second, third, forth and fifth restriction equations respectively of maximization problem (C.44).

Replacing equation (C.54) with (C.55), the following yields

$$\begin{aligned}
\lambda^1_{t+l} &= \beta\,\mathbb{E}_t\,\frac{1}{\xi_{t+l+1}}\left[\lambda^1_{t+l+1}\right],\\
\frac{1}{P_{t+l}}\,C^{-\sigma}_{t+l} &= \beta\,\mathbb{E}_t\,\frac{1}{\xi_{t+l+1}}\,\frac{1}{P_{t+l+1}}\,C^{-\sigma}_{t+l+1},\\
C^{-\sigma}_{t+l} &= \beta\,\mathbb{E}_t\,\frac{1}{\xi_{t+l+1}}\,\frac{P_{t+l}}{P_{t+l+1}}\,C^{-\sigma}_{t+l+1},\\
C_{t+l}(h) &= \beta\,\mathbb{E}_t\,(1+R_{t+l})\,C_{t+l+1}(h). \qquad \text{(C.56)}
\end{aligned}$$

In which R_t is the risk-free real interest rate, representing the rate of return on an asset paying one unit of output in every state of nature at time $t+l$.

It is important to note that given the complete markets assumption, consumption is identical across households, oo that household identifiers can be omitted.

$$C_{t+l} = \beta\,\mathbb{E}_t\,(1+R_{t+l})\,C_{t+l+1}. \qquad \text{(C.57)}$$

C.2.4. *Euler equation for investment*

Turning now to solve for the Euler's Condition for investment in the s generic sector[36],

$$\begin{aligned}
(-\lambda^1_{t+l})P_{s,\,t+l}&\left(1+(\phi^s_I/2)\left(\frac{I_{s,\,t+l}}{I_{s,\,t+l-1}}-1\right)^2+\phi^s_I\left(\frac{I_{s,\,t+l}}{I_{s,\,t+l-1}}\right)\left(\frac{I_{s,\,t+l}}{I_{s,\,t+l-1}}-1\right)\right)\\
&+\lambda^s_{t+l}+\beta\left[(-\lambda^1_{t+l+1})\,P_{s,\,t+l+1}\,I_{s,\,t+l+1}\,\phi^s_I\left(\frac{I_{s,\,t+l+1}}{I_{s,\,t+l}}\right)^2\left(\frac{I_{s,\,t+l+1}}{I_{s,\,t+l}}-1\right)\right]=0,\\
-&\left(1+(\phi^s_I/2)\left(\frac{I_{s,\,t+l}}{I_{s,\,t+l-1}}-1\right)^2+\phi^s_I\left(\frac{I_{s,\,t+l}}{I_{s,\,t+l-1}}\right)\left(\frac{I_{s,\,t+l}}{I_{s,\,t+l-1}}-1\right)\right)\\
&+\frac{\lambda^s_{t+l}}{\lambda^1_{t+l}\,P_{s,\,t+l}}-\beta\left[\frac{\lambda^1_{t+l+1}\,P_{s,\,t+l+1}}{\lambda^1_{t+l}\,P_{s,\,t+l}}\,I_{s,\,t+l+1}\,\phi^s_I\left(\frac{I_{s,\,t+l+1}}{I_{s,\,t+l}}\right)^2\left(\frac{I_{s,\,t+l+1}}{I_{s,\,t+l}}-1\right)\right]=0,\\
-&\left(1+(\phi^s_I/2)\left(\frac{I_{s,\,t+l}}{I_{s,\,t+l-1}}-1\right)^2+\phi^s_I\left(\frac{I_{s,\,t+l}}{I_{s,\,t+l-1}}\right)\left(\frac{I_{s,\,t+l}}{I_{s,\,t+l-1}}-1\right)\right)\\
&+q_{s,\,t+l}-\beta\left[\frac{\lambda^1_{t+l+1}\,P_{s,\,t+l+1}}{\lambda^1_{t+l}\,P_{s,\,t+l}}\,\phi^s_I\left(\frac{I_{s,\,t+l+1}}{I_{s,\,t+l}}\right)^2\left(\frac{I_{s,\,t+l+1}}{I_{s,\,t+l}}-1\right)\right]=0,
\end{aligned}$$

Leading to the following sectoral Euler Equation for Investment

$$\begin{aligned}
q_{s,\,t+l} = 1+(\phi^s_I/2)&\left(\frac{I_{s,\,t+l}}{I_{s,\,t+l-1}}-1\right)^2+\phi^s_I\left(\frac{I_{s,\,t+l}}{I_{s,\,t+l-1}}\right)\left(\frac{I_{s,\,t+l}}{I_{s,\,t+l-1}}-1\right)\\
&-\beta\left[\frac{\Lambda_{s,\,t+l+1}}{\Lambda_{s,\,t+l}}\,\phi^s_I\left(\frac{I_{s,\,t+l+1}}{I_{s,\,t+l}}\right)^2\left(\frac{I_{s,\,t+l+1}}{I_{s,\,t+l}}-1\right)\right]. \qquad \text{(C.58)}
\end{aligned}$$

C.2.5. *Wage setting problem*

The sectoral wage aggregator is given by

$$W^{1-\epsilon}_{s,\,t} = \int_0^1 W_{s,\,t}(h)^{1-\epsilon^W}\,\mathrm{d}\,h. \qquad \text{(C.59)}$$

Given that the sectoral probability of not adjusting wages α^W_s can be interpreted as the proportion of households that cannot change wages in a given period, the integer can be partitioned

$$W^{1-\epsilon^W}_{s,\,t} = \int_0^1 W_{s,\,t}(h)^{1-\epsilon^W}\,\mathrm{d}\,h,$$

[36]Noting that the problem is symmetric across households, we abstract from using the household identifier h in the following procedure.

$$
\begin{aligned}
&= \int_0^{\alpha_s^W} \left(W_{s,\,t-1}(h)(1+\pi_{t-1})^{l_W^s}\right)^{1-\epsilon^W} \mathrm{d}\ h \ + \int_{\alpha_s^W}^1 W_{s,\,t}^*(j)^{1-\epsilon^W}\ \mathrm{d}\ j,\\
&= \int_0^{\alpha_s^W} \left(W_{s,\,t-1}(h)(1+\pi_{t-1})^{l_W^s}\right)^{1-\epsilon^W} \mathrm{d}\ h \ + \ (1-\alpha_s^W)\,(W_{s\,t}^*)^{1-\epsilon^W},\\
&= \alpha_s^W \cdot \left(W_{s,\,t-1}(h)(1+\pi_{t-1})^{l_W^s}\right)^{1-\epsilon^W} \ + \ (1-\alpha_s^W)\cdot (W_{s,\,t}^*)^{1-\epsilon^W},\\
\left(\frac{W_{s,\,t}}{W_{s,\,t-1}}\right)^{1-\epsilon^W} &= \alpha_s \cdot \left(\frac{W_{s,\,t-1}}{W_{s,\,t-1}}(1+\pi_{t-1})^{l_W^s}\right)^{1-\epsilon^W} \ + \ (1-\alpha_s)\cdot \left(\frac{W_{s,\,t}^*}{W_{s,\,t-1}}\right)^{1-\epsilon},\\
(\Pi_{s,\,t}^W)^{1-\epsilon^W} &= \alpha_s^W \left((1+\pi_{t-1})^{l_W^s}\right)^{1-\epsilon^W} \ + \ (1-\alpha_s^W)\cdot \left(\frac{W_{s,\,t}^*}{W_{s,\,t-1}}\right)^{1-\epsilon^W}. \qquad \text{(C.60)}
\end{aligned}
$$

Log-linearizing the RHS of equation (C.60)

$$
\begin{aligned}
(\Pi_{s,\,t}^W)^{1-\epsilon^W} &\approx (\Pi_s^W)^{1-\epsilon^W} + (1-\epsilon^W)\,(\Pi_s^W)^{-\epsilon^W}\cdot\left(\Pi_{s,\,t}-\Pi_s^W\right),\\
&\approx 1 + (1-\epsilon^W)\cdot(\Pi_{s,\,t}^W - 1),\\
&\approx 1 + (1-\epsilon^W)\cdot(1+\pi_{s,\,t}^W - 1),\\
&\approx 1 + (1-\epsilon^W)\cdot\pi_{s,\,t}^W. \qquad \text{(C.61)}
\end{aligned}
$$

Note that $\Pi_{s,\,t}^W = 1+\pi_{s,\,t}^W$, where $\Pi_{s,\,t}^W$ is the gross wage inflation rate and $\pi_{s,\,t}^W$ is the net wage inflation rate. Additionally, we have used that the steady state value of the gross wage inflation rate is one. Log-linearizing the LHS of equation (C.60),

$$
\begin{aligned}
&\alpha_s^W \left((1+\pi_{t-1})^{l_W^s}\right)^{1-\epsilon^W} + (1-\alpha_s^W)\cdot\left(\frac{W_{s,\,t}^*}{W_{s,\,t-1}}\right)^{1-\epsilon^W}\\
&\qquad \approx \left(\alpha_s^W (1^{l_W^s})^{1-\epsilon^W} + (1-\alpha_s^W)\left(\frac{W_s}{W_s}\right)^{1-\epsilon^W}\right)\\
&\qquad + \left(\alpha_s^W(1-\epsilon^W)\,l_W^s\,(1^{l_W^s-1})^{-\epsilon^W}\,(\pi_{t-1}-\pi)\right)\\
&\qquad + \left((1-\alpha_s^W)\,(1-\epsilon^W)\,\left(\frac{W_s}{W_s}\right)^{-\epsilon^W}\,\frac{1}{W_s}\,(W_{s,\,t}^*-W_s)\right)\\
&\qquad + \left((1-\alpha_s^W)\,(1-\epsilon^W)\,\left(\frac{W_s}{W_s}\right)^{-\epsilon} W_s\,(-1)\,\frac{1}{W_s^2}\,(W_{s,\,t-1}-W_s)\right),\\
&\qquad \approx (\alpha_s^W + (1-\alpha_s^W)\,) + (1-\alpha_s^W)\,(1-\epsilon^W)\left[w_{s,\,t}^* - w_{s,\,t-1}\right] + \alpha_s^W(1-\epsilon^W)\,l_W^s\,\pi_{t-1},\\
&\qquad \approx 1 + (1-\alpha_s^W)\,(1-\epsilon^W)\,(w_{s,\,t}^* - w_{s,\,t-1}) + \alpha_s^W(1-\epsilon^W)\,l_W^s\,\pi_{t-1}. \qquad \text{(C.62)}
\end{aligned}
$$

Combining both results[37],

$$
\begin{aligned}
1 + (1-\epsilon^W)\,\pi_{s,\,t}^W &= 1 + (1-\alpha_s^W)\,(1-\epsilon^W)\,(w_{s,\,t}^* - w_{s,\,t-1}) + \alpha_s^W(1-\epsilon^W)\,l_W^s\,\pi_{t-1},\\
(1-\epsilon^W)\,\pi_{s,\,t}^W &= (1-\alpha_s^W)\,(1-\epsilon^W)\,(w_{s,\,t}^* - w_{s,\,t-1}) + \alpha_s^W(1-\epsilon^W)\,l_W^s\,\pi_{t-1},\\
\pi_{s,\,t}^W &= (1-\alpha_s^W)\,(w_{s,\,t}^* - w_{s,\,t-1}) + \alpha_s^W\,l_W^s\,\pi_{t-1}. \qquad \text{(C.63)}
\end{aligned}
$$

with the later as the sectoral wage inflation, resulted by the fact that there exists a proportion of households that change wages to their optimum ones, and the remaining that index their wages to aggregate inflation.

Households that are able to adjust sectoral wages do so by choosing $W_{s,\,t}^*$ that maximize the expected discounted value of the utility functional subject to the usual constraints, i.e. demand functions and budget constraints.

The problem for sector s wage optimization can be summarized by

$$
\max_{W_{s,\,t}^*} \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha_s^W)^l \Big[W_{s,\,t+l}(h)\,N_{s,\,t+l}(h) - MRS_{s,\,t+l}(h)\,N_{s,\,t+l}(h)\Big], \qquad \text{(C.64)}
$$

[37]Lower case variables represent the first-order expansion around the steady state of the upper case variable.

$$S.t.\ N_{s,\,t+l}(h) = \left(\frac{W_{s,\,t+l}(h)}{W_{s,\,t+l}}\right)^{-\frac{1+\theta_n}{\theta_n}} N_{s,\,t+l}.$$

Using that $W_{s,\,t+l}(h) = W^*_{s,\,t}\prod_{k=0}^{l-1}(1+\pi_{t+j})^{l^s_W}$[38], the problem can be reduced to

$$\max_{W^*_{s,\,t}} \mathbb{E}_t \sum_{l=0}^{\infty} (\beta\,\alpha^W_s)^l \left[W^*_{s,\,t}\mathcal{I}_{t,\,l} - MRS_{s,\,t+l}(h)\right] \left(\frac{W^*_{s,\,t}\mathcal{I}_{t,\,l}}{W_{s,\,t+l}}\right)^{-\frac{1+\theta_n}{\theta_n}} N_{s,\,t+l}. \tag{C.65}$$

solving for the optimal wage in the First Order Condition

$$\sum_o^{\infty}(\beta\,\alpha^W_s)^l \left[\left(-\frac{1}{\theta_n}\right)(W^*_{s,\,t})^{-\frac{1}{\theta_n}-1}\mathcal{I}_{t,l} + \left(\frac{1+\theta_n}{\theta_n}\right) MRS_{s,\,t+l}(h)\, W_{s,\,t}^{-\frac{1+\theta_n}{\theta_n}-1}\right]\times$$

$$\left(\frac{\mathcal{I}_{t,l}}{W_{s,\,t+l}}\right)^{-\frac{1+\theta_n}{\theta_n}} N_{s,\,t+l} = 0,$$

$$\sum_o^{\infty}(\beta\,\alpha^W_s)^l \left[\frac{\mathcal{I}_{t,l}}{1+\theta_n} - MRS_{s,\,t+l}(h)\,\frac{1}{W^*_{s,\,t}}\right] N_{s,\,t+l}(h) = 0,$$

$$\frac{W^*_{s,\,t}}{1+\theta_n}\sum_o^{\infty}(\beta\,\alpha^W_s)^l\, N_{s,\,t+l}(h) = \sum_o^{\infty}(\beta\,\alpha^W_s)^l\, N_{s,\,t+l}(h)\frac{MRS_{s,\,t+l}(h)}{\mathcal{I}_{t,l}}. \tag{C.66}$$

which yields

$$W^*_{s,\,t} = (1+\theta_n)\,\frac{\sum_o^{\infty}(\beta\,\alpha^W_s)^l\, N_{s,\,t+l}(h)\frac{MRS_{s,\,t+l}(h)}{\mathcal{I}_{t,l}}}{\sum_o^{\infty}(\beta\,\alpha^W_s)^l\, N_{s,\,t+l}(h)} \tag{C.67}$$

using the ratio decomposition proceeding for log-linearizing of Appendix C.1.2, the following result can be obtained for the log-linearized version of equation (C.67)

$$w^*_{s,t} = (1-\beta\alpha^W_s)mrs_{s,t} + \beta\alpha^W_s E_t w^*_{s,t+1} - \beta\alpha^W_s l^s_W \pi_t. \tag{C.68}$$

With the last term coming from the log-linearization of the indexation factor, $\mathcal{I}_{t,l}$.

Combining equation (C.68) and (C.60),

$$\begin{aligned}
\pi^W_{s,t} &= (1-\alpha^W_s)\,(w^*_{s,t} - w_{s,t-1}) + \alpha^W_s\, l^s_W\, \pi_{t-1},\\
\pi^W_{s,t} &= (1-\alpha^W_s)\left[(1-\beta\alpha^W_s)\,mrs_{s,t} + \beta\,\alpha^W_s\, E_t w^*_{s,t+1} - \beta\,\alpha^W_s\, l^s_W\,\pi_t - w_{s,t-1}\right]\\
&\quad + \alpha^W_s\, l^s_W\,\pi_{t-1},\\
\pi^W_{s,t} &= (1-\alpha^W_s)\left[(1-\beta\alpha^W_s)\,mrs_{s,t} + \beta\,\alpha^W_s\, E_t w^*_{s,t+1} - \beta\,\alpha^W_s\, l^s_W\,\pi_t - w_{s,\,t} + \pi^W_{s,\,t}\right]\\
&\quad + \alpha^W_s\, l^s_W\,\pi_{t-1},\\
\pi^W_{s,t} &= \left(\frac{1-\alpha^W_s}{\alpha^W_s}\right)\left[(1-\beta\alpha^W_s)\,mrs_{s,t} + \beta\,\alpha^W_s\, E_t w^*_{s,t+1} - \beta\,\alpha^W_s\, l^s_W\,\pi_t - w_{s,\,t}\right]\\
&\quad + \alpha^W_s\, l^s_W\,\pi_{t-1},\\
\pi^W_{s,t} &= \left(\frac{1-\alpha^W_s}{\alpha^W_s}\right)\left[(1-\beta\alpha^W_s)\,mrs_{s,t} + \beta\,\alpha^W_s\, E_t w^*_{s,t+1} - \beta\,\alpha^W_s\, l^s_W\,\pi_t - w_{s,\,t}\right]\\
&\quad + l^s_W\,\pi_{t-1}.
\end{aligned} \tag{C.69}$$

Using equation (C.60) in $t+1$ and solving for $\mathbb{E}_t w^*_{s,\,t+1}$,

$$\pi^W_{s,t+1} = (1-\alpha^W_s)\,(\mathbb{E}_t w^*_{s,t+1} - w_{s,t}) + \alpha^W_s\, l^s_W\,\pi_t,$$

$$\mathbb{E}_t w^*_{s,\,t+1} = \left(\frac{1}{1-\alpha^w_s}\right)\left(\mathbb{E}_t\pi^W_{s,\,t+1} - \alpha^W_s\, l^s_W\,\pi_t\right) + w_{s,\,t}.$$

replacing the previous result in equation (C.69)

$$\pi^W_{s,t} = \left(\frac{1-\alpha^W_s}{\alpha^W_s}\right)\left[(1-\beta\alpha^W_s)\,mrs_{s,t} + \beta\,\alpha^W_s\left(\left(\frac{1}{1-\alpha^w_s}\right)\left(\mathbb{E}_t\pi^W_{s,\,t+1} - \alpha^W_s\, l^s_W\,\pi_t\right) + w_{s,\,t}\right)\right.$$

$$\left. - \beta\,\alpha^W_s\, l^s_W\,\pi_t - w_{s,\,t}\right] + l^s_W\,\pi_{t-1},$$

[38]For notation purposes, we are going to be using the following equivalence $W_{s,\,t+l}(h) = W^*_{s,\,t}\prod_{k=0}^{l-1}(1+\pi_{t+j})^{l^s_W} \equiv W_{s,\,t+l}(h) = W^*_{s,\,t}\mathcal{I}_{t,\,l}$

$$\pi^W_{s,t} = \left(\frac{1-\alpha^W_s}{\alpha^W_s}\right)\Bigg[(1-\beta\alpha^W_s)\left(mrs_{s,t} - w_{s,\,t}\right) + \beta\,\alpha^W_s\left(\left(\frac{1}{1-\alpha^w_s}\right)\left(\mathbb{E}_t\pi^W_{s,\,t+1} - \alpha^W_s\,l^s_W\,\pi_t\right)\right) - \beta\,\alpha^W_s\,l^s_W\,\pi_t\Bigg] + l^s_W\,\pi_{t-1},$$

$$\pi^W_{s,t} = \left(\frac{1-\alpha^W_s}{\alpha^W_s}\right)\left[(1-\beta\alpha^W_s)\left(mrs_{s,t} - w_{s,\,t}\right) + \left(\frac{\beta\,\alpha^W_s}{1-\alpha^W_s}\right)\mathbb{E}_t\pi^W_{s,\,t+1} - \left(\frac{\alpha^W_s}{1-\alpha^W_s}\right)\beta\,l^s_W\,\pi_t\right] + l^s_W\,\pi_{t-1},$$

From which the sectoral Hybrid Wage Phillips Curve is derived

$$\pi^W_{s,\,t} - l^s_W\,\pi_{t-1} = \beta\big(\mathbb{E}_t\pi^W_{s,\,t+1} - l^s_W\,\pi_t\big) + \kappa^W_s\left(mrs_{s,\,t} - w_{s,\,t}\right). \tag{C.70}$$

where $\kappa^W_s = (1-\alpha^W_s)\,(1-\beta\,\alpha^W_s)/\alpha^W_s$.

Appendix D. Calibration

Table D.1. Summary of Parameters

	Value	Comments
Shock-Size Block		
Credibility of the Mon. Authority (k_g)	$[0.15, 0.95]$	Learning speed of private agents.
Monetary Policy Horizon (ρ_p)	$[0.65, 0.85]$	From 3 to 8 years.
Shock Pass-Through (ϕ_z)	$[0.01, 0.3]$	Overall relevance of shocks on mon. policy.
Activation Threshold (τ)	2.0	In S.D. units.
Monetary Policy		
Taylor Rule Inertia (γ_i)	0.5	AR(1) coefficient of the Taylor Rule.
Inflation-Gap Coefficient (ϕ_π)	2.0	Inflation-Gap relevance in mon. policy.
Output-Gap Coefficient (ϕ_y)	0.5	Output-Gap relevance in mon. policy.
Firms		
Calvo Price Probabilities (α_S, α_G)	(0.75, 0.5)	Sectoral price contract duration.
Share of Back. Looking Firms (ω_S, ω_G)	(0.3, 0.3)	Sectoral proportion of backward-looking firms.
Price Indexation of Non-Adj. (l_S, l_W)	(0.9, 0.9)	Sectoral degree of lagged inflation indexation.
Elas. of Factor Substitution (σ_S, σ_G)	(0.5, 0.8)	Sectoral factor complementary in production.
Sectoral Labor Share	(60%, 40%)	Sectoral labor share in production.
Investment Adj. Costs (ϕ_I^S, ϕ_I^G)	(5.0, 5.0)	Convex investment adjustment costs.
Depreciation Rate (δ_S, δ_G)	(0.025, 0.025)	Quarterly capital depreciation rate.
Households		
Discount Factor (β)	0.99	Utility discount factor.
Inverse EIS (σ)	1.5	Inverse elasticity of intertemporal consumption.
Inverse Frisch Elasticity (χ)	2.0	Inverse elasticity of labor supply.
Calvo Wage Probabilities (α_S^W, α_G^W)	(0.4, 0.75)	Sectoral wage contract duration.
Wage Indexation of Non-Ad. (l_S^W, l_G^W)	(0.5, 0.5)	Degree of indexation of non-adjusted wages.
Elas. of Sub. of Labor Types (η_n)	1.8	Substitutability of labor across sectors.
Elas. of Sub. of Consumption Types (η_c)	0.9	Substitutability of consumption across sectors.

Appendix E. Robustness Exercise

Table E.1: Robustness Exercise: Complete Statistics

		Aggregate		Sectoral Outcomes				Sacrifice Ratios			Policy stance			
Shock	Scen.	$\sum \pi_t$	$\sum y_t$	$\sum \pi_{S,t}$	$\sum \pi_{G,t}$	$\sum y_{S,t}$	$\sum y_{G,t}$	SR^{rec}	SR^{vol}	$H^*_{5\%}$	$\sum i_t$	$\sum r_t$	$\sum \pi^T_t$	$\sum \pi^e_t$
Monetary policy reaction														
$\gamma_i = 0.00$														
Prod.	S1	1.469	−2.393	1.453	1.498	−1.503	−4.469	−1.629	1.968	26	1.741	0.479	0.000	1.262
Prod.	S2	1.568	−2.245	1.552	1.598	−1.318	−4.409	−1.432	1.480	24	1.793	0.455	0.232	1.338
Prod.	S3	1.561	−2.305	1.545	1.591	−1.386	−4.450	−1.477	1.669	25	1.824	0.481	0.290	1.344
CP	S1	0.403	−0.703	0.398	0.411	−0.435	−1.329	−1.747	2.018	33	0.453	0.112	0.000	0.341
CP	S2	0.501	−0.557	0.496	0.510	−0.252	−1.270	−1.113	0.731	25	0.505	0.088	0.230	0.417
CP	S3	0.494	−0.617	0.489	0.503	−0.320	−1.310	−1.248	1.132	28	0.536	0.114	0.287	0.422
$\gamma_i = 0.25$														
Prod.	S1	1.422	−2.410	1.406	1.451	−1.532	−4.459	−1.695	2.086	26	1.636	0.422	0.000	1.214
Prod.	S2	1.519	−2.265	1.503	1.548	−1.350	−4.400	−1.492	1.567	24	1.682	0.394	0.232	1.287
Prod.	S3	1.511	−2.324	1.495	1.541	−1.418	−4.439	−1.538	1.767	25	1.712	0.420	0.290	1.292
CP	S1	0.390	−0.710	0.386	0.398	−0.445	−1.328	−1.818	2.174	33	0.425	0.096	0.000	0.329
CP	S2	0.486	−0.566	0.481	0.495	−0.265	−1.269	−1.165	0.784	25	0.470	0.068	0.230	0.402
CP	S3	0.479	−0.624	0.474	0.488	−0.332	−1.308	−1.304	1.212	28	0.501	0.094	0.287	0.407
$\gamma_i = 0.75$														
Prod.	S1	1.217	−2.419	1.203	1.244	−1.583	−4.370	−1.987	2.298	24	1.200	0.211	0.000	0.990
Prod.	S2	1.304	−2.281	1.289	1.331	−1.411	−4.310	−1.749	1.743	22	1.223	0.170	0.232	1.054
Prod.	S3	1.292	−2.334	1.277	1.320	−1.473	−4.343	−1.806	1.952	23	1.249	0.195	0.290	1.053
CP	S1	0.341	−0.718	0.337	0.348	−0.464	−1.311	−2.108	2.524	30	0.314	0.038	0.000	0.276
CP	S2	0.426	−0.581	0.422	0.435	−0.294	−1.251	−1.363	0.944	24	0.336	−0.003	0.230	0.339
CP	S3	0.415	−0.634	0.411	0.424	−0.356	−1.285	−1.528	1.417	26	0.361	0.023	0.287	0.339
$\phi_\pi = 1.50$														
Prod.	S1	1.921	−1.990	1.904	1.953	−1.002	−4.296	−1.036	0.801	32	1.874	0.224	0.000	1.650
Prod.	S2	1.999	−1.854	1.982	2.032	−0.832	−4.237	−0.927	0.612	29	1.895	0.189	0.232	1.707
Prod.	S3	2.000	−1.899	1.982	2.032	−0.883	−4.269	−0.950	0.676	30	1.929	0.213	0.290	1.717
CP	S1	0.543	−0.605	0.538	0.552	−0.311	−1.290	−1.113	0.915	40	0.508	0.041	0.000	0.467
CP	S2	0.621	−0.470	0.616	0.630	−0.144	−1.232	−0.757	0.342	34	0.529	0.006	0.230	0.523
CP	S3	0.621	−0.515	0.616	0.631	−0.194	−1.263	−0.828	0.496	36	0.563	0.030	0.287	0.533
$\phi_\pi = 2.50$														
Prod.	S1	1.052	−2.668	1.038	1.078	−1.876	−4.516	−2.536	4.238	23	1.289	0.417	0.000	0.873
Prod.	S2	1.152	−2.524	1.137	1.179	−1.695	−4.458	−2.192	3.110	21	1.335	0.386	0.232	0.950
Prod.	S3	1.137	−2.588	1.123	1.164	−1.770	−4.498	−2.276	3.570	22	1.361	0.413	0.290	0.948
CP	S1	0.288	−0.778	0.284	0.296	−0.535	−1.345	−2.700	4.285	27	0.329	0.094	0.000	0.235
CP	S2	0.387	−0.635	0.382	0.395	−0.356	−1.286	−1.643	1.468	21	0.375	0.064	0.230	0.311
CP	S3	0.373	−0.699	0.368	0.381	−0.430	−1.326	−1.875	2.376	24	0.400	0.091	0.287	0.309
$\phi_y = 0.25$														
Prod.	S1	0.902	−2.722	0.889	0.927	−1.958	−4.502	−3.017	4.917	19	1.117	0.394	0.000	0.724
Prod.	S2	1.017	−2.559	1.003	1.043	−1.755	−4.436	−2.517	3.453	17	1.167	0.354	0.232	0.813
Prod.	S3	1.003	−2.626	0.989	1.028	−1.832	−4.478	−2.619	4.005	19	1.197	0.385	0.290	0.812
CP	S1	0.235	−0.798	0.232	0.242	−0.565	−1.340	−3.388	5.141	23	0.269	0.087	0.000	0.183
CP	S2	0.349	−0.637	0.345	0.357	−0.363	−1.275	−1.825	1.518	16	0.319	0.048	0.230	0.271
CP	S3	0.335	−0.703	0.331	0.343	−0.440	−1.316	−2.098	2.544	20	0.348	0.078	0.287	0.270
$\phi_y = 0.75$														
Prod.	S1	1.705	−2.180	1.689	1.736	−1.239	−4.378	−1.278	1.241	29	1.766	0.302	0.000	1.464
Prod.	S2	1.782	−2.055	1.765	1.814	−1.082	−4.324	−1.153	0.971	27	1.794	0.273	0.232	1.520
Prod.	S3	1.778	−2.104	1.761	1.809	−1.138	−4.357	−1.184	1.077	27	1.822	0.296	0.290	1.526

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Table E.1: Robustness Exercise: Complete Statistics *(continued)*

		Aggregate		Sectoral Outcomes				Sacrifice Ratios			Policy stance			
Shock	Scen.	$\sum \pi_t$	$\sum y_t$	$\sum \pi_{S,t}$	$\sum \pi_{G,t}$	$\sum y_{S,t}$	$\sum y_{G,t}$	SR^{rec}	SR^{vol}	$H^*_{5\%}$	$\sum i_t$	$\sum r_t$	$\sum \pi^T_t$	$\sum \pi^e_t$
CP	S1	0.483	−0.649	0.478	0.492	−0.365	−1.310	−1.343	1.318	36	0.476	0.062	0.000	0.414
CP	S2	0.559	−0.524	0.554	0.568	−0.210	−1.257	−0.938	0.542	30	0.504	0.034	0.230	0.470
CP	S3	0.554	−0.573	0.549	0.564	−0.266	−1.290	−1.033	0.793	32	0.532	0.057	0.287	0.475
Price rigidity asymmetry														
$\alpha_S = 0.65$, $\alpha_G = 0.55$														
Prod.	S1	1.586	−2.930	1.580	1.596	−2.671	−3.534	−1.848	2.427	27	1.698	0.355	0.000	1.342
Prod.	S2	1.706	−2.846	1.701	1.715	−2.620	−3.374	−1.669	1.935	25	1.760	0.324	0.232	1.436
Prod.	S3	1.690	−2.885	1.684	1.699	−2.649	−3.437	−1.708	2.123	26	1.782	0.349	0.290	1.434
CP	S1	0.433	−0.846	0.431	0.436	−0.740	−1.093	−1.956	2.551	32	0.439	0.077	0.000	0.362
CP	S2	0.551	−0.763	0.550	0.553	−0.689	−0.935	−1.384	1.168	26	0.500	0.045	0.230	0.455
CP	S3	0.535	−0.802	0.534	0.538	−0.719	−0.997	−1.498	1.601	29	0.523	0.070	0.287	0.453
$\alpha_S = 0.85$, $\alpha_G = 0.45$														
Prod.	S1	0.895	−1.456	0.863	0.954	0.464	−5.936	−1.627	1.581	19	1.055	0.333	0.000	0.722
Prod.	S2	0.943	−1.217	0.910	1.005	0.843	−6.024	−1.290	0.966	17	1.051	0.298	0.232	0.753
Prod.	S3	0.943	−1.293	0.910	1.005	0.742	−6.040	−1.371	1.201	18	1.088	0.326	0.290	0.762
CP	S1	0.254	−0.465	0.245	0.270	0.069	−1.710	−1.829	1.739	28	0.273	0.071	0.000	0.202
CP	S2	0.302	−0.228	0.292	0.321	0.445	−1.798	−0.755	0.268	17	0.270	0.036	0.230	0.234
CP	S3	0.302	−0.303	0.292	0.321	0.344	−1.813	−1.004	0.640	20	0.306	0.064	0.287	0.242
$\alpha_S = 0.625$, $\alpha_G = 0.625$														
Prod.	S1	1.616	−3.018	1.614	1.618	−3.032	−2.986	−1.868	2.547	27	1.714	0.337	0.000	1.376
Prod.	S2	1.739	−2.944	1.739	1.740	−3.021	−2.763	−1.693	2.042	26	1.778	0.305	0.232	1.473
Prod.	S3	1.723	−2.980	1.722	1.724	−3.038	−2.845	−1.730	2.231	27	1.801	0.331	0.290	1.471
CP	S1	0.439	−0.869	0.438	0.440	−0.834	−0.950	−1.979	2.720	33	0.440	0.070	0.000	0.371
CP	S2	0.561	−0.795	0.562	0.561	−0.824	−0.730	−1.417	1.268	27	0.504	0.038	0.230	0.466
CP	S3	0.545	−0.832	0.545	0.545	−0.841	−0.810	−1.526	1.708	30	0.527	0.063	0.287	0.464
$\alpha_S = 0.50$, $\alpha_G = 0.75$														
Prod.	S1	1.749	−3.328	1.766	1.717	−4.440	−0.734	−1.903	2.708	26	1.827	0.345	0.000	1.482
Prod.	S2	1.888	−3.290	1.909	1.850	−4.594	−0.249	−1.742	2.226	25	1.905	0.314	0.232	1.590
Prod.	S3	1.868	−3.315	1.888	1.832	−4.562	−0.405	−1.774	2.400	26	1.927	0.340	0.290	1.587
CP	S1	0.471	−0.948	0.475	0.463	−1.194	−0.376	−2.013	2.910	30	0.465	0.070	0.000	0.395
CP	S2	0.609	−0.910	0.616	0.595	−1.346	0.105	−1.496	1.491	26	0.542	0.040	0.230	0.502
CP	S3	0.589	−0.935	0.595	0.577	−1.314	−0.050	−1.588	1.908	28	0.564	0.065	0.287	0.499
Wage rigidity asymmetry														
$\alpha^W_S = 0.50$, $\alpha^W_G = 0.65$														
Prod.	S1	1.342	−2.401	1.323	1.377	−1.323	−4.916	−1.789	2.199	26	1.475	0.345	0.000	1.129
Prod.	S2	1.435	−2.255	1.415	1.472	−1.102	−4.943	−1.571	1.645	24	1.514	0.314	0.232	1.200
Prod.	S3	1.426	−2.312	1.406	1.462	−1.180	−4.954	−1.622	1.860	25	1.541	0.339	0.290	1.202
CP	S1	0.370	−0.715	0.365	0.379	−0.430	−1.379	−1.933	2.396	32	0.379	0.072	0.000	0.308
CP	S2	0.462	−0.570	0.456	0.473	−0.212	−1.406	−1.234	0.855	25	0.418	0.040	0.230	0.377
CP	S3	0.453	−0.627	0.447	0.463	−0.289	−1.417	−1.386	1.332	28	0.445	0.065	0.287	0.380
$\alpha^W_S = 0.575$, $\alpha^W_G = 0.575$														
Prod.	S1	1.335	−2.381	1.315	1.373	−1.201	−5.134	−1.783	2.178	26	1.471	0.349	0.000	1.122
Prod.	S2	1.428	−2.231	1.406	1.468	−0.956	−5.205	−1.562	1.622	24	1.511	0.319	0.232	1.192
Prod.	S3	1.418	−2.290	1.397	1.458	−1.043	−5.201	−1.615	1.840	25	1.537	0.343	0.290	1.194
CP	S1	0.369	−0.715	0.363	0.378	−0.424	−1.395	−1.941	2.430	31	0.376	0.070	0.000	0.306
CP	S2	0.460	−0.566	0.454	0.472	−0.181	−1.465	−1.231	0.858	24	0.416	0.041	0.230	0.375
CP	S3	0.451	−0.625	0.444	0.462	−0.267	−1.461	−1.387	1.351	27	0.442	0.064	0.287	0.377
$\alpha^W_S = 0.75$, $\alpha^W_G = 0.40$														
Prod.	S1	1.283	−2.263	1.258	1.328	−0.841	−5.580	−1.764	2.103	23	1.425	0.357	0.000	1.067

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Table E.1: Robustness Exercise: Complete Statistics *(continued)*

		Aggregate		Sectoral Outcomes				Sacrifice Ratios			Policy stance			
Shock	Scen.	$\sum \pi_t$	$\sum y_t$	$\sum \pi_{S,t}$	$\sum \pi_{G,t}$	$\sum y_{S,t}$	$\sum y_{G,t}$	SR^{rec}	SR^{vol}	$H^*_{5\%}$	$\sum i_t$	$\sum r_t$	$\sum \pi^T_t$	$\sum \pi^e_t$
Prod.	S2	1.367	−2.090	1.340	1.416	−0.524	−5.743	−1.529	1.538	20	1.460	0.331	0.232	1.129
Prod.	S3	1.358	−2.156	1.332	1.407	−0.634	−5.706	−1.587	1.768	21	1.484	0.352	0.290	1.133
CP	S1	0.362	−0.710	0.356	0.373	−0.406	−1.418	−1.964	2.538	28	0.365	0.067	0.000	0.298
CP	S2	0.445	−0.538	0.437	0.460	−0.092	−1.579	−1.209	0.865	18	0.400	0.041	0.230	0.359
CP	S3	0.436	−0.604	0.429	0.450	−0.201	−1.543	−1.383	1.411	22	0.424	0.062	0.287	0.363
Sectoral output composition														
$\omega_S^{c/y} = 0.625$, $\omega_G^{c/y} = 0.625$														
Prod.	S1	1.357	−2.432	1.339	1.389	−1.495	−4.619	−1.792	2.238	26	1.489	0.344	0.000	1.145
Prod.	S2	1.453	−2.295	1.435	1.486	−1.327	−4.555	−1.580	1.683	24	1.530	0.311	0.232	1.219
Prod.	S3	1.444	−2.350	1.426	1.477	−1.388	−4.596	−1.628	1.894	25	1.559	0.337	0.290	1.221
CP	S1	0.373	−0.717	0.368	0.382	−0.436	−1.375	−1.924	2.402	31	0.384	0.073	0.000	0.311
CP	S2	0.468	−0.582	0.463	0.478	−0.269	−1.312	−1.243	0.877	24	0.425	0.041	0.230	0.384
CP	S3	0.459	−0.637	0.454	0.469	−0.330	−1.352	−1.387	1.341	27	0.453	0.067	0.287	0.386
$\omega_S^{c/y} = 0.60$, $\omega_G^{c/y} = 0.55$														
Prod.	S1	1.325	−2.375	1.310	1.355	−1.480	−4.464	−1.792	2.197	24	1.455	0.344	0.000	1.111
Prod.	S2	1.423	−2.241	1.407	1.452	−1.315	−4.401	−1.575	1.650	22	1.497	0.311	0.232	1.186
Prod.	S3	1.413	−2.295	1.397	1.443	−1.376	−4.440	−1.624	1.859	23	1.525	0.337	0.290	1.188
CP	S1	0.359	−0.697	0.355	0.367	−0.429	−1.321	−1.940	2.386	28	0.367	0.070	0.000	0.296
CP	S2	0.455	−0.563	0.451	0.464	−0.266	−1.258	−1.237	0.863	22	0.409	0.038	0.230	0.371
CP	S3	0.446	−0.617	0.441	0.455	−0.326	−1.297	−1.384	1.328	25	0.437	0.064	0.287	0.373
$\omega_S^{c/y} = 0.80$, $\omega_G^{c/y} = 0.50$														
Prod.	S1	1.360	−2.447	1.346	1.385	−1.651	−4.304	−1.800	2.225	28	1.487	0.340	0.000	1.147
Prod.	S2	1.448	−2.297	1.434	1.474	−1.460	−4.249	−1.587	1.673	25	1.519	0.306	0.232	1.212
Prod.	S3	1.441	−2.356	1.427	1.467	−1.530	−4.283	−1.635	1.884	26	1.549	0.333	0.290	1.217
CP	S1	0.381	−0.728	0.377	0.387	−0.486	−1.294	−1.914	2.336	36	0.393	0.074	0.000	0.319
CP	S2	0.468	−0.580	0.464	0.475	−0.297	−1.240	−1.239	0.838	28	0.425	0.041	0.230	0.384
CP	S3	0.461	−0.638	0.457	0.468	−0.366	−1.273	−1.386	1.292	31	0.455	0.067	0.287	0.388
$\omega_S^{c/y} = 0.55$, $\omega_G^{c/y} = 0.70$														
Prod.	S1	1.367	−2.445	1.347	1.404	−1.438	−4.794	−1.788	2.259	26	1.503	0.346	0.000	1.157
Prod.	S2	1.466	−2.313	1.446	1.504	−1.278	−4.727	−1.577	1.700	24	1.547	0.313	0.232	1.234
Prod.	S3	1.456	−2.367	1.436	1.494	−1.337	−4.770	−1.625	1.912	25	1.575	0.339	0.290	1.236
CP	S1	0.375	−0.722	0.369	0.387	−0.421	−1.423	−1.923	2.449	31	0.387	0.073	0.000	0.313
CP	S2	0.473	−0.591	0.467	0.485	−0.263	−1.357	−1.249	0.905	24	0.430	0.041	0.230	0.389
CP	S3	0.464	−0.645	0.457	0.475	−0.321	−1.400	−1.391	1.375	27	0.458	0.067	0.287	0.391
Production labor shares														
$B_S = 0.55$, $B_G = 0.45$														
Prod.	S1	1.431	−2.582	1.423	1.447	−1.949	−4.062	−1.804	2.300	28	1.562	0.345	0.000	1.217
Prod.	S2	1.523	−2.440	1.514	1.540	−1.769	−4.005	−1.602	1.754	26	1.598	0.312	0.232	1.286
Prod.	S3	1.515	−2.497	1.507	1.532	−1.835	−4.041	−1.648	1.960	27	1.628	0.338	0.290	1.290
CP	S1	0.391	−0.741	0.388	0.397	−0.463	−1.390	−1.894	2.295	35	0.409	0.081	0.000	0.328
CP	S2	0.482	−0.600	0.479	0.489	−0.285	−1.334	−1.244	0.869	26	0.444	0.047	0.230	0.396
CP	S3	0.475	−0.656	0.471	0.481	−0.351	−1.370	−1.383	1.310	30	0.474	0.074	0.287	0.400
$B_S = 0.50$, $B_G = 0.50$														
Prod.	S1	1.511	−2.733	1.510	1.512	−2.358	−3.606	−1.809	2.362	31	1.645	0.351	0.000	1.294
Prod.	S2	1.602	−2.591	1.601	1.605	−2.179	−3.553	−1.617	1.824	28	1.679	0.317	0.232	1.363
Prod.	S3	1.595	−2.648	1.593	1.597	−2.246	−3.587	−1.661	2.025	29	1.711	0.344	0.290	1.367
CP	S1	0.407	−0.758	0.405	0.410	−0.472	−1.428	−1.864	2.249	35	0.431	0.089	0.000	0.341
CP	S2	0.498	−0.619	0.495	0.502	−0.294	−1.375	−1.243	0.888	26	0.465	0.055	0.230	0.409

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Table E.1: Robustness Exercise: Complete Statistics *(continued)*

		Aggregate		Sectoral Outcomes				Sacrifice Ratios			Policy stance			
Shock	Scen.	$\sum \pi_t$	$\sum y_t$	$\sum \pi_{S,t}$	$\sum \pi_{G,t}$	$\sum y_{S,t}$	$\sum y_{G,t}$	SR^{rec}	SR^{vol}	$H^*_{5\%}$	$\sum i_t$	$\sum r_t$	$\sum \pi^T_t$	$\sum \pi^e_t$
CP	S3	0.490	−0.675	0.488	0.494	−0.360	−1.409	−1.377	1.314	29	0.496	0.082	0.287	0.413
$B_S = 0.70$, $B_G = 0.30$														
Prod.	S1	1.192	−2.107	1.165	1.240	−0.880	−4.971	−1.768	2.005	20	1.322	0.345	0.000	0.977
Prod.	S2	1.289	−1.970	1.263	1.338	−0.709	−4.911	−1.528	1.462	19	1.365	0.313	0.232	1.052
Prod.	S3	1.279	−2.025	1.252	1.328	−0.772	−4.947	−1.584	1.676	20	1.391	0.338	0.290	1.053
CP	S1	0.327	−0.650	0.321	0.337	−0.445	−1.127	−1.989	2.600	23	0.326	0.057	0.000	0.269
CP	S2	0.423	−0.514	0.417	0.433	−0.277	−1.068	−1.215	0.845	20	0.369	0.026	0.230	0.343
CP	S3	0.413	−0.568	0.407	0.423	−0.339	−1.104	−1.378	1.354	22	0.395	0.051	0.287	0.344
$B_S = 0.40$, $B_G = 0.60$														
Prod.	S1	1.650	−2.986	1.667	1.619	−3.223	−2.431	−1.809	2.433	35	1.796	0.371	0.000	1.425
Prod.	S2	1.744	−2.852	1.760	1.715	−3.050	−2.389	−1.635	1.921	31	1.831	0.336	0.232	1.496
Prod.	S3	1.736	−2.906	1.752	1.707	−3.115	−2.418	−1.674	2.111	32	1.863	0.363	0.290	1.500
CP	S1	0.424	−0.764	0.426	0.420	−0.500	−1.379	−1.801	2.213	27	0.463	0.107	0.000	0.355
CP	S2	0.517	−0.631	0.518	0.515	−0.328	−1.337	−1.221	0.944	20	0.497	0.073	0.230	0.425
CP	S3	0.509	−0.685	0.510	0.507	−0.393	−1.366	−1.345	1.345	22	0.528	0.100	0.287	0.429
Consumption substitution														
$\eta_c = 0.50$														
Prod.	S1	1.436	−2.607	1.424	1.460	−2.018	−3.983	−1.815	2.313	31	1.559	0.337	0.000	1.222
Prod.	S2	1.532	−2.472	1.519	1.556	−1.858	−3.904	−1.614	1.761	29	1.598	0.304	0.232	1.294
Prod.	S3	1.523	−2.527	1.510	1.547	−1.920	−3.945	−1.659	1.970	30	1.628	0.330	0.290	1.298
CP	S1	0.396	−0.768	0.392	0.403	−0.584	−1.197	−1.940	2.482	37	0.404	0.071	0.000	0.333
CP	S2	0.491	−0.634	0.487	0.498	−0.426	−1.119	−1.292	0.949	30	0.443	0.038	0.230	0.406
CP	S3	0.482	−0.689	0.478	0.489	−0.487	−1.159	−1.429	1.416	33	0.473	0.064	0.287	0.409
$\eta_c = 1.25$														
Prod.	S1	1.240	−2.205	1.221	1.275	−1.033	−4.940	−1.779	2.140	20	1.371	0.344	0.000	1.027
Prod.	S2	1.330	−2.057	1.311	1.366	−0.840	−4.898	−1.547	1.589	19	1.406	0.311	0.232	1.095
Prod.	S3	1.322	−2.114	1.302	1.357	−0.906	−4.933	−1.600	1.802	20	1.435	0.337	0.290	1.098
CP	S1	0.342	−0.654	0.337	0.352	−0.307	−1.465	−1.914	2.259	25	0.354	0.074	0.000	0.280
CP	S2	0.431	−0.508	0.426	0.442	−0.115	−1.423	−1.177	0.769	20	0.389	0.041	0.230	0.347
CP	S3	0.423	−0.564	0.417	0.433	−0.182	−1.458	−1.335	1.222	22	0.417	0.067	0.287	0.350
Labor substitution														
$\eta_n = 1.20$														
Prod.	S1	1.292	−2.286	1.273	1.327	−0.874	−5.582	−1.770	2.142	46	1.427	0.339	0.000	1.088
Prod.	S2	1.381	−2.136	1.362	1.417	−0.661	−5.578	−1.547	1.588	44	1.461	0.306	0.232	1.155
Prod.	S3	1.373	−2.193	1.353	1.409	−0.728	−5.612	−1.598	1.805	45	1.490	0.332	0.290	1.158
CP	S1	0.346	−0.655	0.340	0.358	−0.159	−1.810	−1.890	2.197	55	0.360	0.072	0.000	0.288
CP	S2	0.435	−0.506	0.428	0.447	0.051	−1.806	−1.164	0.726	50	0.394	0.039	0.230	0.355
CP	S3	0.426	−0.562	0.419	0.439	−0.015	−1.840	−1.319	1.189	53	0.423	0.065	0.287	0.358
$\eta_n = 2.50$														
Prod.	S1	1.362	−2.453	1.348	1.388	−1.695	−4.221	−1.801	2.240	25	1.489	0.342	0.000	1.147
Prod.	S2	1.456	−2.312	1.441	1.482	−1.524	−4.151	−1.589	1.690	23	1.527	0.309	0.232	1.218
Prod.	S3	1.447	−2.369	1.433	1.474	−1.588	−4.190	−1.637	1.899	24	1.557	0.335	0.290	1.221
CP	S1	0.378	−0.730	0.374	0.385	−0.516	−1.228	−1.931	2.402	31	0.388	0.073	0.000	0.315
CP	S2	0.471	−0.591	0.467	0.478	−0.347	−1.159	−1.255	0.887	24	0.426	0.040	0.230	0.386
CP	S3	0.462	−0.647	0.458	0.470	−0.411	−1.197	−1.399	1.346	27	0.455	0.066	0.287	0.389
Backward-looking price setters														
$\omega_S = 0.15$, $\omega_G = 0.15$														
Prod.	S1	1.356	−2.409	1.341	1.385	−1.532	−4.454	−1.776	2.033	21	1.500	0.402	0.000	1.098

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Table E.1: Robustness Exercise: Complete Statistics *(continued)*

		Aggregate		Sectoral Outcomes				Sacrifice Ratios			Policy stance			
Shock	Scen.	$\sum \pi_t$	$\sum y_t$	$\sum \pi_{S,t}$	$\sum \pi_{G,t}$	$\sum y_{S,t}$	$\sum y_{G,t}$	SR^{rec}	SR^{vol}	$H^*_{5\%}$	$\sum i_t$	$\sum r_t$	$\sum \pi^T_t$	$\sum \pi^e_t$
Prod.	S2	1.450	−2.264	1.434	1.479	−1.351	−4.395	−1.562	1.547	19	1.539	0.375	0.232	1.164
Prod.	S3	1.441	−2.322	1.425	1.471	−1.417	−4.433	−1.611	1.740	20	1.568	0.398	0.290	1.170
CP	S1	0.374	−0.712	0.370	0.383	−0.449	−1.326	−1.902	2.102	26	0.390	0.091	0.000	0.299
CP	S2	0.467	−0.569	0.462	0.476	−0.269	−1.269	−1.219	0.793	20	0.429	0.064	0.230	0.365
CP	S3	0.458	−0.626	0.454	0.467	−0.335	−1.307	−1.367	1.217	22	0.457	0.087	0.287	0.370
$\omega_S = 0.45$, $\omega_G = 0.45$														
Prod.	S1	1.340	−2.440	1.325	1.367	−1.598	−4.403	−1.821	2.437	34	1.451	0.279	0.000	1.172
Prod.	S2	1.432	−2.301	1.417	1.460	−1.426	−4.342	−1.607	1.809	31	1.485	0.239	0.232	1.246
Prod.	S3	1.424	−2.356	1.409	1.452	−1.489	−4.378	−1.655	2.035	32	1.516	0.269	0.290	1.247
CP	S1	0.369	−0.720	0.365	0.377	−0.464	−1.318	−1.951	2.679	40	0.375	0.054	0.000	0.320
CP	S2	0.460	−0.582	0.456	0.469	−0.293	−1.258	−1.265	0.926	33	0.409	0.015	0.230	0.394
CP	S3	0.452	−0.637	0.448	0.461	−0.356	−1.293	−1.408	1.415	36	0.439	0.044	0.287	0.395
$\omega_S = 0.45$, $\omega_G = 0.15$														
Prod.	S1	1.356	−2.417	1.342	1.382	−1.578	−4.376	−1.783	2.191	27	1.494	0.350	0.000	1.144
Prod.	S2	1.450	−2.277	1.436	1.477	−1.409	−4.303	−1.570	1.631	25	1.534	0.317	0.232	1.218
Prod.	S3	1.442	−2.333	1.427	1.468	−1.471	−4.345	−1.619	1.838	26	1.563	0.343	0.290	1.220
CP	S1	0.373	−0.712	0.369	0.381	−0.456	−1.311	−1.909	2.328	32	0.387	0.076	0.000	0.311
CP	S2	0.467	−0.573	0.463	0.475	−0.288	−1.239	−1.227	0.818	26	0.427	0.043	0.230	0.383
CP	S3	0.458	−0.629	0.454	0.466	−0.350	−1.281	−1.373	1.256	28	0.455	0.069	0.287	0.386
$\omega_S = 0.15$, $\omega_G = 0.45$														
Prod.	S1	1.340	−2.431	1.324	1.370	−1.555	−4.476	−1.814	2.303	25	1.457	0.330	0.000	1.127
Prod.	S2	1.431	−2.288	1.415	1.462	−1.370	−4.430	−1.598	1.744	23	1.491	0.296	0.232	1.195
Prod.	S3	1.424	−2.345	1.407	1.454	−1.438	−4.462	−1.647	1.959	24	1.521	0.323	0.290	1.198
CP	S1	0.370	−0.720	0.366	0.378	−0.457	−1.332	−1.944	2.469	33	0.377	0.068	0.000	0.309
CP	S2	0.461	−0.578	0.456	0.470	−0.275	−1.286	−1.256	0.913	25	0.411	0.035	0.230	0.376
CP	S3	0.453	−0.634	0.448	0.462	−0.341	−1.318	−1.401	1.393	28	0.441	0.061	0.287	0.380
Price indexation														
$l_S = 0.70$, $l_G = 0.70$														
Prod.	S1	1.323	−2.367	1.307	1.351	−1.510	−4.365	−1.790	2.196	25	1.454	0.345	0.000	1.108
Prod.	S2	1.414	−2.221	1.398	1.443	−1.329	−4.303	−1.571	1.643	23	1.489	0.312	0.232	1.177
Prod.	S3	1.406	−2.278	1.390	1.434	−1.395	−4.340	−1.621	1.854	24	1.519	0.338	0.290	1.180
CP	S1	0.365	−0.700	0.360	0.373	−0.442	−1.304	−1.920	2.338	31	0.376	0.074	0.000	0.302
CP	S2	0.455	−0.556	0.450	0.464	−0.262	−1.242	−1.222	0.826	24	0.412	0.041	0.230	0.370
CP	S3	0.447	−0.612	0.442	0.455	−0.327	−1.279	−1.371	1.284	27	0.441	0.067	0.287	0.374
$l_S = 1.00$, $l_G = 1.00$														
Prod.	S1	1.364	−2.454	1.348	1.392	−1.589	−4.470	−1.799	2.234	26	1.492	0.341	0.000	1.151
Prod.	S2	1.458	−2.314	1.442	1.487	−1.414	−4.413	−1.587	1.684	24	1.530	0.307	0.232	1.223
Prod.	S3	1.449	−2.370	1.433	1.478	−1.479	−4.450	−1.636	1.894	25	1.559	0.334	0.290	1.226
CP	S1	0.376	−0.724	0.372	0.384	−0.463	−1.334	−1.925	2.378	33	0.387	0.073	0.000	0.314
CP	S2	0.469	−0.586	0.465	0.478	−0.289	−1.277	−1.248	0.869	26	0.425	0.040	0.230	0.385
CP	S3	0.461	−0.642	0.456	0.469	−0.354	−1.314	−1.393	1.328	29	0.454	0.066	0.287	0.388
$l_S = 1.00$, $l_G = 0.70$														
Prod.	S1	1.367	−2.457	1.353	1.391	−1.700	−4.225	−1.798	2.237	26	1.496	0.343	0.000	1.153
Prod.	S2	1.460	−2.317	1.447	1.485	−1.529	−4.155	−1.587	1.686	24	1.534	0.310	0.232	1.223
Prod.	S3	1.452	−2.373	1.438	1.477	−1.594	−4.191	−1.635	1.895	25	1.563	0.337	0.290	1.227
CP	S1	0.377	−0.726	0.374	0.384	−0.495	−1.264	−1.924	2.383	33	0.388	0.073	0.000	0.315
CP	S2	0.470	−0.587	0.466	0.477	−0.326	−1.195	−1.248	0.871	26	0.426	0.040	0.230	0.385
CP	S3	0.462	−0.643	0.458	0.469	−0.391	−1.231	−1.392	1.329	28	0.455	0.067	0.287	0.389
$l_S = 0.70$, $l_G = 1.00$														

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Table E.1: Robustness Exercise: Complete Statistics *(continued)*

		Aggregate		Sectoral Outcomes				Sacrifice Ratios			Policy stance			
Shock	Scen.	$\sum \pi_t$	$\sum y_t$	$\sum \pi_{S,t}$	$\sum \pi_{G,t}$	$\sum y_{S,t}$	$\sum y_{G,t}$	SR^{rec}	SR^{vol}	$H^*_{5\%}$	$\sum i_t$	$\sum r_t$	$\sum \pi^T_t$	$\sum \pi^e_t$
Prod.	S1	1.320	−2.364	1.303	1.352	−1.405	−4.602	−1.791	2.193	25	1.450	0.342	0.000	1.107
Prod.	S2	1.412	−2.219	1.394	1.444	−1.219	−4.552	−1.572	1.641	23	1.486	0.309	0.232	1.177
Prod.	S3	1.403	−2.275	1.385	1.436	−1.284	−4.589	−1.622	1.852	24	1.515	0.336	0.290	1.180
CP	S1	0.364	−0.699	0.359	0.373	−0.411	−1.370	−1.921	2.332	31	0.375	0.073	0.000	0.302
CP	S2	0.454	−0.555	0.449	0.464	−0.227	−1.321	−1.222	0.825	24	0.411	0.040	0.230	0.370
CP	S3	0.446	−0.611	0.441	0.456	−0.291	−1.358	−1.371	1.283	27	0.440	0.066	0.287	0.373
Wage indexation														
$l^W_S = 0.25$, $l^W_G = 0.25$														
Prod.	S1	1.350	−2.422	1.338	1.373	−1.705	−4.093	−1.794	2.197	25	1.481	0.347	0.000	1.134
Prod.	S2	1.443	−2.279	1.430	1.466	−1.544	−3.992	−1.580	1.651	23	1.518	0.315	0.232	1.204
Prod.	S3	1.435	−2.336	1.422	1.458	−1.606	−4.041	−1.628	1.859	24	1.548	0.341	0.290	1.207
CP	S1	0.372	−0.715	0.369	0.379	−0.492	−1.237	−1.922	2.342	31	0.384	0.074	0.000	0.309
CP	S2	0.464	−0.574	0.460	0.471	−0.332	−1.138	−1.237	0.844	25	0.420	0.042	0.230	0.379
CP	S3	0.456	−0.631	0.452	0.463	−0.393	−1.186	−1.383	1.298	27	0.450	0.068	0.287	0.382
$l^W_S = 0.75$, $l^W_G = 0.75$														
Prod.	S1	1.349	−2.427	1.331	1.383	−1.421	−4.773	−1.798	2.247	26	1.477	0.337	0.000	1.140
Prod.	S2	1.443	−2.286	1.424	1.477	−1.227	−4.755	−1.584	1.690	24	1.514	0.303	0.232	1.211
Prod.	S3	1.434	−2.342	1.415	1.468	−1.296	−4.780	−1.633	1.903	25	1.543	0.329	0.290	1.213
CP	S1	0.372	−0.717	0.367	0.382	−0.420	−1.409	−1.925	2.387	33	0.383	0.072	0.000	0.311
CP	S2	0.465	−0.577	0.459	0.475	−0.228	−1.392	−1.242	0.865	26	0.420	0.038	0.230	0.382
CP	S3	0.456	−0.633	0.450	0.466	−0.297	−1.417	−1.388	1.329	29	0.448	0.064	0.287	0.384
$l^W_S = 0.75$, $l^W_G = 0.25$														
Prod.	S1	1.362	−2.447	1.350	1.384	−1.760	−4.049	−1.796	2.212	26	1.493	0.346	0.000	1.147
Prod.	S2	1.456	−2.307	1.444	1.479	−1.605	−3.943	−1.584	1.666	24	1.531	0.314	0.232	1.218
Prod.	S3	1.448	−2.363	1.436	1.470	−1.665	−3.993	−1.632	1.873	25	1.561	0.340	0.290	1.221
CP	S1	0.375	−0.722	0.372	0.382	−0.506	−1.226	−1.923	2.359	32	0.387	0.074	0.000	0.313
CP	S2	0.469	−0.583	0.465	0.475	−0.353	−1.121	−1.245	0.860	25	0.425	0.041	0.230	0.383
CP	S3	0.460	−0.639	0.457	0.467	−0.412	−1.171	−1.389	1.313	28	0.454	0.068	0.287	0.387
$l^W_S = 0.25$, $l^W_G = 0.75$														
Prod.	S1	1.338	−2.402	1.319	1.372	−1.370	−4.812	−1.796	2.231	26	1.465	0.338	0.000	1.127
Prod.	S2	1.430	−2.259	1.410	1.465	−1.170	−4.799	−1.580	1.675	24	1.501	0.304	0.232	1.197
Prod.	S3	1.421	−2.315	1.402	1.456	−1.241	−4.823	−1.629	1.888	25	1.530	0.331	0.290	1.200
CP	S1	0.369	−0.711	0.364	0.379	−0.407	−1.419	−1.924	2.370	33	0.380	0.072	0.000	0.308
CP	S2	0.460	−0.568	0.455	0.471	−0.209	−1.406	−1.234	0.850	25	0.416	0.039	0.230	0.377
CP	S3	0.452	−0.624	0.446	0.463	−0.279	−1.430	−1.382	1.315	28	0.445	0.065	0.287	0.380
Aggregate output weight														
$\omega_y = 0.50$														
Prod.	S1	1.572	−2.913	1.556	1.602	−1.412	−4.415	−1.853	2.488	32	1.677	0.329	0.000	1.348
Prod.	S2	1.673	−2.790	1.657	1.704	−1.226	−4.355	−1.668	1.944	29	1.721	0.295	0.232	1.426
Prod.	S3	1.664	−2.843	1.647	1.694	−1.294	−4.393	−1.709	2.146	30	1.750	0.322	0.290	1.428
CP	S1	0.436	−0.861	0.432	0.445	−0.408	−1.315	−1.974	2.705	37	0.438	0.068	0.000	0.370
CP	S2	0.536	−0.739	0.531	0.545	−0.223	−1.255	−1.379	1.156	30	0.481	0.034	0.230	0.447
CP	S3	0.527	−0.792	0.522	0.536	−0.291	−1.293	−1.503	1.629	33	0.511	0.061	0.287	0.450
$\omega_y = 0.20$														
Prod.	S1	1.246	−2.196	1.231	1.274	−1.634	−4.444	−1.762	2.079	23	1.387	0.348	0.000	1.039
Prod.	S2	1.336	−2.045	1.321	1.364	−1.460	−4.386	−1.531	1.528	21	1.422	0.316	0.232	1.106
Prod.	S3	1.328	−2.104	1.312	1.356	−1.524	−4.422	−1.585	1.741	22	1.451	0.342	0.290	1.109
CP	S1	0.343	−0.648	0.338	0.350	−0.478	−1.328	−1.893	2.179	30	0.358	0.076	0.000	0.282
CP	S2	0.431	−0.499	0.427	0.440	−0.307	−1.270	−1.158	0.709	23	0.392	0.043	0.230	0.349

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Table E.1: Robustness Exercise: Complete Statistics *(continued)*

		Aggregate		Sectoral Outcomes				Sacrifice Ratios			Policy stance			
Shock	Scen.	$\sum \pi_t$	$\sum y_t$	$\sum \pi_{S,t}$	$\sum \pi_{G,t}$	$\sum y_{S,t}$	$\sum y_{G,t}$	SR^{rec}	SR^{vol}	$H^*_{5\%}$	$\sum i_t$	$\sum r_t$	$\sum \pi^T_t$	$\sum \pi^e_t$
CP	S3	0.423	−0.557	0.419	0.432	−0.370	−1.306	−1.317	1.152	26	0.421	0.069	0.287	0.352
$\omega_y = 0.40$														
Prod.	S1	1.458	−2.663	1.443	1.487	−1.489	−4.425	−1.826	2.358	29	1.575	0.336	0.000	1.240
Prod.	S2	1.555	−2.530	1.539	1.585	−1.307	−4.365	−1.627	1.809	27	1.616	0.302	0.232	1.314
Prod.	S3	1.546	−2.585	1.530	1.576	−1.373	−4.402	−1.672	2.015	27	1.645	0.329	0.290	1.316
CP	S1	0.404	−0.787	0.399	0.412	−0.432	−1.319	−1.950	2.540	35	0.410	0.070	0.000	0.340
CP	S2	0.499	−0.655	0.495	0.508	−0.252	−1.260	−1.313	1.004	27	0.450	0.037	0.230	0.413
CP	S3	0.491	−0.710	0.486	0.499	−0.318	−1.297	−1.447	1.473	30	0.479	0.063	0.287	0.416
$\omega_y = 0.70$														
Prod.	S1	1.818	−3.451	1.801	1.850	−1.249	−4.395	−1.898	2.729	37	1.898	0.316	0.000	1.583
Prod.	S2	1.928	−3.350	1.911	1.960	−1.054	−4.334	−1.737	2.202	35	1.949	0.281	0.232	1.669
Prod.	S3	1.918	−3.398	1.900	1.950	−1.125	−4.373	−1.772	2.392	36	1.979	0.308	0.290	1.670
CP	S1	0.507	−1.021	0.503	0.516	−0.356	−1.306	−2.013	3.005	42	0.500	0.063	0.000	0.437
CP	S2	0.616	−0.920	0.611	0.626	−0.162	−1.245	−1.494	1.459	35	0.550	0.028	0.230	0.522
CP	S3	0.606	−0.968	0.601	0.615	−0.232	−1.284	−1.599	1.928	38	0.579	0.055	0.287	0.524
Aggregate inflation weight														
$\omega_\pi = 0.50$														
Prod.	S1	1.362	−2.437	1.340	1.384	−1.571	−4.457	−1.789	2.146	24	1.497	0.364	0.000	1.133
Prod.	S2	1.456	−2.296	1.433	1.478	−1.394	−4.399	−1.577	1.628	22	1.535	0.332	0.232	1.204
Prod.	S3	1.447	−2.352	1.424	1.469	−1.459	−4.435	−1.626	1.830	23	1.564	0.357	0.290	1.207
CP	S1	0.377	−0.720	0.371	0.383	−0.458	−1.332	−1.912	2.227	29	0.390	0.081	0.000	0.309
CP	S2	0.470	−0.581	0.463	0.476	−0.283	−1.275	−1.237	0.841	23	0.428	0.049	0.230	0.379
CP	S3	0.461	−0.637	0.454	0.467	−0.348	−1.311	−1.382	1.280	26	0.456	0.074	0.287	0.382
$\omega_\pi = 0.65$														
Prod.	S1	1.374	−2.449	1.346	1.390	−1.579	−4.479	−1.782	2.077	23	1.515	0.386	0.000	1.130
Prod.	S2	1.469	−2.309	1.439	1.484	−1.403	−4.423	−1.572	1.589	21	1.554	0.354	0.232	1.201
Prod.	S3	1.459	−2.365	1.430	1.475	−1.468	−4.458	−1.621	1.782	22	1.582	0.379	0.290	1.204
CP	S1	0.381	−0.725	0.373	0.385	−0.461	−1.340	−1.901	2.107	27	0.397	0.089	0.000	0.308
CP	S2	0.475	−0.586	0.466	0.479	−0.287	−1.285	−1.235	0.830	21	0.436	0.058	0.230	0.378
CP	S3	0.465	−0.641	0.457	0.470	−0.351	−1.320	−1.378	1.251	24	0.463	0.082	0.287	0.381
Aggregate labor weight														
$\omega_n = 0.50$														
Prod.	S1	1.277	−2.277	1.263	1.304	−1.436	−4.241	−1.783	2.129	26	1.407	0.338	0.000	1.070
Prod.	S2	1.367	−2.128	1.352	1.394	−1.252	−4.173	−1.557	1.574	24	1.441	0.305	0.232	1.137
Prod.	S3	1.359	−2.186	1.344	1.386	−1.318	−4.211	−1.608	1.788	25	1.471	0.331	0.290	1.140
CP	S1	0.338	−0.647	0.334	0.345	−0.395	−1.235	−1.917	2.214	34	0.349	0.069	0.000	0.279
CP	S2	0.426	−0.500	0.422	0.434	−0.213	−1.168	−1.172	0.711	26	0.382	0.036	0.230	0.346
CP	S3	0.419	−0.557	0.414	0.426	−0.279	−1.206	−1.331	1.162	29	0.412	0.063	0.287	0.349
$\omega_n = 0.65$														
Prod.	S1	1.213	−2.147	1.199	1.238	−1.324	−4.069	−1.770	2.038	27	1.344	0.333	0.000	1.010
Prod.	S2	1.299	−1.991	1.285	1.325	−1.134	−3.992	−1.533	1.480	25	1.374	0.300	0.232	1.074
Prod.	S3	1.292	−2.050	1.278	1.318	−1.201	−4.032	−1.587	1.697	26	1.405	0.326	0.290	1.078
CP	S1	0.307	−0.586	0.303	0.313	−0.341	−1.157	−1.911	2.056	36	0.317	0.066	0.000	0.252
CP	S2	0.392	−0.432	0.388	0.399	−0.153	−1.081	−1.101	0.580	27	0.348	0.033	0.230	0.315
CP	S3	0.385	−0.490	0.381	0.392	−0.220	−1.120	−1.273	1.013	31	0.378	0.059	0.287	0.319
Factor substitution asymmetry														
$\sigma_S = 0.60$, $\sigma_G = 0.70$														
Prod.	S1	1.304	−2.336	1.285	1.339	−1.397	−4.528	−1.792	2.187	24	1.432	0.341	0.000	1.091

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Table E.1: Robustness Exercise: Complete Statistics *(continued)*

		Aggregate		Sectoral Outcomes				Sacrifice Ratios			Policy stance			
Shock	Scen.	$\sum \pi_t$	$\sum y_t$	$\sum \pi_{S,t}$	$\sum \pi_{G,t}$	$\sum y_{S,t}$	$\sum y_{G,t}$	SR^{rec}	SR^{vol}	$H^*_{5\%}$	$\sum i_t$	$\sum r_t$	$\sum \pi^T_t$	$\sum \pi^e_t$
Prod.	S2	1.396	−2.190	1.376	1.431	−1.211	−4.475	−1.569	1.631	22	1.468	0.309	0.232	1.160
Prod.	S3	1.387	−2.248	1.368	1.423	−1.278	−4.510	−1.620	1.844	23	1.498	0.334	0.290	1.163
CP	S1	0.371	−0.722	0.366	0.379	−0.520	−1.194	−1.948	2.481	29	0.377	0.069	0.000	0.309
CP	S2	0.461	−0.577	0.457	0.470	−0.336	−1.141	−1.252	0.890	23	0.414	0.036	0.230	0.377
CP	S3	0.453	−0.635	0.448	0.462	−0.403	−1.176	−1.401	1.367	26	0.442	0.062	0.287	0.381
$\sigma_S = 0.65$, $\sigma_G = 0.65$														
Prod.	S1	1.284	−2.298	1.264	1.322	−1.327	−4.562	−1.789	2.169	23	1.412	0.342	0.000	1.070
Prod.	S2	1.375	−2.150	1.354	1.413	−1.137	−4.511	−1.563	1.612	21	1.448	0.310	0.232	1.138
Prod.	S3	1.367	−2.208	1.346	1.405	−1.206	−4.546	−1.616	1.826	22	1.477	0.335	0.290	1.142
CP	S1	0.370	−0.727	0.366	0.378	−0.554	−1.130	−1.961	2.550	28	0.375	0.066	0.000	0.309
CP	S2	0.460	−0.580	0.456	0.469	−0.366	−1.080	−1.260	0.912	23	0.411	0.034	0.230	0.377
CP	S3	0.452	−0.638	0.448	0.461	−0.434	−1.114	−1.410	1.399	25	0.439	0.059	0.287	0.380
$\sigma_S = 0.80$, $\sigma_G = 0.50$														
Prod.	S1	1.233	−2.196	1.208	1.279	−1.152	−4.634	−1.781	2.107	20	1.362	0.349	0.000	1.013
Prod.	S2	1.322	−2.043	1.297	1.369	−0.952	−4.590	−1.545	1.552	19	1.397	0.318	0.232	1.079
Prod.	S3	1.314	−2.103	1.289	1.361	−1.023	−4.621	−1.600	1.767	20	1.425	0.343	0.290	1.083
CP	S1	0.372	−0.745	0.368	0.379	−0.662	−0.940	−2.005	2.803	26	0.368	0.057	0.000	0.311
CP	S2	0.460	−0.593	0.456	0.468	−0.464	−0.896	−1.290	0.998	22	0.404	0.026	0.230	0.378
CP	S3	0.452	−0.652	0.448	0.460	−0.535	−0.927	−1.444	1.520	24	0.432	0.051	0.287	0.381
Investment adjustment costs														
$\phi^I_S = 3.00$, $\phi^I_G = 3.00$														
Prod.	S1	1.356	−2.527	1.340	1.386	−1.595	−4.702	−1.864	2.526	29	1.439	0.291	0.000	1.148
Prod.	S2	1.447	−2.374	1.431	1.478	−1.404	−4.638	−1.640	1.881	26	1.479	0.262	0.232	1.217
Prod.	S3	1.440	−2.437	1.423	1.471	−1.476	−4.680	−1.693	2.128	27	1.506	0.285	0.290	1.221
CP	S1	0.376	−0.760	0.371	0.385	−0.479	−1.418	−2.022	2.869	36	0.368	0.052	0.000	0.316
CP	S2	0.466	−0.609	0.461	0.476	−0.289	−1.354	−1.305	1.009	28	0.407	0.023	0.230	0.384
CP	S3	0.459	−0.671	0.454	0.468	−0.360	−1.396	−1.463	1.572	31	0.435	0.047	0.287	0.388
$\phi^I_S = 7.00$, $\phi^I_G = 7.00$														
Prod.	S1	1.344	−2.353	1.330	1.371	−1.543	−4.243	−1.751	2.053	24	1.504	0.375	0.000	1.129
Prod.	S2	1.438	−2.218	1.424	1.466	−1.375	−4.187	−1.542	1.552	22	1.540	0.339	0.232	1.200
Prod.	S3	1.429	−2.272	1.415	1.457	−1.436	−4.221	−1.589	1.743	23	1.570	0.367	0.290	1.203
CP	S1	0.369	−0.687	0.365	0.377	−0.442	−1.258	−1.859	2.100	30	0.393	0.086	0.000	0.306
CP	S2	0.463	−0.553	0.458	0.471	−0.275	−1.202	−1.195	0.769	23	0.428	0.051	0.230	0.378
CP	S3	0.454	−0.606	0.450	0.462	−0.335	−1.236	−1.334	1.176	26	0.458	0.078	0.287	0.380
$\phi^I_S = 7.00$, $\phi^I_G = 3.00$														
Prod.	S1	1.338	−2.464	1.322	1.369	−1.484	−4.750	−1.841	2.420	29	1.436	0.303	0.000	1.133
Prod.	S2	1.433	−2.324	1.416	1.464	−1.315	−4.679	−1.622	1.811	27	1.475	0.270	0.232	1.204
Prod.	S3	1.424	−2.380	1.407	1.455	−1.377	−4.723	−1.672	2.040	28	1.503	0.296	0.290	1.207
CP	S1	0.367	−0.729	0.363	0.376	−0.424	−1.439	−1.983	2.657	36	0.367	0.059	0.000	0.309
CP	S2	0.461	−0.590	0.456	0.470	−0.256	−1.369	−1.280	0.943	29	0.406	0.027	0.230	0.379
CP	S3	0.452	−0.646	0.447	0.462	−0.318	−1.412	−1.429	1.450	32	0.434	0.052	0.287	0.382
$\phi^I_S = 3.00$, $\phi^I_G = 7.00$														
Prod.	S1	1.364	−2.432	1.349	1.390	−1.672	−4.205	−1.784	2.188	24	1.503	0.357	0.000	1.146
Prod.	S2	1.455	−2.283	1.440	1.482	−1.482	−4.153	−1.569	1.645	22	1.540	0.325	0.232	1.215
Prod.	S3	1.447	−2.343	1.432	1.474	−1.553	−4.187	−1.619	1.856	23	1.569	0.351	0.290	1.218
CP	S1	0.378	−0.724	0.375	0.386	−0.502	−1.241	−1.913	2.342	30	0.392	0.077	0.000	0.314
CP	S2	0.469	−0.576	0.464	0.477	−0.313	−1.190	−1.229	0.850	23	0.429	0.046	0.230	0.383
CP	S3	0.461	−0.636	0.457	0.469	−0.384	−1.223	−1.379	1.316	26	0.457	0.071	0.287	0.386

Note: All cumulative variables are computed over the fixed 40-period horizon, except for the convergence statistic $H^*_{5\%}$. The convergence horizon $H^*_{5\%}$ denotes the first horizon at which aggregate inflation falls to at most 5% of its initial response and remains there for four consecutive periods. The policy-stance columns report the cumulative nominal interest rate i_t, real interest rate r_t, target component π^T_t, and implied expected inflation $\pi^e_t = i_t - r_t$. $S1$ denotes the inactive mechanism under high credibility and the short policy horizon. $S2$ denotes the active mechanism under high credibility and the short policy horizon. $S3$ denotes the active mechanism under low credibility and the extended policy horizon. The recession sacrifice ratio is $SR^{rec} = \sum_t \min(y_t, 0) / \sum_t \pi_t$. $SR^{vol} = \sum_t y_t^2 / \sum_t \pi_t^2$. "Prod." denotes the productivity shock and "CP" denotes the cost-push shock.